

System Startup Guide Zip Drive

SW11-614

**Implementation
Startup & Reconfiguration - 1**

***System Startup Guide
Zip Drive***

**SW11-614
Release 681
9/08**

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About This Publication

The Zip drive is part of the **TotalPlant** Solution (TPS) system. This publication provides an example of the startup of a TPS system for Release 650 and later, using Honeywell-supplied Zip disks.

The tasks presented are intended to show you how a system is initially configured. You should have filled in configuration forms based on the system you are working with, so you know exactly what information needs to be entered.

If you are upgrading to a newer release of software, consult the *Customer Release Guide* for procedures. If you are reconfiguring an existing LCN network, refer to the *Network Data Entry* manual (see References).

You should know how to use the Command Processor, the Data Entity Builder, the Picture Editor, the Text Editor, etc., before attempting to install a new system. This information is explained in the manuals listed in the References section of this manual.

IMPORTANT

- A. The Zip disk should be removed from the drive when not in use to avoid unnecessary shortening of the drive life.
- B. Write-Protect for the root volume on the Zip disk is accomplished by using the Protect (PROTV) or the Unprotect (UNPTV) command. When these commands are used in EC files, the -NQ option is required to prevent the system from waiting for user input.
- C. Inserting and removing a Zip disk:
Always insert the Zip disk in a smooth fashion. Upon inserting a Zip disk into the drive, wait for the following events to occur:
 - 1. Yellow light blinks several times
 - 2. Yellow light stops blinkingTo remove a Zip disk from the drive:
 - 1. Push the Eject button.
 - 2. Wait until the yellow light stops flashing (is extinguished) and the Zip disk pops partway out with a "click." If the Zip disk resists removal, push the Ejection button again.

WARNING

Attempted removal of the disk without following the above procedure can result in serious damage to both the disk/disk drive.

Support and Other Contacts

Location	Organization	Address	Phone/Facsimile
United States and Canada	Honeywell Solution Support Center	Honeywell TAC, MS P13, 2500 West Union Hills Drive, Phoenix, AZ, 85027	Phone: 1-800 822-7673 602- 313-5558 (In Arizona) Facsimile: (602) 313-3293
Note: Calls are answered by dispatcher between 6:00 A.M. and 4:00 P.M. Mountain Standard Time. Emergency calls outside normal working hours are received by an answering service and returned within one hour.			
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Pacific	Honeywell Global TAC – Pacific	Honeywell Limited Australia 5 Kitchener Way Burswood 6100, Western Australia Email: GTAC@honeywell.com	Phone: 1300-300-4822 (toll free within Australia) +61-8-9362-9559 (outside Australia) Facsimile: +61-8-9362-9564
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Elsewhere	Call your nearest Honeywell office.		
World Wide Web	Honeywell Solution Support Online: http://www.honeywell.com/ps		
Training Classes	Honeywell Automation College: http://www.automationcollege.com		

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Section 1 – Introduction

1.1 Scope of Coverage and Suggestions

This "how-to" document covers the engineering tasks necessary to initially configure a History Module-based **TotalPlant** Solution (TPS) system. Its step-by-step instructions cover all file and Zip disk manipulations necessary for initial system startup. It does not cover all permutations of data that can be entered. For instance, in developing the guide, we configured one Unit, one Area, a few points, and one Group; otherwise, the step-by-step instructions would have been far too bulky.

During Tasks 6-11 you should configure as much of the network as possible. At the very least you will have to use the Online Network Reconfiguration procedures if you later wish to change the Network Configuration. Pay particular attention to accurately and completely configuring your History Modules; otherwise, you will have to perform time-consuming procedures if you have to change the History Module's volume configuration.

Refer to the *Customer Release Guide* to determine exactly what functions are included in the current software.

The procedures described in this publication require at least one Universal Station with dual Zip drives from which configuration operations are performed.

Universal Station^X, Micro TDC, and Universal Workstation

After the system is configured, any Universal Station^X (U^XS) can easily be loaded as described in Task 30.

If the system contains at least one Universal Station^X with an Engineering Keyboard and dual Zip drives, it should be possible to perform the procedures described in this publication, but they have not been tested for that situation.

U^XS station preparation — The U^XS station must contain software that is compatible with the system software (this software is updated with DAT cartridges). Refer to the *Universal Station^X User's Guide* for the loading procedure.

An Engineer's Workstation or Micro TDC station can also be used, if the slight differences are taken into account.

See *System Startup Guide – CD-ROM* when using GUS.

1.2 How This Startup Guide Is Organized

This guide is organized as follows:

Section (Number and Title)	You Should Use the Information to...
1. INTRODUCTION	Learn what this document is all about
2. OVERVIEW	Understand the overall System Startup process
3. INSTRUCTIONS General Instructions System Startup Task Record Procedures One or more Tasks Multiple Instructions	Take action necessary to start up your system Understand the format of standard instructions Check off tasks completed and record information Know major objective for each set of tasks Know objective represented by each task Know exact steps to accomplish each task
4. ERROR RECOVERY PROCEDURES	Solve a problem that you have encountered
A. APPENDIX	Load the Operating Personality into a second Universal Station
B. APPENDIX	Save Hiway Gateway checkpoint data in the History Module and on Zip disks
C. APPENDIX	Save Application Module checkpoint data in the History Module and on Zip disks
D. APPENDIX	Build points by exception
E. APPENDIX	Create exception build files from IDFs

1.3 How to Use the Procedures in This Document

The procedures in this manual were developed on a sample system. Use them as a guide for starting up a system.

One person can easily perform these procedures, but if two people are available, the most efficient method is to have one person read the step-by-step instructions while another person keys in the data.

1.4 What to Do before Starting-Up the System

Format History Modules

The system's hardware must be installed and checked out before performing system startup. Each History Module (HM) must be formatted before system startup. This is normally done at the factory. See your maintenance technician or Honeywell support engineer if there is any question.

Have Reference Material Available

You should have access to the publications listed under References during the entire system startup process.

Understand Various Responses after the Enter Key Is Pressed

In the Command Processor function, when the <ENTER> key is pressed, the command entered is redisplayed with all characters in the command in uppercase. In the Network Configuration functions, when the <ENTER> key is pressed, the data is entered and reappears in upper case and in cyan. Also in the Network Configuration functions, when a target (pick) is selected, it usually fills with full intensity white. After the <ENTER> key is pressed, the target reappears filled with half-intensity white. In Point Build, HG Library, and Area Database Parameter Entry Displays (PEDs), when the <ENTER> key is pressed the data entered reappears in cyan and in uppercase.

Understand Responses when Using Zip Disks

When a Zip disk is inserted, the system usually reads the volume name for use in subsequent displays. Wait for the steady green light on the front of the drive before performing the next instruction (see the notice regarding handling of Zip disks at the front of this manual).

Understand the Function of the HELP key

When you are entering configuration data, help information is available. If you get a red error message or if a data entry region reappears in red after you press the <ENTER> key, press the <HELP> key and any available help information is displayed.

Understand the Concept of Work Files

Work files accumulate configuration data that you enter until you perform an "install" function, which renames the work file to become your permanent site-specific configuration file.

1.4 What to Do before Starting-Up the System (continued)

Understand Time Delays

As you perform some of the procedures in this guide, time delays can occur. You may have to wait up to several minutes for a few of the responses listed in the instructions.

Printer Use

It is an advantage to have a printer attached to the Universal Station where the configuration tasks are being performed. This allows you to perform screen-print and file-print operations on that printer.

Understand How to Respond to Messages

The *TPS Messages Directory* gives you a definition of the message and error recovery procedures, if available.

Know What Files You Are Copying or Deleting

If you add "-D" at the end of command lines such as copy, delete, etc., the software lets you see exactly which files are being copied, deleted, etc. Using this feature allows you to know that all file operations are completed successfully.

Function Keys on US

To access the function key, hold down <CTL> and press <#>. For example:

F2 function -> hold down <CTL> and press <2>

F10 function -> hold down <CTL> and press <0>

1.5 References

Publication Title	Binder
<i>Advanced Process Manager Control Function and Algorithms</i>	Implementation/Advanced Process Manager – 1
<i>Advanced Process Manager Parameter Reference Dictionary</i>	Implementation/Advanced Process Manager – 2
<i>Application Module^X User Guide</i>	Application Module ^X
<i>Button Configuration Data Entry</i>	Implementation/Engineering Operations – 1
<i>Command Processor Operation</i>	Implementation/Startup & Reconfiguration – 1
<i>Control Language/Application Module Data Entry</i>	Implementation/Application Module – 3
<i>Control Language/MC Reference Manual</i>	Implementation/Hiway Gateway – 2
<i>Control Language/MC Data Entry</i>	Implementation/Hiway Gateway – 2
<i>Configuration Data Collection Guide</i>	Implementation/Startup & Reconfiguration – 2
<i>Data Entity Builder Manual</i>	Implementation/Engineering Operations – 1
<i>Engineer's Reference Manual</i>	Implementation/Startup & Reconfiguration – 2
<i>Free Format Log Data Entry</i>	Implementation/Engineering Operations – 1
<i>High-Performance Process Manager Control Function and Algorithms</i>	Implementation/High-Performance Process Manager – 1
<i>High-Performance Process Manager Parameter Reference Dictionary</i>	Implementation/High-Performance Process Manager – 2
<i>History Module Service</i>	LCN Service – 2
<i>HM History Group Form Instructions</i>	Implementation/Engineering Operations – 1
<i>Logic Manager Control Functions</i>	Implementation/Logic Manager
<i>Logic Manager Parameter Reference Dictionary</i>	Implementation/Logic Manager
<i>Messages Directory</i>	Implementation/Startup & Reconfiguration – 3
<i>Network Form Instructions</i>	Implementation/Startup & Reconfiguration – 1
<i>Network Gateway Implementation and Operation</i>	Implementation/Network Gateway
<i>Picture Editor Data Entry</i>	Implementation/Engineering Operations – 3
<i>PLC Gateway Control Functions</i>	Implementation/PLC Gateway
<i>Process Manager Control Functions and Algorithms</i>	Implementation/Process Manager – 1
<i>Process Manager Parameter Reference Dictionary</i>	Implementation/Process Manager – 2
<i>Control Language/Process Manager Data Entry</i>	Implementation/Process Manager – 2
<i>Safety Manager Module Control Functions</i>	Implementation/Safety Manager

1.5 References (continued)

Publication Title	Binder
<i>Safety Manager Module Parameter Reference Dictionary</i>	Implementation/Safety Manager
<i>Text Editor Operation</i>	Implementation/Engineering Operations – 3
<i>Universal Work Station Specification and Technical Data</i>	System Summary – 2

1.6 Sample System Document Is Based Upon

Figure 1 shows the system that this guide is based upon. Please memorize the definitions and be aware that your node numbers may be different. A Programmable Logic Controller Gateway (PLCG) and its associated controllers was also temporarily connected.

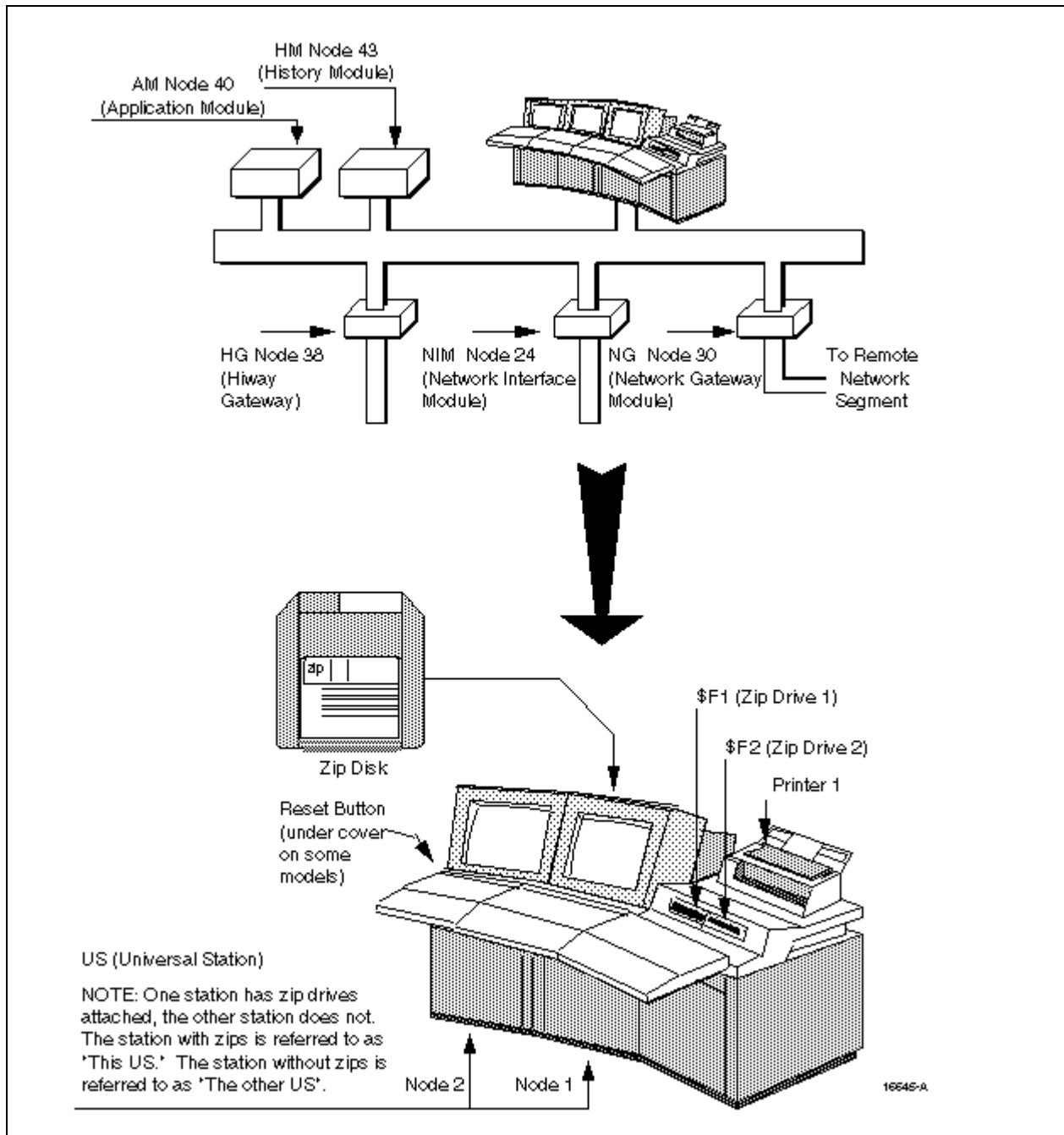


Figure 1 Sample System This Document is Based Upon

Section 2 – Overview

Figure 2 shows an overview of the build and modification sequences involved in the **TotalPlant** Solution (TPS) system startup process.

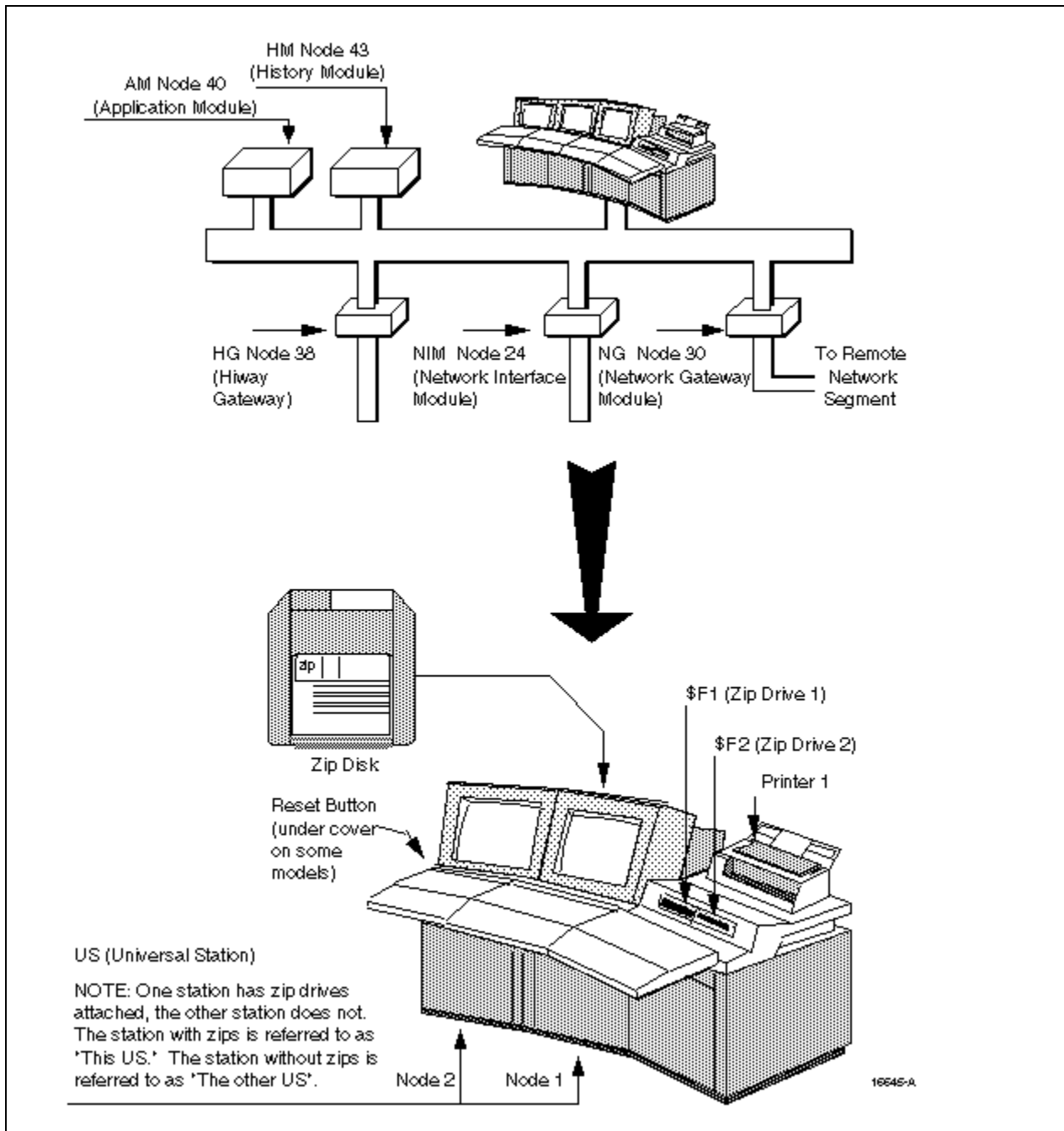


Figure 2 System Startup Process

Section 3 – Instructions

3.1 General Instructions

The process of starting up a system is really the process of following step-by-step instructions that tell you how you use configuration programs and configuration data to specify the specifics of your unique site.

Program and Data Sources—During Tasks 2 through 15, the program and data sources are Zip Disks. After the History Module (HM) has been loaded in Task 15, if it is necessary to restart the system, either the Universal Personality or the Operator Personality must be loaded from removable media; however, most other data and program sources can be read from the History Module by entering N (for NET) when the source is requested. Some infrequently used software is not loaded onto the HM, but can be accessed from Zip disks (i.e., removable media), if needed.

After an operating personality (Operator's or Universal) is running, Zip disks are accessed by their logical-device numbers (for example, \$F1 for Drive 1 and \$F2 for Drive 2, etc.).

Special Software—Two types of software are available for most node types. If a node contains an HMPU, K2LCN, or HPK2 printed wire board, you must load 68020 type software into that node. Also, in Task 10 you must specify the software used by these nodes as TYPE II (for example, HG II, USUNP II, etc.). If you do not know which nodes contain these boards, refer to Figure 3-2 for help. If a node contains a K4LCN printed wire board, you must load 68040 type software into that node. In Task 10 you must specify the software used by these nodes as TYPE IV (for example, NIM IV, AM IV).

Organization—The organization of this section is based on the following hierarchy:

Procedure

 One or more Tasks

 Multiple Instructions

Eight procedures are made up of 31 individual tasks. The first seven procedures (Tasks 1-22) must be performed in sequence the first time you configure a system. The procedure 8 tasks (Tasks 23-28) are prerequisite for changing the database after your system is up and running on-process. Some tasks instruct you to perform a task previously performed; therefore, the instructions in a given task are generally written as free-standing or re-entrant.

To make it easier for you to find a particular task, a task number has been printed in the upper left-hand and right-hand corner of each page. You can use the Table of Contents to first find Task Numbers and then find instructions.

Instructions—Instructions are presented as Action/Response pairs. Figure 3 shows the standard format of a sample task within a procedure. You must understand the parts of the standard format before performing the instructions that follow.

Examples—Examples are used extensively. They are based upon the sample system and example values used for developing this guide. If you decide to use the examples, you must substitute **your site-specific values** for our example values in most cases, but **be consistent**. Use the same Area and Console IDs throughout the procedure.

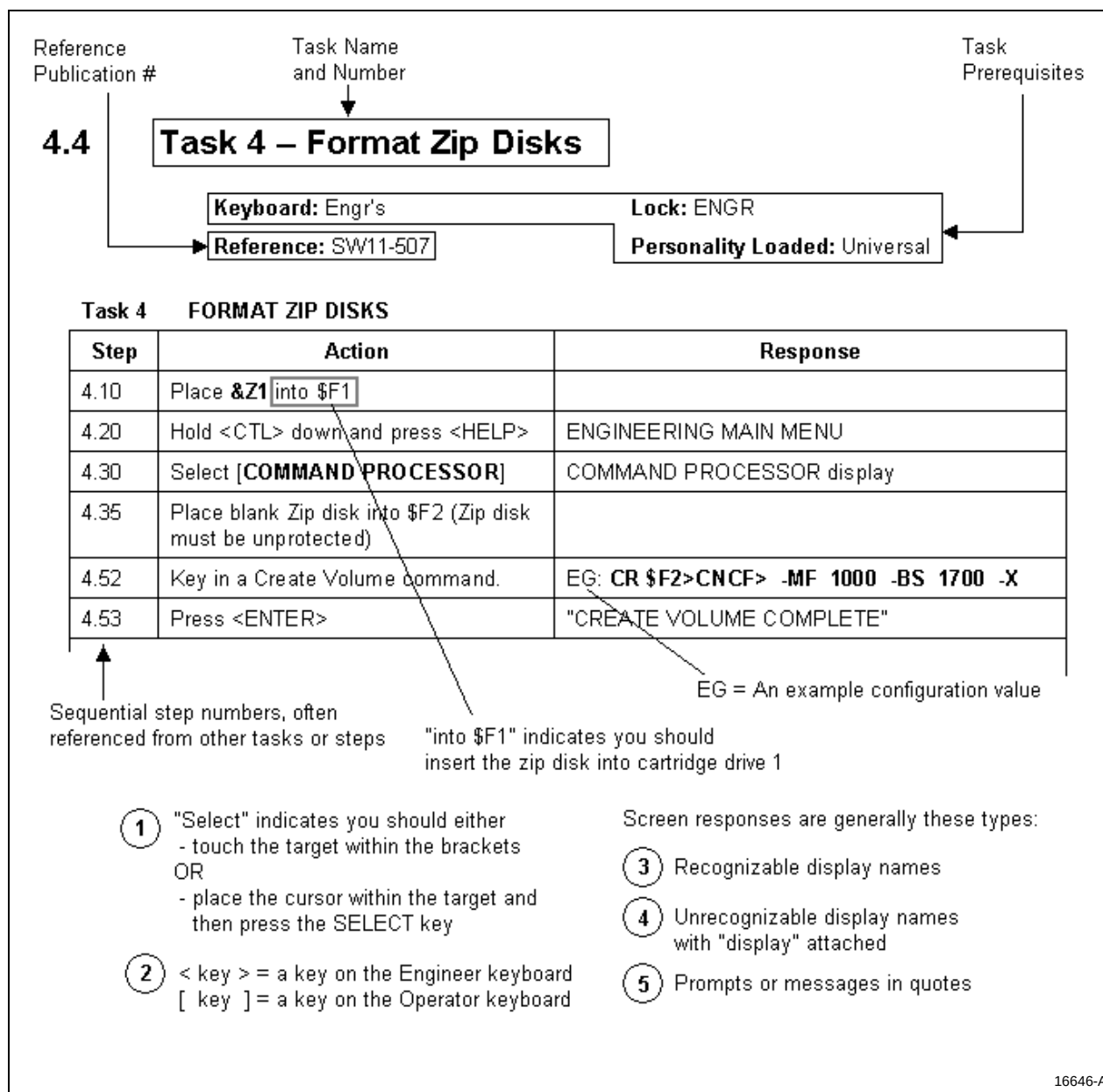


Figure 3 Standard Format of System Startup Instructions

3.2 System Startup Task Record

As each task is performed and completed you should record the indicated information and check the associated task box. You will find the recorded information useful to have when performing subsequent tasks.

Task # Information to be recorded or checked off when configured. *As you reset Nodes, check them off; also record Node Nos.*

1	US	AMs	HMs	HGs	NIMs	CIUs	PLCGs
NGs	U ^X Ss	A ^X Ms	GUSs	APP	PHD	BRG	

☐ 2 Physical Device Nos:

Cartridge Drive Nos: _____, _____, _____, _____, _____, _____

Zip Drive Nos: _____, _____, _____, _____

Printers: _____, _____, _____, _____

☐ 3 ☐ 4 ☐ 5 ☐ 6

☐ 7 Unit IDs: _____; _____; _____; _____; _____; _____; _____; _____; _____; _____; _____; _____; _____

Area Name: _____; Console Name: _____

☐ 8

* For configuration tasks in this document, CIU, CG and CM broadly refer to the same thing.

3.2 System Startup Task Record (continued)

☐ 9 Nodes: Type/Numbers : / , / , / , / , / , /
 / , / , / , / , / , /
 Hiway Numbers: , , , ,
 UCN Numbers: , , , ,

☐ 10 Area Names: _____; _____; _____

User's Volume IDs: _____, _____, _____

11 ☐ 12 ☐ 13 ☐ 14 ☐ 15 ☐

16 ☐ 17 ☐ 18 ☐

☐ 19 Hiway Box Types/Box Nos/Slot Nos:

☐ 20 Point-IDF Names: _____; _____; _____; _____

Point IDs: _____

☐ 21 Logic Block Conf.

☐ 22 Node (PMM) No./Primary (P),Redundant (R):

☐ 23 Picture Filenames: _____

☐ 24 Button Filenames:

Picture _____/Button # _____; Picture _____/Button # _____

Picture _____/Button # _____; Picture _____/Button # _____

Picture _____/Button # _____; Picture _____/Button # _____

☐ 25 HG Library Names: _____

3.2 System Startup Task Record (continued)

☐ 26 Free Format Log Filenames

☐ 27 HM History Groups

☐ 28 CL Configuration

☐ 29 Area Database—IDFs: _____

Areas Installed: _____

☐ 30 Node Numbers of USs Loaded: _____

3.3 Node Identification

Figure 4 shows how to determine what type of nodes you have, their node numbers, and which ones require 68020 software. Many modules have a tag to identify the module type. You will need this information when loading software, and for Task 10 configuration.

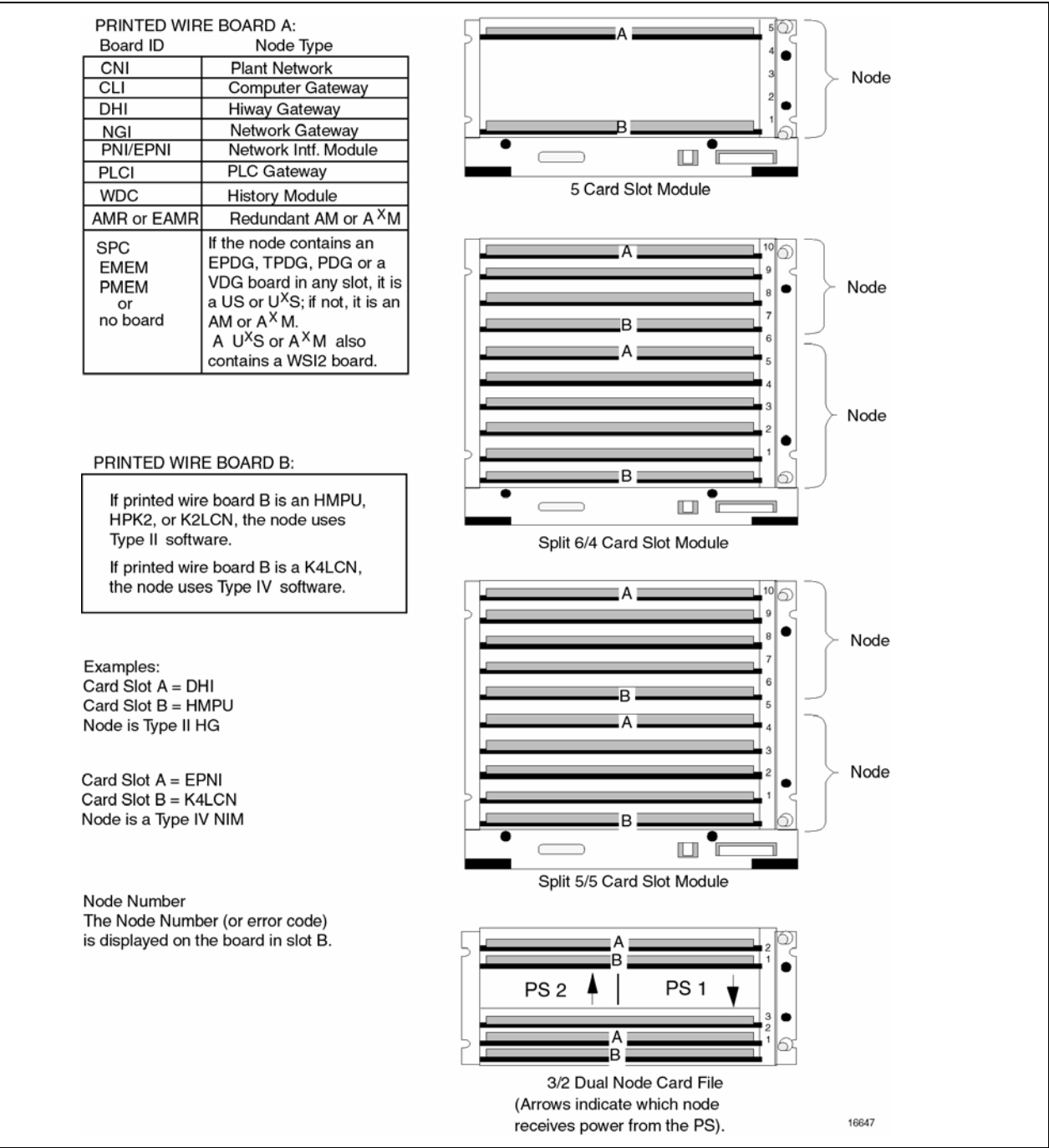


Figure 4 Node Identification

Table 1 shows how Honeywell Zip disks are labeled, their contents and Honeywell Part number.

Table 1 Standard Software and Zip Disk Names

Zip Disk	Control No.	Directories	
&Z1 Personalities and Support Files	MP-M(n*)ZIP1	&AMG	AM GDFs
		&ARG	Area GDFs, initial area 00 files
		&CIO	Computing Module Personality
		&CUS	Custom backplane Modules
		&DS2	HM Resident Displays backup
		DIA1	Tool Kit Diagnostics
		&DSY	Standard displays & display abstracts
		&EC	Honeywell-supplied Executables
		&HGG	HG GDFs
		&HGO	Hiway Gateway Personality
		&HMO	History Module Personality
		&HMI	History Module Initialization Personality
		&LDR	Boot loaders and QLT personalities
		&UCN	UCN Device Personality
		&NMG, &NM2	NIM GDFs
		&NGO	Network Gateway Personality
		&NMO	NIN On-Process Personality
		&OP1	UNP Overlays
		&OPR	US Oper Personality
		&OVG	Equipment List Builder overlay
&Z3 Application Module	MP-M(n*)ZIP3	TLK1	Tool Kit Diagnostics
		&UNP	Universal Personality
		&AMO	AM On-Process Personality
		&LDR	Boot loaders and QLT personalities
		&CUS	AM backplane Modules
		&EC	Honeywell-supplied Executables
		TLK2	CL file read/write option package
		&CLX	AM Sets Syntax files

* Where (n) is specific to the series release (for example, R600 MP6ZIP1).

Table 1 Standard Software and Zip Disk Names

Zip Disk	Control No.	Directories
&Z5 HVTS	MP-M(n*)ZIP5	Test Diagnostic Software
&Z6 Customer Startup Files	MP-M(n*)ZIP6	&ASY and &D01 null Area01, and Button file &I01 - &I20 checkpoint files &AMC Application Module null checkpoint &CIC Computing Module null checkpoints Note: TLK1 and DIA1 were moved to &Z1 and &KFO moved to &Z16
&Z8 Micro TDC	MP-M(n*)ZIP8	&ASY &D01, &I01 , &I02, &I03, &I04, &I05, &I06, &I07, &I08, &I09, &I10, &I11, &I12, &I13, &I14, &I15, &I16, &I17, &I18, &I19, &I20, &AMC, &CIC, &H01, &H03, &H05, &H07, &H09, &E01, &F02, &F04, &F06, &F08, PIC, CL
&Z11 YH Support Files	MP-M(n*)ZIP11	&NMG, &NM2, &KJF, &UCN, &EC (NOTE: TLK1 AND DIA1 now in &Z1)
&Z12	MP-M(n*)ZIP12	GUS Files Only
&Z13 USIM	MP-M(n*)ZIP13	Simulators &UCN, &EC
&Z14 MS	MP-M(n*)ZIP14	Multiple Schematics &CUS, &EC
&Z15 RULA	MP-M(n*)ZIP15	&CUS, &EC
&Z16 New	MP-M(n*)ZIP16	&KFO Software Options Key-file
&Z17 New	MP-M(n*)ZIP17	&UCN Triconex Safety Manager Personality

* Where (n) is specific to the series release (for example, R600 MP6ZIP1).

Note: TLK2 CL file read/write is an optional package.

Section 4 – Procedure 1/Prepare For Network Configuration

4.1 Task 1 – Reset Nodes

Keyboard: N/A

Lock: N/A

Reference: This Document

Personality Loaded: N/A

Task 1 RESET NODES

Step	Action	Response
1.10	For all LCN nodes: press the <RESET> button on the electronic chassis; Power Off/On Univ. Work Stations	Nodes go to a reset state Note: After reset, the indicators near the reset button display the Node Number
1.15	Record all node numbers	
1.20	Do NOT perform this step if the HM has the LCN Extender Hardware. Turn power off to all HMs	

NOTE

The first US is started on Area 01.

4.2 Task 2 – Load the Universal Personality

Task 2 loads the Universal Personality to a station with Zip disks and an engineering keyboard.

Reference: This Document

Personality Loaded: None; reset

Task 2 LOAD THE UNIVERSAL PERSONALITY

Step	Action	Response
2.10	Place &Z1 into \$F1 and &Z6 into \$F2 or CNCF into \$F2	
2.20	Press (RESET) (see Figure 1), or Power Off/On the Universal Work Station (not necessary if coming from Task 1)	Wait for the > sign to appear.
2.25	Place keyswitch in ENG position	Engineering Keylock enabled
2.30	Press (LOAD) on the US Operator's Keyboard (or <LOAD> on the Universal Work Station Engineering Keyboard)	"N,1,2,3,4,X?"
2.40	Press (1) and then press (ENTER)	"OPR, UNP?"
2.50	Press the ALPHA SHIFT key until lit, enter U and press (ENTER)	Series of messages, ending with "NODE STARTING UP"... "NCF? N,1,2,3,4,X"
2.51	&Z6 (CNCF if coming from Task 12) into \$F2	
2.52	Press (2) and then press (ENTER)	"ABST N,1,2,3,4,X" is displayed
2.56	Press (1) and then press (ENTER)	"BUTT N, 1,2,3,4,X"
2.60	Press (F2) and then press (ENTER)	SYSTEM STATUS Display appears
2.64	Select "MOUNT STD ABST FOR USnn"	Source targets appear
2.68	Select [ALTERNATE SOURCE]	Additional targets appear
2.72	Select drive that contains &Z1	<u>EG:</u> [1]
2.74	Select [EXECUTE COMMAND]	Abstract data is read from &Z1
2.78	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
2.101	Select [SUPPORT UTILITIES]	SUPPORT UTILITY MENU
2.102	Select [MODIFY VOLUME PATHS] display	MODIFY DEFAULT VOL PATHNAMES
2.103	Select [SET DEVICE PATH TO REM. MEDIA]	Paths are changed from NET to \$Fn, \$Fn+1 (EG: \$F1 and \$F2)
2.104	Select [MAIN MENU]	ENGINEERING MAIN MENU
2.110	Select [CONSOLE STATUS] display	CONSOLE STATUS AND ASSIGNMENT

Task 2 LOAD THE UNIVERSAL PERSONALITY

Step	Action	Response
2.120	Record device numbers for the printer and the drives.	PRINTER DRIVES ACCESS ----- -----, ----- ENGR Note: If the drive numbers are other than 1 and 2, remember to substitute the correct numbers in place of \$F1 and \$F2 wherever these are used in the tasks that follow.

4.3 Task 3 – Set Time and Date

Keyboard: Engr's

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 3 SET TIME AND DATE

Step	Action	Response
3.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
3.20	Select [SYSTEM STATUS]	SYSTEM STATUS FROM display
3.30	Select [TIME/DATE]	"ENTER TIME: HH MM" appears at top left
3.50	Key in hours-space-minutes	<u>EG</u> : 14 25
3.60	Press <ENTER>	<u>EG</u> : 14:25 in upper-right corner, "ENTER DATE: DD MMM YY" in upper left corner
3.80	Key in days-space-month-space-year	<u>EG</u> : 15 JUN 95
3.90	Press <ENTER>	<u>EG</u> : 15 Jun 95 in upper right corner

4.4 Task 4 – Format Zip Disks

Keyboard: Engr's

Reference: SW11-507

Lock: ENGR

Personality Loaded: Universal

Task 4 FORMAT ZIP DISKS

Step	Action	Response
4.10	Place &Z1 into \$F1	
4.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
4.30	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
4.35	Place blank Zip disk into \$F2 (Zip disk must be unprotected)	
4.52	Key in a Create Volume command.	CR \$F2>CNCF> -MF 1000 -BS 1700 -X
4.53	Press <ENTER>	"CREATE VOLUME COMPLETE"
NOTE A Zip disk can hold a great number of files; however, some operations will be slowed as a function of the size of the directory; this example specifies 1000 files.		
4.54	Key in a Create Directory command for the &ASY volume.	CD \$F2>CNCF> &ASY
4.55	Press <ENTER>	"CREATE DIRECTORY COMPLETE"
4.58	Repeat steps 4.54 and 4.55 to create &KFO volume.	CD \$F2>CNCF> &KFO
4.65	Remove the formatted ZIP DISK from \$F2 and write the date and Volume ID (for example, "CNCF") on the label.	This will be the system-specific NCF ZIP disk.
4.66	Repeat steps 4.35 – 4.53 (using same commands) and label it (Task 28, "Backup CNCF")	These volumes are used for backing up customer databases as suggested in the note at the end of Task 28.
4.70	Repeat steps 4.35 – 4.55 except use Volume/Directory names below to create the two additional volumes for future use. 4.70.1 Area Database for area button file for custom files Repeat step 4.70.1 if you have additional areas. 4.70.2 User volumes for data points for custom graphics for CL	CR \$F2>AR01> -MF 1000 -BS 1700 -X CD \$F2>AR01> &D01 CD \$F2>AR01> BUTT CD \$F2>AR01> &CUS <u>EG:</u> CD \$F2>AR02> &D02 CD \$F2>AR03> &D03 <u>EG:</u> CR \$F2>USER> -MF 1000 -BS 1700 -X CD \$F2>USER> HMV1 CD \$F2>USER> PICT CD \$F2>USER> CL
4.80	Repeat steps 4.35 – 4.53 to create a checkpoint volume, and label it	CR \$F2>CKPT> -MF 1000 -BS 1700 -X

Task 4 **FORMAT ZIP DISKS**

Step	Action	Response
4.90	Mount &Z6 in \$F1. Repeat the Create Volume steps to create a checkpoint volume.	For HG or NIM: CD \$F2>CKPT> &Ixx where xx = Hiway or Process Network number
	Copy Null checkpoint files to the CKPT directories.	For HG or NIM: CP \$F1>&Ixx>*. * \$F2> &Ixx>= -D UNPT \$F2>&Ixx>*. * For CG: CD \$F2>CKPT> &Hxx where xx is the CG unit index number from Unit Names CP \$F1>&CIU>CIUxx MAS.CP \$F2>&Hxx>= -D For AM: CD \$F2>CKPT> &Fxx where xx is the AM unit index number from Unit Names CP \$F1> &AMC>AMxxxMAS.CP \$F2>&Fxx>= -D UNPT \$F2>&Fxx>*. * where xxx represents the AM Unit Name position number

4.5 Task 5 – Copy Files

Keyboard: Engr's

Reference: SW11-507

Lock: ENGR

Personality Loaded: Universal

Task 5 COPY FILES

Step	Action	Response
5.10	If the COMMAND PROCESSOR display is still on the US go to step 5.20; if not, hold <CTL> down and press <HELP>. Then, Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
5.20	Place &Z6 into \$F1	
5.30	Load the CNCF Zip disk created in steps 4.50 – 4.65 into \$F2. Key in a copy command for the NCF files.	CPV \$F1>&ASY> \$F2>&ASY> -D where \$F1 and \$F2 are the drive numbers assigned
5.35	Press <ENTER>	Lists files copied, followed by: "COPY VOLUME COMPLETE"
5.40	Key in a copy volume for the &D01 File (&D01 contains null Area 1 and Button Files).	CPV \$F1>&D01> \$F2>&D01> -D
5.42	Press <ENTER>	"COPY VOLUME COMPLETE"
5.44	Place &Z16 into \$F1	
5.46	Key in a copy volume command for the Key File Option file.	Lists files copied, followed by: CPV \$F1>&KFO> \$F2>&KFO> -D
5.48	Press <ENTER>	"COPY VOLUME COMPLETE"
5.60	If you intend to assign this US to any area 2 - 10, go perform steps 5.64 – 5.78. &Z6 should be \$F1 and CNCF in \$F2.	Note: This US started up on Area 1. You will need the proper area database to reload if you intend to configure this US on another area.
5.64	Create a &Dnn directory for whatever Area you will assign this US to (nn = 02 - 10).	<u>EG:</u> CD \$F2>CNCF> &D03 where nn = 03 Area Database for this example
5.66	Press <ENTER>	Create Directory Complete
5.70	Copy/Rename the &D01 volume for whatever Area (Dnn) you will assign this US to.	<u>EG:</u> CPV \$F1>&D01> \$F2>&D03> -D where nn = 03 for this example
5.74	Press <ENTER>	"COPY VOLUME COMPLETE"
5.76	Rename the added area file within &Dnn	<u>EG:</u> RN \$F2>&D03>AREA01.DA AREA03
5.78	Press <ENTER>	RENAME COMPLETE

Section 5 – Procedure 2/Configure the Network

Figure 5 shows a graphic representation of the Tasks (6-11) that must be performed to Configure the Network.

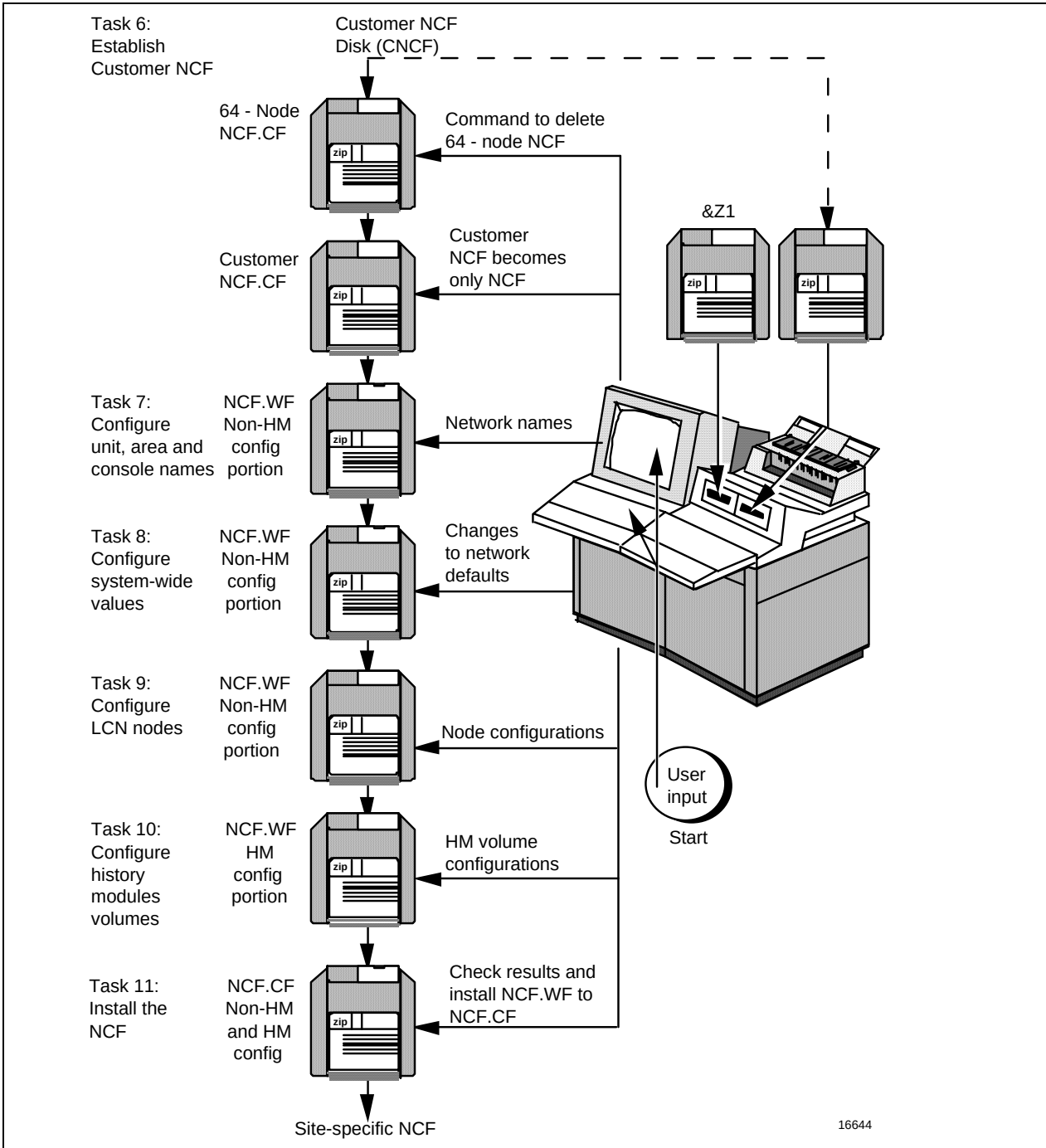


Figure 5 Overview of Configuring the Network

5.1 Task 6 – Establish Customer NCF

Keyboard: Engr's

Lock: ENGR

Reference: SW11-605

Personality Loaded: Universal

Task 6 ESTABLISH CUSTOMER NCF

Step	Action	Response
6.10	Place &Z1 into \$F1 and CNCF into \$F2	CNCF is the primary copy made in Task 5
6.20	If the COMMAND PROCESSOR display is on screen, go to step 6.50; if not, hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
6.30	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
6.50	Use \$F2 device number from step 2.120 and delete the NCF files	DL \$F2>&ASY>NCF.* -D
6.60	Press <ENTER>	"DELETE COMPLETE"
6.70	Key in a rename command (Rename customer NCF file)	RN \$F2>&ASY>CSTMRNCF.CF NCF
6.80	Press <ENTER>	"RENAME COMPLETE"
6.90	Go to step 7.20. Be sure to read the warning below.	

WARNING

YOU SHOULD PERFORM TASKS 7 THROUGH 11 SEQUENTIALLY WITHOUT INTERRUPTION. DO NOT INTERRUPT THE SUGGESTED SEQUENCE AND DO NOT REMOVE THE CNCF DISK FROM THE DRIVE UNLESS YOU DO ONE OF THE FOLLOWING FIRST:

1. Return to Engineering Main Menu by holding <CTL> down and pressing <HELP>
2. Perform the Install NCF function (Task 11)

5.2 Task 7 – Configure Unit, Area, Console Names

Keyboard: Engr's

Lock: ENGR

Reference: SW11-605

Personality Loaded: Universal

NOTE

If you are not entering data from filled in configuration forms or if you want a printout of your entries, press <PRT SCN> (hold <CTL> down and press <CLR-ENT>) to get a screen print after each configured page has been entered in Tasks 7-10.

With R681, NCF print option enables you to print NCF to a text file. Initiate a print using F4 function key from any NCF form to print the complete NCF as a text file. It will be stored in NET>&ASY by default with the file name as NCF.XX. If the default filename is not desired, specify a text file name using the extensions ".X?" or ".Y?" or ".Z?".

Where,

? can be any alphabet.

If &ASY has memory shortage, you can print the NCF.XX to a different user volume.

You can use the existing functionality to abort printing – simultaneously press the keys CTL + Break. The printing is aborted and the partially printed text file is retained with a comment PRINT OPERATION ABORTED.

For an NCF parameter with multiple values, the values are printed continuously separated by a space. If the number of characters exceeds 80, the next line begins with &\, indicating it is a continuation of the previous line.

Note that in large networks having more than 64 nodes, the NCF text file takes approximately 30 minutes to print.

Task 7 CONFIGURE UNIT, AREA, CONSOLE NAMES

Step	Action	Response
7.10	Place &Z1 into \$F1 and CNCF into \$F2	
7.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
7.30	Select [UNIT NAMES]	EDIT UNIT NAMES—PAGE 1 display
7.32	If the words OFF LINE appear at the end of the 2nd line on the screen header, skip to step 7.40. If not, perform step 2.35.	
7.35	Press <F3>	Warning Message: "OFF LINE"
7.38	Press <ENTER>	EDIT UNIT NAMES display
7.40	Key in a UNIT ID and DESCRIPTION for each line item as required.	<u>EG:</u> 01 and UNIT 1 for line 1
7.50	When finished, press <ENTER>	Values and ports reappear in cyan
7.52	Page forward, if necessary, and continue entries. Press <ENTER> after each page.	
7.60	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU

Task 7 CONFIGURE UNIT, AREA, CONSOLE NAMES

Step	Action	Response
7.70	Select [AREA NAMES]	EDIT AREA NAMES is displayed
7.80	Key in a AREA NAME and DESCRIPTOR for each line item as required. Do not leave blank lines between entries. Note: An Area Name cannot contain spaces.	<u>EG:</u> BOILER1 for line 1 Area Name <u>EG:</u> WEST AREA for Area 1 Description <u>EG:</u> REACTOR for line 2 Area Name <u>EG:</u> AREA 2 for Area 2 Description
7.90	When finished, press <ENTER>	Values and ports appear in cyan
7.100	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
7.110	Select [CONSOLE NAMES]	EDIT CONSOLES NAMES display
7.120	Key in a Console Description for each line item as required.	<u>EG:</u> WEST AREA CONSOLE for line item #1.
7.130	When finished, press <ENTER>	Values and ports appear in cyan

5.3 Task 8A – Configure System Wide Values

Keyboard: Engr's

Lock: ENGR

Reference: SW11-505

Personality Loaded: Universal

Task 8A CONFIGURE SYSTEM WIDE VALUES

Step	Action	Response
8.10	Ensure &Z1 is in \$F1 and CNCF is in \$F2	
8.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
8.30	Select [SYSTEM WIDE VALUES]	SYSTEM WIDE VALUES MENU display
8.33	Select [SYSTEM ID]	SYSTEM ID display
8.36	Key into SYSTEM DESCRIPTOR port	<u>EG:</u> PLANT NUMBER 1
8.39	Key into SYSTEM ID port	
8.42	Press <ENTER>	Values and ports reappear in cyan
8.45	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.48	Select [CLOCK SOURCE]	CLOCK SOURCE display
8.51	Key into CLOCK LOCATIONS port (lowest number goes in left port)	<u>EG:</u> 24 and 38
8.54	Press <ENTER>	Values and ports reappear in cyan
8.57	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.60	Select [USER AVERAGE PERIOD]	USER AVERAGE PERIOD display
8.63	Select a User Average Period	<u>EG:</u> 5 MINUTES
8.66	Press <ENTER>	Values and ports reappear in cyan
8.69	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.72	Select [SHIFT DATA]	SHIFT DATA display
8.75	Key in changes as required	
8.78	Press <ENTER>	Values and ports reappear in cyan
8.81	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.84	Select [CONSOLE DATA]	CONSOLE DATA PAGE 1 display
8.87	Key in/select changes as desired	<u>EG:</u> (on page 3) Alarm Summary Freeze Timeout (seconds): 180
8.90	Press <ENTER>	Values and ports reappear in cyan
8.95	Press <PAGE FWD>	CONSOLE DATA—Page n display
8.110	Repeat steps 8.87 – 8.95 until pages 1 through 8 have been viewed and modified as required.	Note: Pages 9 and beyond are reserved. Do not change them without specific instructions from Honeywell.

Task 8A CONFIGURE SYSTEM WIDE VALUES

Step	Action	Response
8.111	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.114	Select [SOFTWARE OPTIONS]	SOFTWARE OPTIONS display
8.118	Ensure CNCF in is \$F2	
8.122	Hold down <CTL> and press <0>	MOUNT TARGETS appears
8.124	Select [ALTERNATE]	MOUNT port appears
8.128	Key into MOUNT port	<u>EG</u> : \$F2
8.132	Press <ENTER>	Authorized Options appear in cyan (Do <u>Not</u> press F1 or F2 at this time)
8.136	Press <ENTER>	"BUSY"— SOFTWARE OPTIONS display
8.138	Hold <CTL> down and then press <PAGE BACK> If you do <u>not</u> have a Network Gateway, skip to step 8.172	SYSTEM WIDE VALUES MENU Note that NG Local System Ids must be configured before NG LCN Nodes
8.140	Select [NG LOCAL SYSTEM ID]	NG LOCAL SYSTEM ID display
8.144	Key into LOCAL SYSTEM NUMBER port	<u>EG</u> : 1
8.148	Key into SYSTEM TITLE port (the first character must be an alpha)	<u>EG</u> : P1
8.152	Key into SYSTEM DESCRIPTION port	<u>EG</u> : Plant Number 1
8.156	Select a PIN ADDRESSING FORM option	<u>EG</u> : LOCAL
8.164	Press <ENTER>	Values and ports reappear in cyan
8.168	If you have other NGs repeat steps 8.144 through 8.164 for each.	System Wide Value configuration for NGs continues in Task 8B after node configuration and HM volume configuration.
8.172	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.176	Select [TAG NAME OPTIONS]	TAG NAMES display
8.180	Select a Tag Name option	<u>EG</u> : LONG
8.184	If you selected LONG, make a Formatting choice.	<u>EG</u> : YES
8.188	Press <ENTER>	Values and ports reappear in cyan
8.192	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.196	Select [AM MULTIPLE PRIMMOD ALARMING OPTIONS]	AM MULTIPLE PRIMMOD ALARMING OPTIONS DISPLAY

Task 8A CONFIGURE SYSTEM WIDE VALUES

Step	Action	Response
8.200	Do either Step A or Step B: A. Select ENABLED EXCLUSIVE MODE target to cause alarms to be reflected in alarm groups for those \$MPROD parameters which have values. B. Select ENABLED INCLUSIVE MODE target to cause alarms to be reflected in alarm groups for those \$MPROD parameters which have values and also in the alarm group for PRIMMOD parameter.	
8.204	Press <ENTER>	Values reappear in cyan
8.208	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.212	Select [HG SP/OP CLAMP OPTION]	HG SP/OP CLAMP OPTION DISPLAY
8.216	Select ENABLE OPTION target to enable the HG SP/OP Clamp Option functionality.	
8.220	Press <ENTER>	Values reappear in cyan
8.224	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.228	Hold <CTL> down and then press <PAGE BACK>	ENGINEERING MAIN MENU

5.4 Task 8B – Configure System Wide Values

Keyboard: Engr's (Network Gateway Completion)

Lock: ENGR

Reference: SW11-605

Personality Loaded: Universal

If you do **not** have a Network Gateway, skip to Task 9. The following section completes NG configuration.

Task 8B CONFIGURE SYSTEM WIDE VALUES

Step	Action	Response
8.944	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
8.948	Select [SYSTEM WIDE VALUES]	SYSTEM WIDE VALUES MENU
8.950	Select [NG REMOTE SYSTEMS]	NG REMOTE SYSTEM display
8.952	Key into # port	<u>EG</u> : 2
8.953	Key into TITLE port	<u>EG</u> : P2
8.954	Key into DESCRIPTOR port	<u>EG</u> : Plant NO. 2
8.956	Select a CABLE FOR OUTPUT	<u>EG</u> : A
8.958	Key into RESP NG port	<u>EG</u> : 30
8.960	Key into REMOTE SERVERS RESP LOCAL port.	<u>EG</u> : 16
8.962	Key in other ports as desired.	
8.964	Press <ENTER>	Values and ports reappear in cyan
8.965	Page forward and continue to make entries if necessary. Press <ENTER> after each page.	
8.966	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.968	Select [NG SECURITY DATABASE]	NG SECURITY DATABASE AUTH. display
8.970	Key in a REMOTE SYSTEM NUMBER or select ALL SYSTEMS.	<u>EG</u> : ALL SYSTEMS
8.972	Select [DEFINE ACCESS RESTRICTIONS]	NG SECURITY DATABASE AUTHORIZ. display
8.974	Select a default access for System Vol/Dir.	<u>EG</u> : READ ONLY
8.976	Select a default access for USER Vol/Dir.	<u>EG</u> : READ /WRITE
8.977	Key in a SYSTEM VOL name under EXCEPTIONS if desired and select the access level.	EG: no entry
8.978	Key in a USER VOL name under EXCEPTIONS if desired.	<u>EG</u> : CSW1
8.980	Select default access for User Vol/Direc	<u>EG</u> : READ ONLY

Task 8B CONFIGURE SYSTEM WIDE VALUES

Step	Action	Response
8.982	Page Forward to Parameter Access Defaults	<u>EG</u> : Page 2
8.984	Select a default access for Parameters	<u>EG</u> : READ ONLY
8.985	Select a NODE TYPE for EXCEPTIONS	<u>EG</u> : AM
8.986	Select a HIWAY/UNIT	<u>EG</u> : 01
8.987	Select Exceptions for Parameter Access	<u>EG</u> : READ/WRITE
8.988	Continue entires for other node types. When all entries are complete, press <ENTER>	Values and ports reappear in cyan
8.990	Hold <CTL> down and then press <PAGE BACK>	NG SECURITY DATABASE AUTHORIZ. display
8.994	Hold <CTL> down and then press <PAGE BACK> Unless Honeywell has requested that you change NG Modem parameters, skip the NG MODEM DEFINITION configuration target.	SYSTEM WIDE VALUES MENU

5.5 Task 9 – Configure LCN Nodes

Keyboard: Engr's

Lock: ENGR

Reference: SW11-605

Personality Loaded: Universal

WARNING

You must configure the Universal Station you are using for system startup; otherwise, you will not be able to successfully start up your system. During Network Configuration, each display where you enter data must say OFF LINE at the end of line 2.

Task 9 CONFIGURE LCN NODES

Step	Action	Response
9.10	Ensure &Z1 is in \$F1 and CNCF is in \$F2	
9.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
9.30	Select [LCN NODES] If necessary, <PAGE FWD> until you see the node number for one of your USSs.	LCN NODE CONFIG—Page 1 display <u>EG:</u> NODE 1 LCN NODE CONFIG—Page n display
9.50	Select a US Node Number	(<u>EG:</u> 1) SELECT NODE TYPE display
9.60	Select [UNIVERSAL STATION]	UNIVERSAL STATION NODE display
9.70	Key into CONSOLE # port	<u>EG:</u> 01 (for nodes 1 and 2)
9.80	Key into STATION # port	<u>EG:</u> 1 for node 1 (2 for node 2)
9.90	Key into DEFAULT AREA port	<u>EG:</u> BOILER1 (for nodes 1 and 2)
9.95	Key into ZIP DISK NUMBERS ports	<u>EG:</u> 1 and 2 for node 1 <u>EG:</u> on next pass: 3 and 4 for node 2, and so on
9.100	Key into FLOPPY DISK NUMBERS ports, if applicable	
9.110	Key into PRINTER NUMBER ports (Printer Assignment function allows screen prints to be made from any station, if at least one printer is present within the console)	<u>EG:</u> 1 for node 1, 2 for node 2, and so on (configure a printer on any station that has a printer and stations from which you wish to make screen prints.
9.120	Key into TREND PEN NUMBER ports	<u>EG:</u> 1 (Blank if none implemented)
9.130	Select ACCESS LEVEL option	<u>EG:</u> [OPER] for nodes 1 and 2
9.134	Select DEFAULT LOAD PERS. option	<u>EG:</u> [UP] for nodes 1 and 2
9.140	Select OPERATORS KEYBOARD option	<u>EG:</u> [YES] for nodes 1 and 2
9.150	Select TOUCH SCREEN option	<u>EG:</u> [YES] for nodes 1 and 2 (also YES for UXS with mouse or trackball)
9.160	Select ENGINEERS KB option	<u>EG:</u> [YES] for nodes 1 and 2

Task 9 CONFIGURE LCN NODES

Step	Action	Response
9.175	<PAGE FWD> to enter values on the second page of the Universal Station display (fill in only if extra schematic memory or External Load Module software options are needed). This can be done later.	DO NOT PRESS <ENTER> until all pages of the LCN node are configured. Obtain entry details from the External Load Module documentation. Caution! make sure of adequate Memory for external Load Modules.
9.176	After last page is entered, press <ENTER>	Values and ports reappear in cyan
9.180	If more Universal Stations need to be configured, you can hold <CTL> down, press <PAGE BACK>, and go to	<u>EG: 2</u> (configure as before, except Station 2, and so on) Note that U ^X S and GUS stations are configured like a US, except as mentioned. DAT (tape) drives do not require configuration entries.
9.190	Hold <CTL> down, press <PAGE BACK>	LCN NODE CONFIG—Page 1 display
9.195	<PAGE FWD> look for NIM node number (if no NIM on system, skip to 9.254)	LCN NODE CONFIG—Page n display
9.200	Select a NIM Node Number. If no NIM on system, skip to 9.254.	<u>EG: 24</u> , SELECT NODE TYPE display
9.210	Select [NETWORK INTERFACE MODULE]	NETWORK INTERFACE MODULE Display
9.220	Key into REDUNDANT MEMBER ID port	<u>EG: 25</u> or 0 if none
9.230	Key into UNIV. CONT. NETWORK port	<u>EG: 3</u>
9.244	After last data is entered, press <ENTER>	Values and ports reappear in cyan
9.245	Repeat 9.190 – 9.244 for each NIM node pair.	
9.250	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG—Page n display
9.254	<PAGE FWD> (if necessary) to NG node number. If no NG on system, skip to 9.284.	LCN NODE CONFIG—Page n display
9.258	Select a Network Gateway Node Number	<u>EG: 30</u> , SELECT NODE TYPE display
9.260	Select [NETWORK GATEWAY]	NETWORK GATEWAY MODULE DISPLAY
9.264	Key into PIN LOCAL ADDRESS port	<u>EG: 8</u>
9.266	Key into PIN GLOBAL ADDRESS port (first half Honeywell provided global address, if any)	<u>EG:</u> No entry
9.268	Select CABLES FOR OUTPUT	<u>EG: A</u>
9.276	After last data is entered, press <ENTER>	Values and ports reappear in cyan
9.278	Repeat 9.250 – 9.276 for each NG node pair.	
9.280	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG—Page n display

Task 9 CONFIGURE LCN NODES

Step	Action	Response
9.284	<PAGE FWD> (if necessary) to HG node number. If the system doesn't have an HG or a PLCG, skip to step 9.318.	LCN NODE CONFIG—Page n display
9.288	Select an HG or PLCG Node Number Note: HG and PLCG LCN nodes are configured exactly the same, but must be on different Data Hiways.	(<u>EG</u> : 38) SELECT NODE TYPE display
9.290	Select [HIWAY GATEWAY]	HIWAY GATEWAY NODE display
9.294	Key into REDUNDANT MEMBER ID port	<u>EG</u> : 39 (using the next highest number for the REDUNDANT ID is recommended or 0 if none.)
9.298	Key into HIWAY NUMBER port	<u>EG</u> : 2
9.308	Press <ENTER> after last page.	
9.310	Repeat 9.280 - 9.308 for each HG node pair.	
9.314	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG—Page n display
9.318	<PAGE FWD>; look for AM node No. (If the system does not have an AM, skip to step 9.372).	LCN NODE CONFIG—Page n display
9.322	Select an AM or A ^X M Node Number	<u>EG</u> : 40 SELECT NODE TYPE display
9.326	Select [APPLICATION MODULE]	APPLICATION MODULE NODE display
9.328	REDUNDANT MEMBER Node ID port	<u>EG</u> : 0 (Enter node number or enter 0 if no redundant node)
9.340	Select a STARTUP MODE option	<u>EG</u> : [Cold]
9.344	Key into ASSIGNED UNITS port (the units must have been previously configured) Do not press Enter yet.	<u>EG</u> : 01 (multiple units must be separated by commas or spaces for example: 01,A1,)
9.346	If AM/CL Background functions are used, or if this is an A ^X M, or if Custom AM Software will be used Go to the next step— If not, skip to step 9.360	

Task 9 CONFIGURE LCN NODES

Step	Action	Response
9.350	<PAGE FWD> to page 2. Type in Page 2 information. Obtain from: <i>AM Implementation Manual</i> (Background/CL) <i>A^XM System Administration Manual (A^XM)</i> or Custom AM Software instructions. These numbers may vary according to system requirements. Do not press Enter yet.	USER MEMORY ALLOCATION display. Note that Page 2 and 3 data can be entered or changed later with the On-Line Reconfiguration feature if you are uncertain. EG: Typical entries for A ^X M/Background/CL are: # BACKGROUND CL TASKS 10 # CONCURRENT DATA ACCESS 4 BACKGROUND TASK SIZE 15000 CVB SIZE (Default) 2000 INCLUDE INTERNETWORK PROC. YES USER MEMORY RESERVED 1
9.354	<PAGE FWD> to page 3. Enter information per same manuals as used in the previous steps. A combination of entries is allowable for multiple functions.	EXTERNAL LOAD MODULE NAMES display Typical entries for Background/CL are: EG: NAME PERSONALITY FILE AMO CONV AMO AMCL03 AMO Typical entries for A ^X M are: NAME PERSONALITY AMCL06 AMO XACCES AMO
9.360	Press <ENTER>	Page 1 and (if necessary Page 2/3) complete
9.365	Repeat 9.314- 9.360 for each AM or A ^X M	
9.366	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG—Page n display
9.370	<PAGE FWD>; look for HM node No. (If the system does not have an HM, skip to step 9.372).	LCN NODE CONFIG—Page n display
9.372	Select an HM Node Number	<u>EG</u> : 43 SELECT NODE TYPE display
9.376	Select [HISTORY MODULE]	HISTORY MODULE NODE display
9.388	Key into NODE PAIR NUMBER port	<u>EG</u> : 1
9.402	Press <ENTER> after last page.	
9.406	Repeat 9.370 – 9.402 for each HM	
9.410	Hold down <CTL>, press<PAGE BACK>	LCN NODE CONFIGURATION display
9.412	<PAGE FWD> look for CM node number (if no CM on system, skip to step 9.454)	LCN NODE CONFIG—Page n display
9.416	Select a CM Node Number	<u>EG</u> : 51 , SELECT NODE TYPE display
9.420	Select [COMPUTING MODULE]	COMPUTING MODULE NODE display

Task 9 CONFIGURE LCN NODES

Step	Action	Response
9.422	Key in ASSIGNED UNITS port	<u>EG</u> : 01 02 03 04
9.430	If you have External Load Modules that affect the CM, press <PAGE FWD> (if External Load Modules not used, skip the next step)	EXTERNAL LOAD MODULE display This can be done later with the On-Line Reconfiguration feature.
9.436	Key in data, select items as necessary. Press <ENTER> after last page.	Obtain entry information from the External Load Module documentation.
9.440	Repeat 9.370 – 9.436 for each CM	
9.446	Hold <CTL> down, press <PAGE BACK>	LCN NODE CONFIG—Page n display
9.454	To print the LCN Node Config. Hold <CTL> down, then press <F4>. Select printer target, then press <ENTER>.	(US must be connected to printer) Print instructions appear. LCN configuration is printed.

5.6 Task 10 – Configure HM Volumes

Keyboard: Engr's

Lock: ENGR

Reference: SW11-605, HM13-500

Personality Loaded: Universal

Task 10 CONFIGURE HM VOLUMES

Step	Action	Response
10.005	Place &Z1 into \$F1 and CNCF in \$F2	
10.010	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
10.015	Select [VOLUME CONFIGURATION]	HM PAIR SELECTION MENU
10.020	Select a Node Pair number VOLUME CONFIGURATION display	<u>EG:</u> 1
10.025	Key in the # of Winchester disk drives in the History Module	<u>EG:</u> 1 (each redundant pair counts as 1)
10.030	Key in the type or size of disks in the HM <u>EG:</u> 4 or 215040 for 210 MB SCSI drive <u>EG:</u> 5 or 454130 for 445 MB SCSI drive <u>EG:</u> 6 or 896046 for 875 MB SCSI drive <u>EG:</u> 7 for 1.9 Gigabyte drive	<u>EG:</u> 3 for Wren 3
10.035	Select configuration choices on the left side of the menu: HM INIT PERSONALITY SYSTEM VOLUME GDF FILES INCLUDED FILE MANAGER DESCRIPTORS	<u>EG:</u> [YES]—target changes in intensity [YES]—target changes in intensity [YES]—target changes in intensity [NO]
10.040	Key in AREA SOE JOURNAL size Numbers used in this section were for a Wren 3. Increase them for larger drives.	<u>EG:</u> 2000 (If this HM will collect SOE events). For SOE, Process Unit Journals must also be configured on the same HM in step 10.515.
10.045	Key in BURST BUFFER size	<u>EG:</u> 7500
10.050	Select SYS. UNIT JOURNALS option—YES if you want this HM assigned for journals	<u>EG:</u> YES —Added ports appear. If you select NO, skip to step 10.60.
10.055	Key in STATUS CHANGE port	<u>EG:</u> 2048
10.060	Key in ERROR port	<u>EG:</u> 2048
10.065	Key in MAINT port	<u>EG:</u> 1024
10.070	Press <ENTER>	Values and ports reappear in cyan
10.075	Select [PROGRAM IMAGE]	PROG IMAGE VOLUME CONFIG. display
10.080	Select NODE TYPE options. Select TYPE IV version for any node that has an K4LCN board, or TYPE II for others (see Figure 4). If both Type IV and II nodes exist for any function, select both versions.	<u>EG:</u> [HG II] [USOPR II] [USUNP II] [NG II] [AM IV] [NIM IV] The system must have the options you select. Select HGII or HG IV if the system has a PLCG.

Task 10 CONFIGURE HM VOLUMES

Step	Action	Response
10.085	Key in DUMP VOLUME SIZE	EG: 36000 or 0 for none
10.090	Press <ENTER>	Values and ports reappear in cyan
10.095	Hold <CTL> down, press<DISP BACK>	VOLUME CONFIGURATION display
10.100	Select [AREA]	AREA DATA VOLUME CONFIG. display
10.105	Select a File Manager Descriptor option	<u>EG</u> : NO
10.110	Key into ports for each Area line item as required.	EG: A Name V Size Files BOILER1 5000 150
10.115	Press <ENTER>	Values and ports reappear in cyan
10.120	Hold <CTL> down, press<DISP BACK>	VOLUME CONFIGURATION display
10.125	Select [CHECKPOINT]	P.N. FOR CHECKPOINT CONFIG display
10.130	Key into ports for each Node Pair Number line item as required. Note 1: Enter zero for HG or PLCG Vol. Size & No. of Files Configure an A ^X M the same as an AM. Note 2: Redundant node members—enter sequentially with Size/Files = 0	<u>EG</u> : NODE VOLUME NUMBER <u>NO.</u> <u>SIZE</u> <u>FILES</u> NIM 24 40000 100 NIM 25 0 0 PLCG 31 0 0 HG 38 0 0 HG 39 0 0 AM 40 30000 200 Numbers used were for a Wren 3. Increase them for larger HM drives (refer to SW09-505).
Note: Refer to the <i>Engineer's Reference Manual</i> for volume size guidelines.		
10.135	Press <ENTER>	Values and ports appear in cyan
10.140	Hold <CTL> down, press <DISP BACK>	VOLUME CONFIGURATION display
10.145	Select [CONTINUOUS HISTORY]	CONT. HIST. UNIT AND GROUPS display
10.150	Key into UNIT ID port	<u>EG</u> : 01
10.155	Key into NBR GROUPS port	<u>EG</u> : 60
10.160	Continue entries as required.	
10.165	Press <ENTER>	Values and ports reappear in cyan
10.170	Press <PAGE FWD> and repeat steps 10.150 – 10.165 until all required entries are complete.	
10.175	Select [GROUP OPTIONS] (You can select the [HM/UNIT OPTIONS] display target if desired for a display that allows you to assign options for all.)	CONT. HISTORY GROUP OPTIONS
10.180	Select a SAVE RATE (you must select a save rate to get <u>any</u> history).	<u>EG</u> : 20

Task 10 CONFIGURE HM VOLUMES

Step	Action	Response
10.185	Select either [SNAPSHOTS], [USER AVG] or both (you must select SNAPSHOTS to get trend data).	<u>EG</u> : Select both [SNAPSHOTS] and [USER AVG]
10.190	Key into PREARCHIVE HOURS port for SNAPSHOTS.	<u>EG</u> : 48
10.195	Key into PREARCHIVE HOURS port for USER AVG.	<u>EG</u> : 48
10.200	ARCHIVE [YES] [NO] Function not released	Leave [NO] options selected
10.205	Repeat steps 10.180 – 10.200 for all required entries.	
10.210	Press <ENTER>	Values and ports reappear in cyan
10.215	<PAGE FWD> and repeat steps 10.180 – 10.210 for all required entries.	
10.220	Hold <CTL> down, press <PAGE BACK>	CONT. HISTORY UNIT AND GROUPS display
10.225	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display
10.230	Select [JOURNALS]	PROCESS UNIT JOURNAL CONFIG. Display
10.235	Key into UNIT ID port	<u>EG</u> : 01 Note that Unit numbers you specify here are also the units from which SOE events are collected (if you specified an SOE Journal size for this HM node pair).
10.240	Key into the PROCESS ALARM port	<u>EG</u> : 15000
10.245	Key into the PROCESS CHANGE port	<u>EG</u> : 1000
10.250	Key into the OPERATOR MESSG port	<u>EG</u> : 1000
10.255	Repeat steps 10.235 – 10.550 for other applicable Units.	
10.260	Press <ENTER>	Values and ports reappear in cyan
10.265	Press <PAGE FWD>	Page n of screen forms appears
10.270	Repeat steps 10.235 – 10.265 for all applicable Units.	
10.275	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display
10.280	Select [CL STORAGE]	CL STORAGE VOLUME CONFIGURATION display
10.285	Select a HIWAY FILE MGR Description choice.	<u>EG</u> : NO
10.290	Select a NETWORK FILE MGR Description choice.	<u>EG</u> : NO
10.295	Key into HIWAY/NETWORK NBR. port	<u>EG</u> : 02

Task 10 CONFIGURE HM VOLUMES

Step	Action	Response																
10.300	Key into NUMBER OF FILES port	<u>EG:</u> 30																
10.305	Key into VOLUME SIZE port	<u>EG:</u> 100																
10.310	Repeat steps 10.295-10.305 for other Process Networks.																	
10.315	Press <ENTER>	Values and ports reappear in cyan																
10.320	Hold <CTL> down, press <DISP BACK>	VOLUME CONFIGURATION display																
10.325	Select [USER FILE STORAGE]	USER FILE STORAGE CONFIG. display																
10.330	Key into Node Pair Number n ports for each line item as required (HNV1 is a general HM user volume, CL is for CL programs, CSW1 is for custom programs). Numbers used were for a Wren 3. Increase them for larger drives.	<u>EG:</u> <table><tr><td><u>Vol ID</u></td><td><u># of Files</u></td><td><u>Vol Size</u></td><td><u>Descriptor</u></td></tr><tr><td>HNV1</td><td>1000</td><td>15000</td><td>FAST</td></tr><tr><td>CL</td><td>200</td><td>3000</td><td>NO</td></tr><tr><td>CSW1</td><td>200</td><td>3000</td><td>NO</td></tr></table>	<u>Vol ID</u>	<u># of Files</u>	<u>Vol Size</u>	<u>Descriptor</u>	HNV1	1000	15000	FAST	CL	200	3000	NO	CSW1	200	3000	NO
<u>Vol ID</u>	<u># of Files</u>	<u>Vol Size</u>	<u>Descriptor</u>															
HNV1	1000	15000	FAST															
CL	200	3000	NO															
CSW1	200	3000	NO															
10.335	Press <ENTER>	Values and ports reappear in cyan																
10.340	Hold <CTL> down, press <DISP BACK>	VOLUME CONFIGURATION display																
CHECK HM VOLUME CONFIGURATION																		
10.345	Press <F1>. This performs the CHECK function.	Messages: "NCF FILE IS BEING PACKED INTO MEMORY...NEW NCF.WF IS BEING COPIED....HM CHECKER RUNNING."																
10.350	To configure volumes for another HM, repeat all steps in Task 10.																	
10.355	If errors are indicated, go to step 10.360; otherwise, skip ahead to Task 8B.																	
10.360	Press <F8>	NCF.ER file displayed																
10.365	If you want a printed copy, press <F4>. Make sure the printer is on.	Print Instructions appear																
10.370	Select printer target, press <ENTER>	NCF.ER file is printed																
10.375	If total HM size exceeds 100% -> If the number of virtual volumes is exceeded ->	Go to step 10.330 and reduce the size of the User Volume. Go to step 10.330 and delete one or more User volumes, or delete the dump volume (10.085). Or else call a Honeywell support engineer.																
10.380	Hold <CTL> down and press <PAGE BACK>	EDIT VOLUME CONFIG display																
10.385	Go repeat steps 10.325 – 10.345 until errors are cleared.																	
10.390	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU																

5.7 Task 11 – Install the NCF

Keyboard: Engr's

Reference: SW11-505

Lock: ENGR

Personality Loaded: Universal

Task 11 INSTALL THE NCF

Step	Action	Response
11.01	Ensure &Z1 is in \$F1 and CNCF is in \$F2.	
11.30	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
11.30	Select [UNIT NAMES]	Units—Page 1 display
11.40	Press <ENTER>	
11.50	Press <F1>	"NCF FILE IS BEING PACKED INTO MEMORY...NEW NCF.WF IS BEING COPIED HM CHECKER RUNNING"
11.52	Press <F2>	"WARNING:...PRESS ENTER"
11.60	Press <ENTER>	"OFFLINE INSTALL COMPLETE—DATE/TIME)"
11.65	After the "INSTALL COMPLETE" message appears, remove the CNCF Zip disk from \$F2 and write the date on the label.	
11.66	Make a backup copy of your CNCF . Primary CNCF into \$F1 and the backup CNCF disk from step 4.66 into \$F2.	
11.67	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
11.68	Key in a disk copy command.	<u>EG:</u> FCP \$F1 \$F2
11.70	Press <ENTER>	"Media Copy Complete" If your primary CNCF disk becomes corrupted, you have a backup copy.
11.80	Remove both Zips from the drives.	

Section 6 – Procedure 3/Prepare History Module(s)

Figure 6 shows a graphic representation of the Tasks (12-16) that must be performed to prepare the History Module.

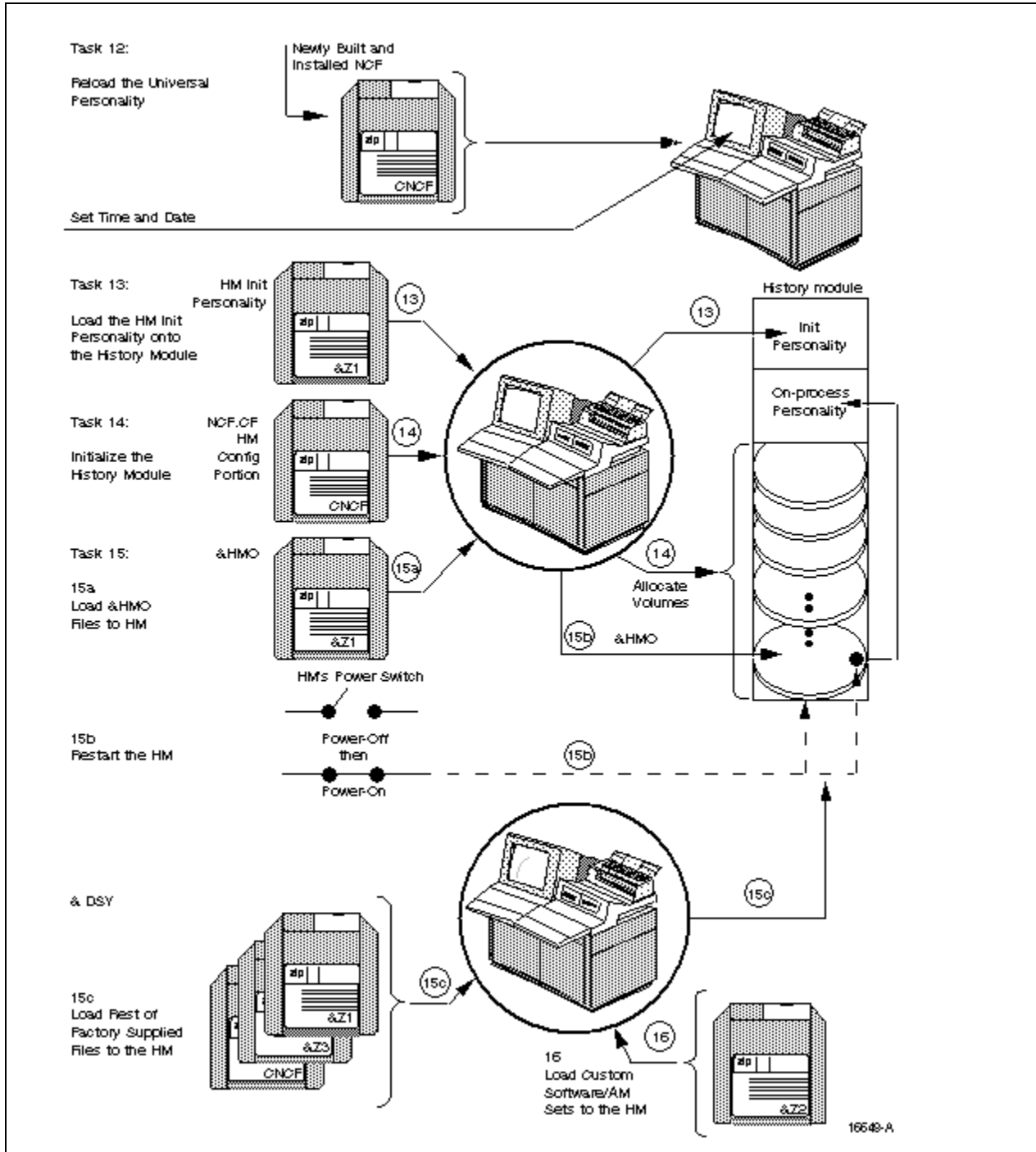


Figure 6 Prepare the History Modules

6.1 Task 12 – Reload Universal Personality/Set Date

Keyboard: N/A

Lock: N/A

Reference: This Document

Personality Loaded: N/A

Task 12 RELOAD UNIVERSAL PERSONALITY/SET DATE

Step	Action	Response
12.05	Go perform Task 1 (Reset all nodes)	Note: Not required for nodes previously reset.
12.10	Go perform Task 2 using the CNCF from Task 11 and return here.	You must reload the US using your site-specific CNCF (from Task 11).
12.20	Go perform Task 3 and return here	Current time/date in "this US"
12.30	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU

6.2 Task 13 – Load History Module with HM Initialization Personality

Keyboard: Both

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 13 LOAD HISTORY MODULE WITH HM INITIALIZATION PERSONALITY

Step	Action	Response
13.05	Turn power <u>ON</u> at all HMs. RECORD THE TIME _____	HM Power was turned off in Task 1
13.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
13.15	Select [SYSTEM STATUS]	SYSTEM STATUS display
13.18	Select [DE-SELECT]	All nodes de-selected
13.20	Select the HISTORY MODULE Node	HM Node box selected
13.24	Select [NODE STATUS]	HISTORY MODULES display (Ignore HM status)

CAUTION

The HMs must have had power applied for at least 5 minutes before you press an HM's Reset button **in the next step**. Make sure you press it only once.

13.30	Reset each HM by pressing white RESET button on HM module.	Wait until the HM Status changes from FAIL or PWR-OFF to POWER_ON.
13.35	If any HM does not show POWER_ON status, turn its power off for a few seconds; then turn its power on and repeat step 14.29.	
13.40	Insert &Z1 in \$F1, CNCF in \$F2	
13.50	Select Number under NODE heading on the HISTORY MODULE display.	<u>EG</u> : 43 (Number changes color)
13.60	Select [LOAD/DUMP]	LOAD/DUMP targets appear
13.64	Select [MANUAL LOAD]	PGM targets appear
13.70	Select [INIT PROGRAM]	PGM SOURCE FOR NODE NN display
13.80	Select [ALTERNATE SOURCE]	<u>EG</u> : Boxes [1] and [2] appear
13.90	Select [1] (if this is the number for the first Zip drive entered in step 9.95/9.100)	"VOL ID: &Z1" and targets appear
13.100	Select [EXECUTE COMMAND]	DATA SOURCE FOR NODE NN display
13.110	Select [ALTERNATE SOURCE]	<u>EG</u> : Boxes [1] and [2] appear
13.120	Select [2]	"VOL ID: CNCF" and targets appear
13.130	Select [EXECUTE COMMAND]	HISTORY MODULES display

Task 13 LOAD HISTORY MODULE WITH HM INITIALIZATION PERSONALITY

Step	Action	Response
13.140	Press <ENTER>	Messages appear at the top of the screen "SINGLE LOAD; BUSY: HM NN, STATUS: LOAD, ...TYPE: HMOF, . . . OK" This can take several minutes to complete.
13.150	Repeat 13.50 - 13.140 to load the HM Init Personality to other HMs.	

NOTE

1. If you get an error message, refer to the TotalPlant Solution (TPS) system Messages Directory for possible error recovery procedures.
2. From this point on, the Zip drive numbers will be as entered in Step 9.95/9.100 and may differ from the examples.

6.3 Task 14 – Initialize the History Module

Keyboard: Engr's

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 14 INITIALIZE THE HISTORY MODULE

Step	Action	Response
14.10	Insert &Z1 in \$F1, CNCF in \$F2	
14.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
14.30	Select [VOLUME CONFIGURATION]	HISTORY MODULE PAIR SEL. Menu
14.40	Select Node Pair Number	<u>EG: 1</u> (Node 43), VOLUME CONFIGURATION display
14.50	Press <F6> Note: Requires several minutes to complete.	"HM INITIALIZATION RUNNING... DISC SURFACE HAS BEEN INITIALIZED AND A NEW LOCAL-NVCF FILE HAS BEEN WRITTEN"....
NOTE If you get an error message, refer to the <i>TPS Messages Directory</i> publication, for possible error recovery procedures.		
14.60	Hold <CTL> down, press <PAGE BACK>	HISTORY MODULE PAIR SELECTION Menu
14.70	Repeat 14.40 – 14.60 for each HM node pair number.	
14.80	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU

6.4 Task 15 – Copy Software from Zip to History Module

Keyboard: Engr's

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 15 COPY SOFTWARE FROM ZIP TO HISTORY MODULE

Step	Action	Response
HISTORY MODULE PREPARATION		
15.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
15.20	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
15.30	Check that the HM Local Vol. file is on the HM by keying in a Catalog command.	<u>EG:</u> CAT PN:NN>!9np>*. * where NN = HM Node Number and np = node pair. (CAT PN:43>!901 for this sample system)
15.40	Press <ENTER>	"CATALOG COMPLETE"
15.50	If the L01_NVCF.MM file is listed on the display, continue with step 15.60; else call for help.	Your Honeywell engineer
15.60	Assure printer is powered on.	To obtain a record of the files transferred in the upcoming steps.
15.70	Key in " PD ON "	
15.80	Press <ENTER>	"Prompt To Dataout Turned On"
15.90	Key in " DO (Print Device ID)"	<u>EG:</u> DO \$P1
15.100	Press <ENTER>	"DATA OUT COMPLETE" Paper advances in printer Note: Printer must be connected to the station.
15.110	Insert &Z1 in \$F1	
15.120	Key in an EC command for the Local Volume.	<u>EG:</u> EC \$F1>&EC>LOC_VOLZ.EC \$F1 43 01
15.130	Press <ENTER>	"THIS EC IS USED TO COPY HM PERSONALITIES TO ITS LOCAL VOLUME"

At this point you have initiated an "Execute Command" program to simplify the task of loading a series of files from Zip to the HM. You will be asked a series of questions to which you must respond by keying in a "Y" for yes or an "N" for no, followed by pressing <ENTER>.

NOTES:

1. The &HMI files can be copied to more than one HM.
2. The &HMO files must be copied to all HMs.
3. &Z1 contains both 68020 and 68040 HMO & HMI software. Answers to the EC questions will determine which type is loaded (HMs with K4LCN boards require 68040 software).

Task 15 COPY SOFTWARE FROM ZIP TO HISTORY MODULE

Step	Action	Response
15.140	Shown here are the files that should be transferred during the execution of the above EC file. Check your printout to determine that all required files have been loaded on the HM(s). ___ &LDR ___ &HMI (HMOFxxx) files ___ &HMO (HMONxxx) files When the EC program is complete, the message "Write Boot ...EC Complete" appears.	
15.150	If you have other HMs, return to Step 15.120 to copy the &HMO (&HMI if desired) directory to the additional HM(s).	
NOTE The &ASY directory in the following steps must be copied to only one HM, the system HM.		
15.160	Ensure CNCF in \$F2	
15.170	Copy the custom NCF files to the HM	<u>EG:</u> CP \$F2>&ASY>*. * PN:43>&ASY>= -D
15.180	Press <ENTER>	"COPY COMPLETE"
RESTART THE HISTORY MODULE		
15.190	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
15.200	Select [SYSTEM STATUS]	SYSTEM STATUS display
15.210	Select the HISTORY MODULE node	Node is selected
15.230	Select [SHUT DOWN]	Target changes color
15.240	Press <ENTER>	Status changes to QUALIF
15.250	At the HM module, turn power OFF , wait at least 5 seconds, and then turn power ON .	Status of the HM changes to: LOC LOAD, then READY, and finally OK . HMON On-process personality is now running in the HM This process can take several minutes.
15.255	Return to Task 15.210 to bring additional HM's online.	
NOTE If you get an error message, check the <i>TPS Messages Directory</i> publication (see References), for possible error recovery procedures.		
COPY REMAINING SOFTWARE TO THE HM		
15.260	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
15.270	Select [COMMAND PROCESSOR]	Command Processor display
15.280	Assure printer is on	To obtain a record of the files transferred in the upcoming steps.
15.290	Key in " PD ON "	
15.300	Press <ENTER>	"Prompt To Dataout Turned On"
15.310	Key in " DO (Printer Device ID)"	<u>EG:</u> DO \$P1

Task 15 COPY SOFTWARE FROM ZIP TO HISTORY MODULE

Step	Action	Response
15.320	Press <ENTER>	"DATA OUT COMPLETE" Paper advances in printer
15.330	Ensure &Z1 is in \$F1. If you want to enter files for modules with 68020 microprocessors, key in the EC command for the 68020 personality volume. For 68040 processor files, key in the EC command for the 68040 personality.	<u>EG:</u> EC \$F1>&EC>PRS2VOLZ.EC \$F1 <u>EG:</u> EC \$F1>&EC>PRS4VOLZ.EC \$F1
15.340	Press <ENTER>	"THIS EC IS USED TO COPY PERSONALITIES FILES TO THE SYSTEM HM"
You should answer "Y" for each of the following files that is needed by a node that you have on your system. Answer "Y" only for nodes that use 68020 OR 68040 microprocessors. In either case, to save HM space, decline the copying of files for nodes that you do not have. As part of this procedure, you will be requested to place other Zips into the drive. The EC program you invoke will select the correct software from &Z3 for a Yes or No answer.		

Task 15 COPY SOFTWARE FROM ZIP TO HISTORY MODULE

Step	Action	Response																																																						
15.350	This step provides a list of the files that can be transferred during the execution of the above EC file(s), and the module with which each is associated. The "PRS2VOLZ" EC loads files to nodes with 68020 microprocessors and the "PRS4VOLZ" EC loads files to nodes with 68040 microprocessors. In release R650, the PRS2VOLZ.EC and PRS4VOLZ.EC files have been modified to allow you to optionally copy the APP personality files (&AXM) and GUS personality files (&UXS) into the HM. If the &AXM or &UXS directories do not already exist, they are automatically created by these EC files. Note: &AXM directory occupies 3 MB of space and &UXS directory occupies 5 MB of space on HM. This software is normally copied to <u>only</u> the system HM. Note that file &LDR is loaded along with the other files. <table> <tr> <th>Module</th><th>&Z1</th><th>&Z3</th></tr> <tr> <td>US II</td><td>___ &OPR</td><td></td></tr> <tr> <td>US IV</td><td>___ &OPR^k</td><td></td></tr> <tr> <td>UNP II</td><td>___ &UNP</td><td>(Also requires overlay files &OP1).</td></tr> <tr> <td>UNP IV</td><td>___ &UNP^k</td><td>(Also requires overlay files &OP1).</td></tr> <tr> <td>US</td><td>___ &OVG</td><td></td></tr> <tr> <td>HG II</td><td>___ &HGO</td><td></td></tr> <tr> <td>HG IV</td><td>___ &HGO^k</td><td></td></tr> <tr> <td>NIM II</td><td>___ &NMO</td><td></td></tr> <tr> <td>NIM IV</td><td>___ &NMO^k</td><td></td></tr> <tr> <td colspan="3">&UCN PM, APM, HPM, LM, SM</td></tr> <tr> <td></td><td>___ &UCN</td><td></td></tr> <tr> <td>AM II</td><td></td><td>___ &AMO</td></tr> <tr> <td>AM IV</td><td></td><td>___ &AMO^k</td></tr> <tr> <td>CG II</td><td>___ &CIO</td><td></td></tr> <tr> <td>CG IV</td><td>___ &CIO^k</td><td></td></tr> <tr> <td>NG II</td><td>___ &NGO</td><td></td></tr> <tr> <td>NG IV</td><td>___ &NGO^k</td><td></td></tr> </table> k means type 68040 software (required when the node contains a K4LCN board). When the EC asks if you want to create/copy &OVG, answer Yes if you want the Equipment List functions. When the EC asks where, specify a user volume such as HMV1 or CSW1. When the EC copies the AM files, it asks if you want to create/copy &CUS and CLX. This is optional software but you should probably copy it in unless you know that it isn't needed. If you copy the Computer Gateway files, the EC asks if you want to copy &CUS. You probably should. If the EC asks where you want to create the directories, key in the name of one of your User volumes such as CSW1 or HMV1. When the EC program is complete, the message "EC COMPLETE" appears. Check the printout to assure that all required files have been loaded.	Module	&Z1	&Z3	US II	___ &OPR		US IV	___ &OPR ^k		UNP II	___ &UNP	(Also requires overlay files &OP1).	UNP IV	___ &UNP ^k	(Also requires overlay files &OP1).	US	___ &OVG		HG II	___ &HGO		HG IV	___ &HGO ^k		NIM II	___ &NMO		NIM IV	___ &NMO ^k		&UCN PM, APM, HPM, LM, SM				___ &UCN		AM II		___ &AMO	AM IV		___ &AMO ^k	CG II	___ &CIO		CG IV	___ &CIO ^k		NG II	___ &NGO		NG IV	___ &NGO ^k		
Module	&Z1	&Z3																																																						
US II	___ &OPR																																																							
US IV	___ &OPR ^k																																																							
UNP II	___ &UNP	(Also requires overlay files &OP1).																																																						
UNP IV	___ &UNP ^k	(Also requires overlay files &OP1).																																																						
US	___ &OVG																																																							
HG II	___ &HGO																																																							
HG IV	___ &HGO ^k																																																							
NIM II	___ &NMO																																																							
NIM IV	___ &NMO ^k																																																							
&UCN PM, APM, HPM, LM, SM																																																								
	___ &UCN																																																							
AM II		___ &AMO																																																						
AM IV		___ &AMO ^k																																																						
CG II	___ &CIO																																																							
CG IV	___ &CIO ^k																																																							
NG II	___ &NGO																																																							
NG IV	___ &NGO ^k																																																							
15.360	If necessary, return to Step 15.330 and invoke the other EC.																																																							

Task 15 COPY SOFTWARE FROM ZIP TO HISTORY MODULE

Step	Action	Response
15.370	Ensure &Z1 is in \$F1.	
15.380	Key in an EC command for the System	<u>EG:</u> EC \$F1>&EC>SYS_VOLZ.EC \$F1 43 (the last parameter is the system HM node number)
15.390	Press <ENTER>	
15.400	The files that can be transferred using the above EC command are listed here for comparison with your printout to determine that all required files have been loaded on the HM.	
NOTE To the question "COPY VOLUME &ASY (YES/NO)" answer No ("N"). To the question "DO YOU WISH TO CHANGE TO ANOTHER DRIVE" usually answer No ("N"). To the question "DO YOU WISH TO SKIP &DSY...." usually* answer No ("N"). *This EC loads several small files. If you answer "No," the EC will load all of them. If you answer "Yes," you can choose to load or not load any or all of the following files: __ &DSY __ &HGG __ &AMG __ &ARG __ &NMG & NM2 The system does not have to have the node type to load any of these files.		
15.410	Ensure CNCF is in \$F2.	
15.420	Copy the customer key file to the HM	<u>EG:</u> CPV \$F2>&KFO> NET>&KFO> -D
15.424	Copy the null Area Database file to the HM	<u>EG:</u> CPV \$F2>&D01> NET>&D01> -D
15.430	Press <ENTER>	"COMMAND COMPLETE"
15.443	You can copy certain standard displays (NCF Status, etc.) to a user volume on an HM other than that which contains the &DSY volume. The system hunts for these displays in directory &DS2 on another HM. If the system HM fails. To load the displays, ensure &Z1 is in \$F1. Key in the EC command EC \$F1>&EC>DS2_BKPZ.EC \$F1 . Press <ENTER>. When asked "Do you wish to create &DS2...?", answer Yes. Key in the user volume name when asked.	
15.440	Return to Task 15.330 if any of the above software must be copied to a different HM (as determined earlier during Volume Configuration in Task 10).	
15.450	Key in " PD OFF "	
15.460	Press <ENTER>	"PROMPT TO DATAOUT TURNED OFF"
15.470	Key in " DO "	

6.5 Task 16 – Load Custom Software

Keyboard: Engr's

Lock: ENGR

Reference: Custom Software

Personality Loaded: Universal

Documentation:

Engineer's Reference Manual

Command Processor Operation manual

If you purchased optional custom software (load modules) that were not loaded in Task 15 or if you want to load the Tool Kit or Diagnostic software, perform the following steps. If not, skip to Task 17.

These steps assume the Command Processor is still selected.

Task 16 LOAD CUSTOM SOFTWARE

Step	Action	Response								
16.10	Create directories for the software you intend to load. List the contents of the Zips, if necessary, to find the example directories. <u>EG</u>: Key in a Create Directory command. Press <ENTER>	In Task 10 you may have created a user volume for custom software (CSW1) You can also create a suitable directory under another volume, such as the example user volume HMV1. <u>EG</u>: CD NET>CSW1 &NNN* (where &NNN is custom software)								
*Substitute the volume name of the custom software Zip(s) for &NNN.										
16.80	Copy the optional software into the History Module: <table><tr><td><u>In \$F1</u></td><td><u>Cmd Type</u></td><td><u>Example of command to be keyed</u></td><td><u>Press</u></td></tr><tr><td>&Z__</td><td>Copy</td><td><u>EG</u>: CP \$F1>&NNN>.* NET>&NNN>= -D</td><td><ENTER></td></tr></table>	<u>In \$F1</u>	<u>Cmd Type</u>	<u>Example of command to be keyed</u>	<u>Press</u>	&Z__	Copy	<u>EG</u> : CP \$F1>&NNN>.* NET>&NNN>= -D	<ENTER>	
<u>In \$F1</u>	<u>Cmd Type</u>	<u>Example of command to be keyed</u>	<u>Press</u>							
&Z__	Copy	<u>EG</u> : CP \$F1>&NNN>.* NET>&NNN>= -D	<ENTER>							
16.90	Repeat Task 16 for each HM that you wish to copy custom software.									

NOTE

If the system has redundant disk drives, they can be synchronized now (or later). Refer to the *Engineer's Reference Manual*, Section 7.5 and Section 5 in the *Command Processor Operation manual* for the procedure. The sample system did not have this feature.

Section 7 – Procedure 4/Configure Process Networks and Hiways

Figure 7 shows a graphic representation of the Tasks (17-22) that must be performed to Configure Process Networks and Hiways.

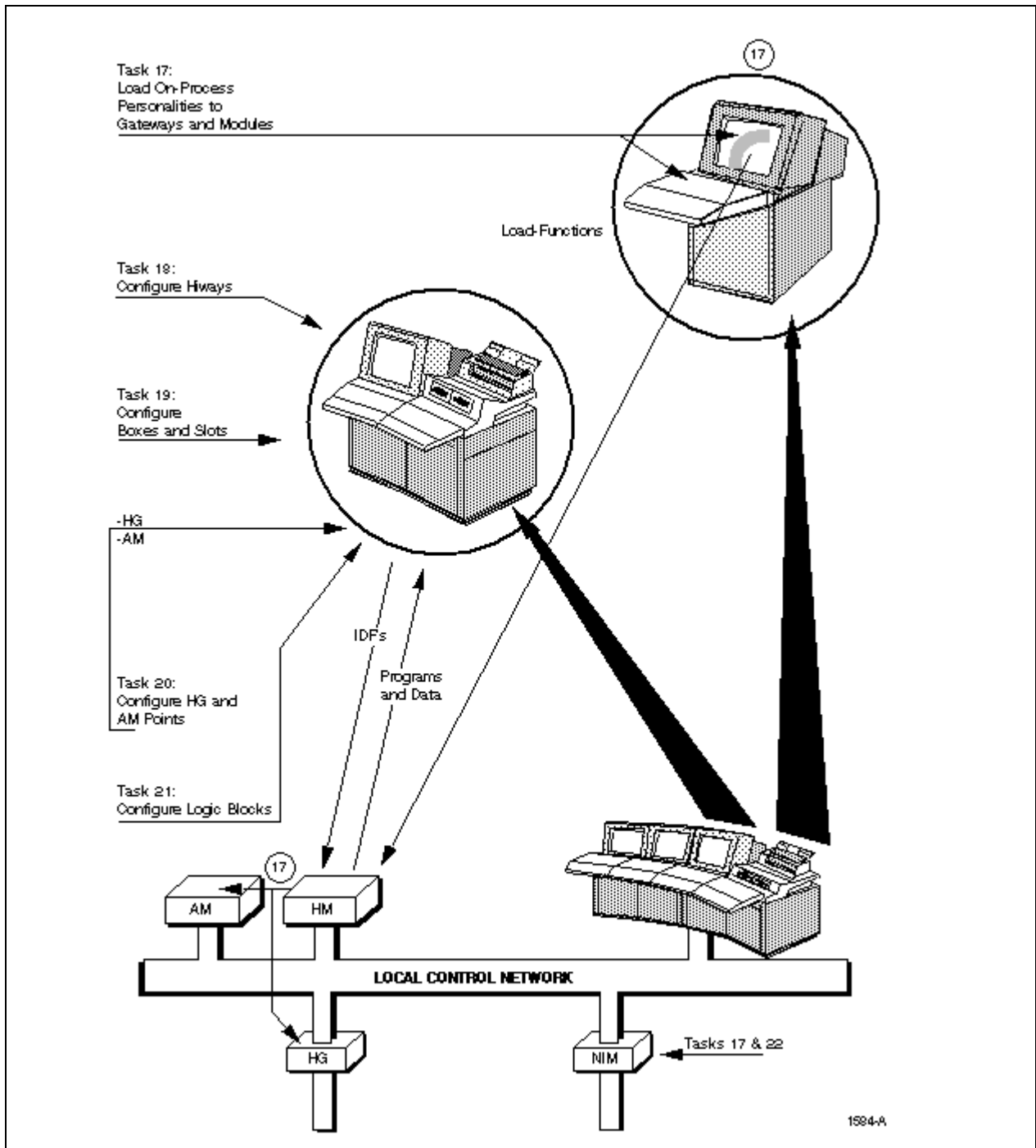


Figure 7 Configure Process Networks and Hiways

7.1 Task 17 – Load On-Process Personalities to Gateways and Modules

Keyboard: Both

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 17 LOAD ON-PROCESS PERSONALITIES TO GATEWAYS AND MODULES

Step	Action	Response
17.01	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
17.03	Select [SUPPORT UTILITIES]	SUPPORT UTILITY MENU
17.04	Select [MODIFY VOLUME PATHS]	MODIFY DEFAULT VOL PATHNAMES display
17.05	Select [SET DEVICE PATH TO "NET"]	LDIDs (Logical Device IDs) are all filled in with NET .
17.06	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
17.10	Select [SYSTEM STATUS]	SYSTEM STATUS display
17.20	Select an AM, NIM, or Gateway Node NIM example: Redundant NIM: NG example: HG/PLCG example, before Tasks 18-19: HG/PLCG example, after Tasks 18-19: AM example: CM example:	Only one node should be selected at a time <u>EG: 24</u> for Network Interface Mod (NIM) <u>EG: 25</u> for Redundant NIM <u>EG: 30</u> for Network Gateway <u>EG: 38</u> for Primary HG or PLCG Module <u>EG: 39</u> for Redundant HG or PLCG Module <u>EG: 40</u> for Application Modules <u>EG: 46</u> for Computing Modules

WARNING

- Do not select a redundant (secondary) HG or PLCG node unless Hiway and Box/Slot Configuration has been done in Tasks 18 and 19.
- The primary HG or PLCG of a redundant pair must be loaded first, using the &Z6 checkpoint disk. After Hiway & Box/Slot configuration, the primary HG or PLCG is checkpointed. The secondary HG or PLCG is loaded later and must be started with a checkpoint volume from the primary node. See Engineer's Reference Manual, Section 8.
- Do not attempt to load a Primary HG or PLCG from one station and a Secondary node from a different station at the same time. The nodes will both fail.
- Always load the primary AM before loading the redundant AM (if any).

17.25	If you will be loading the null checkpoint for an HG, use the Unprotect Zip disk command (UNPT \$Fn) from the command Processor to unprotect the CKPT Zip disk.	Otherwise, you won't be able to complete the HG load.
17.30	Select [NODE STATUS]	Process Networks Node Status Display or Application Modules Node Status Display
17.40	Select [MANUAL LOAD]	NODE LOAD targets appear.

Task 17 LOAD ON-PROCESS PERSONALITIES TO GATEWAYS AND MODULES

Step	Action	Response
17.50	Select a LOAD option	<u>EG</u> : [COLD LOAD] for AM <u>EG</u> : [OPERATOR PROGRAM] for HG, NG, PLCG, or NIM PGM SOURCE FOR NODE NN display
17.60	Select [DEFAULT SOURCE]	[EXECUTE COMMAND] target appears.
17.70	Select [EXECUTE COMMAND]	DATA SOURCE FOR NODE NN display
17.80	Ensure CKPT in \$F1 Make sure Zip is NOT write protected— use Unprotect (UNPTV) command	CKPT contains null checkpoints for all checkpoint nodes (HG, NIM, AM, CG, and PLCG)
17.90	Select [ALTERNATE SOURCE]	Boxes [1], [2], etc. appear.
17.100	Select <u>EG</u> : [1]	VOL ID:CKPT and Targets appear]
17.110	Select [EXECUTE COMMAND]	GATEWAYS or AM display reappears.
17.120	Press <ENTER>	"BUSY: AM or GWY nn or CM" Status goes to LOAD.
17.130	Wait a few minutes.	Status goes to OK when loaded.
17.134	Repeat 17.06-17.130	Note: PMs, APMs, and HPMS are loaded in a later task.
17.138	Select [SAVE DATA]	CHKPT DEST display appears
17.140	Select [DEFAULT SOURCE]	Additional targets appear
17.142	Select [EXECUTE COMMAND]	BUSY AM nn. Node status =SAVE... then OK
17.144	Select [SHUTDOWN]	Node should still be selected
17.145	Press <ENTER>	Node status changes to READY...QUALIF
17.146	Select [LOAD/DUMP]	Additional targets appear
17.147	Select [AUTOLOAD NET]	
17.148	Press <ENTER>	Node status = LOAD...LOC LOAD...OK Node Type changes to AM
17.149	Repeat steps 17.136 - 17.148 for all AMs.	
17.150	Remove the Zip disk from the disk drive.	

7.2 Task 18 – Configure Hiways

Keyboard: Engr's

Lock: ENGR

Reference: SW11-605 and SW11-511

Personality Loaded: Universal

Task 18 CONFIGURE HIWAYS

Step	Action	Response
18.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
NOTE The primary HG or PLCG must be loaded and running with the startup database (loaded in Task 17). If the system does not have an HG or PLCG, skip to task 20.		
\$HIWAYnn Where nn is hiway number		
18.70	Select [HIWAY GATEWAY]	HG-BUILD TYPE AND CONFIG. MENU
18.75	Select [HIWAY CONFIG]	HG-HIWAY POINT ASSIGN. PED display
18.80	Key into HIWAY NUMBER port	<u>EG:</u> 02
18.90	Select HTD ASSIGNMENT option	<u>EG:</u> [THISHG] (HTD is mandatory if a Basic OP Station is on the Hiway)
18.100	Fill in the remaining ports as required and press <ENTER>	PED reappears with values in cyan
18.110	Press <COMND>	COMMAND DISPLAY
NOTE - You should make an IDF with a different name for each HG pair and each PLCG pair. - When building a database create the IDFs in this order: - In this task, create the Hiway Configuration database for the HG pair or PLCG pair. - In Task 19, create the box/slot database for the HG pair, or PLCG pair and then create the database for all the boxes and devices on the Hiway.		
18.120	If this is the <u>first</u> write to this IDF, go to step 18.130; If <u>not</u> , go to step 18.160	IDF for this example is HIWAY2
18.130	Select [WRITE TO IDF]	Pathname ports and target appear
18.140	Key into REFERENCE PATH NAME port	<u>EG:</u> NET>HMV1>
18.150	Key into pathname for IDF port	<u>EG:</u> HIWAY2
18.160	Press <ENTER>	Hiway Config data written to IDF. HG-HIWAY POINT ASSIGNMENT display
18.170	Press <COMND>	COMMAND DISPLAY
18.180	Select [LOAD]	
18.190	Press <ENTER>	Hiway Config data loaded to HG or PLCG. PARAM ENTRY display reappears
18.200	Repeat 18.10 – 18.190 for each other Hiway to be configured.	<u>EG:</u> HIWAY4 for a PLCG

7.3 Task 19 – Configure Hiway Boxes and Slots

Keyboard: Engr's

Lock: ENGR

Reference: SW11-605 and SW11-511

Personality Loaded: Universal

Task 19 CONFIGURE BOXES AND SLOTS

Step	Action	Response
19.05	Press <F9>. If the HG BUILD TYPE MENU appears go to step 19.20; if not, go to 20:10	HG-BUILD TYPE AND CONFIG. MENU
19.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
\$HYxxBnn Where xx is the Hiway number and nn is the box number		
19.15	Select [HIWAY GATEWAY]	HG-BUILD TYPE AND CONFIG. MENU
19.20	Select [BOX/SLOT CONFIG]	HG-BOX POINT ASSIGNMENT PED display
19.30	Key into HIWAY NUMBER port	<u>EG</u> : 02
19.40	Key into BOX NUMBER port	<u>EG</u> : 02 for an HG (Configure HG or PLCG before other Hiway boxes) 03 if you have an HG or PLCG pair 04 if you have a Basic Station 05 for a MC, 10 for CB, etc. (configure 1 box/pass)

NOTE

If this box is the HG or PLCG, the box number of the first node of a redundant pair must be an even number and the second node must be the next odd number.

19.50	Select a BOX TYPE option	<u>EG</u> : [HG] [CB] if box is CB [MC] if box is MC
19.55	Select a BOX ASSIGNMENT option	<u>EG</u> : [THISHG]

NOTE

If an HG or PLCG pair is not your master HG pair for this Hiway, the BOX ASSIGNMENT for that HG or PLCG pair must be either **[REMOTEHG]** or **[ADDEDHG]**.

NOTE

If you selected any options outlined in yellow or if you keyed data into any port outlined in yellow, the content of the PED set or PED changes when you press the <ENTER> key. You should press <ENTER> after each PED page.

19.90	Select a LOAD DESTINATION (if appl.)	<u>EG</u> : [HG_HIWAY]
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Task 19 CONFIGURE BOXES AND SLOTS

Step	Action	Response
19.95	Press <ENTER>	PED reappears with values in cyan. (If PLCG, perform steps in Note at end of Sec. 20 to clear address error).
19.100	Press <PAGE FWD> to next PED page key in data, select options, and press <ENTER> after each page.	HG-BOX CONFIGURATION PED
CHECK THE DATA YOU HAVE ENTERED		
19.110	When you have paged forward and entered all PED pages press <F8>.	PARAM ENTRY DISP SET STATUS display
19.120	If "Display Set Entered?" is Yes, and no errors are indicated, go to step 19.150; else go to 19.129.	Display Set Entered? Yes Errors? NO
19.130	Select a Display Title option, correct data and press <ENTER>.	Applicable PED appears.
19.135	Repeat step 19.120	Go check that data is all OK.
WRITE TO IDF		
19.150	Press <COMND>	COMMAND DISPLAY
19.160	If this is the first write to this IDF, go to step 19.165; if not, go to step 19.190.	IDF for this example is HY2BXSL .
19.165	Select [WRITE TO IDF]	Pathname ports and target appear.
19.170	Key into REFERENCE PATH NAME port	<u>EG:</u> NET>HMV1>
19.180	Key into pathname for IDF port	<u>EG:</u> HY2BXSL
19.190	Press <ENTER>	Hiway Config data written to IDF. HG-BOX POINT ASSIGNMENT PED display
19.195	Repeat steps 19.30 – 19.190 for each redundant HG or PLCG.	
19.200	Repeat steps 19.30 – 19.190 for each other box/slot.	All configured boxes/slots on IDF
19.210	Press <COMND>	COMMAND DISPLAY
19.220	Select [LOAD MULTIPLE]	Additional pathname ports appear
19.230	Key into REFERENCE PATH NAME port	<u>EG:</u> NET>HMV1>
19.240	Key into pathname for IDF port	<u>EG:</u> HY2BXSL
19.245	Press <ENTER>	"WAIT" in yellow; "PROCESSING ..." "OPERATION COMPLETE" (See note below if an error results)
19.250	Press <COMND>	COMMAND DISPLAY
19.260	Select [PRINT ENTITIES]	Path name ports/targets appear. REF. PATH NAME = NET>HMV1>
19.270	Select [PRINT IDF ENTITIES]	

Task 19 CONFIGURE BOXES AND SLOTS

Step	Action	Response
19.280	Use default Print Device ID	<u>EG</u> : \$P1
19.290	Press <ENTER>	Box Slot configuration parameters are printed

NOTE

Entity names in the printout identify Hiway and Box numbers as follows:

"\$HIWAYnn" where nn is the Hiway number.

"\$HYnnBnn" where nn after HY is Hiway number and nn after B is Box number.

Address Error—If an address error results when loading Box/Slot data to a Hiway Box, make sure you saved the data in an IDF. Then proceed as follows (doing this on another station is preferable):

Call up the System Status Display.

Select the appropriate HG Node.

Select [NTWK/HWY STATUS] near the bottom of the display.

Select [HIWAY CMND].

Select [INIT ADDR].

Select [EXECUTE COMMAND]. The address errors should be cleared.

Return to the BOX/SLOT configuration display and repeat the LOAD MULTIPLE steps.

This time the result should be "OPERATION COMPLETE."

PLCG Interface—Certain PLCG interface functions have to be programmed with a Local Loader Station, refer to PLCG documentation for details.

7.4 Task 20 – Configure HG/AM/CM Points

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

Task 20 CONFIGURE HG/AM/CMPOINTS

Step	Action	Response
20.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
20.20	Select [HIWAY GATEWAY], or [APPLICATION MODULE], or [COMPUTING MODULE] (if the system has a PLCG, select [HIWAY GATEWAY]. If the system has none of the above, skip to Task 21).	EG: [HIWAY GATEWAY] HG/AM/CM BUILD TYPE MENU
20.21	Select a Point Build option.	EG: [REGULATORY]

NOTE

1. For detailed information on point building, refer to the *Data Entity Builder Manual*, (see References). For point building tips and precautions, refer to Section 9 in the *Engineer's Reference Manual*.
2. The general scenario of point building is to key into ports, select options on each PED page, press <ENTER> and then <PAGE FWD> to the next PED page. The pages in a particular PED set may change depending upon what type of box, algorithm, or point you are configuring.
3. If you select any **options outlined in yellow** or if you key data into any port outlined in yellow, the content of the PED set or PED changes when you press the <ENTER> key. You should press <ENTER> after each page.
4. Steps 21.30 through 20.420 describe the process of building a typical PID Normal point for a **Basic** Controller. For other types of points see Tables 2 through 4. When building any of the example points, use default values and options for items not on the list.

POINT BUILD

20.30	Key into TAG NAME port	EG: HG1001
20.40	Key into UNIT ID port	EG: 01
20.50	Key into POINT DESCRIPTOR port	EG: CB REG POINT
20.60	Key into POINT KEYWORD port	EG: CBpid
20.64	PRIMARY MODULE POINT ID	-----
20.66	Key into POINT ASSOCIATED DISPLAY	EG: DISP1 (R510 and later systems)
20.70	Key into HIWAY NUMBER port	EG: 02
20.75	Select BOX TYPE option.	EG: [CB]
20.80	Key into BOX NUMBER port.	EG: 10
20.90	Select POINT DISCLOSURE option.	EG: [FULL]

Task 20 CONFIGURE HG/AM/CMPOINTS

Step	Action	Response
20.100	Select LOAD DESTINATION option.	<u>EG</u> : [HG_HIWAY]
20.110	Press <ENTER>	PED reappears with values in cyan
20.120	Press <PAGE FWD>	ALGORITHM Selection display
20.125	Select ALGORITHM option	<u>EG</u> : [PIDNORM]
20.130	Press <ENTER>	PED reappears with values in cyan
20.135	Press <PAGE FWD>	HG-PV CONFIGURATION display
20.140	Key into E.U. DESCRIPTOR port	<u>EG</u> : GAL/HOUR
20.150	Select CHARACTERIZATION option	<u>EG</u> : [SQRROOT]
20.160	Select INPUT CONDITIONING option	<u>EG</u> : [SQRROOT]
20.170	Select PV DECIMAL FORMAT option	<u>EG</u> : [D2] (places point at 999.99)
20.180	Key into OVERVIEW VALUE port	<u>EG</u> : 20
20.190	Select PV RANGE OPTION	<u>EG</u> : [CLMPZERO]
20.200	Select CLAMP VALUE OPTION	<u>EG</u> : [NOCLAMP]
20.210	Press <ENTER>	PED reappears with values in cyan
20.220	Press <PAGE FWD>	HG—ALARMING display
20.225	Select ALARM FORMAT option	<u>EG</u> : [ALFMT00]
20.230	Key into PVLOTP port	<u>EG</u> : 10
20.240	Key into PVHITP port (far right)	<u>EG</u> : 100
20.250	Select ALARM PRIORITY option	<u>EG</u> : [HIGH]
20.260	Press <ENTER>	PED reappears with values in cyan
20.270	Press <PAGE FWD>	HG-OPERATING CONFIGURATION display
20.290	Select PV SIGNAL Type option	<u>EG</u> : [PV]
20.310	Select INITIALIZATION option	<u>EG</u> : [INIT]
20.320	Key into SETPOINT port	<u>EG</u> : 43
20.330	Press <PAGE FWD>	
20.350	Key into OUTPUT HIGH LIMIT port	<u>EG</u> : 100
20.360	Key into INTEGRAL HIGH LIMIT port	<u>EG</u> : 100
20.370	Key into FILTER CONSTANT port	<u>EG</u> : 1
20.380	Press <ENTER>	PED reappears with values in cyan
20.390	Press <PAGE FWD>	HG-MODE CONFIGURATION display
20.400	Select NORMAL MODE option	<u>EG</u> : [CAS]
20.410	Select NRML MODE ATTRIBUTE option	<u>EG</u> : [OPERATOR]
20.420	Press <ENTER>	PED reappears with values in cyan

Task 20 CONFIGURE HG/AM/CMPOINTS

Step	Action	Response
CHECK THE DATA YOU ENTERED		
20.430	When you have paged forward and entered all PED pages, press <F8>.	PARAM ENTRY DISPLAY SET STATUS display
20.440	If "Display Set Entered?" is Yes, and no missing data or errors are indicated, go to step 20.460; else 20.450.	
20.450	Select a Display Title option, correct data and press <ENTER>.	Applicable PED appears. PED set is corrected.
20.455	Repeat step 20.440.	

NOTE

You should make IDFs with different names for the points that you build in each box on the hiway (HG1, CB1, and so on).

WRITE TO IDF		
20.460	Press <COMND>	COMMAND DISPLAY
20.470	If this is the <u>first</u> write to this IDF, go to step 20.475. If not, go to step 20.510.	In this example, the IDF is HGPNTS for HG, AMPNTS for AM, and so on.
20.475	Select [WRITE TO IDF]	Additional port/target appear.
20.480	Key into REFERENCE PATH NAME port	<u>EG:</u> NET>HNV1>
20.500	Key into pathname for IDF port	<u>EG:</u> HGPNTS for HG, AMPNTS for AM
20.510	Press <ENTER>	Point Config data written to IDF PARAMETER ENTRY DISPLAY
20.520	Repeat steps 20.30 – 20.510 for each other point.	All configured points are on IDF for HG, or IDF for AM, and so on.
20.530	Press <COMND>	COMMAND DISPLAY
20.540	Select [LOAD MULTIPLE]	Additional ports/target appear
20.550	Key into Reference Pathname port	<u>EG:</u> NET>HNV1>
20.555	Select [WITH OVERWRITE]	
20.560	Key into Pathname For IDF port	<u>EG:</u> HGPNTS for HG; AMPNTS for AM
20.570	Press <ENTER>	"WAIT" in yellow; "ENTITY NAMES" Point Config data loaded to HG. "OPERATION COMPLETE"
20.572	If errors occur, select the point on the error display. Make corrections; then repeat steps 20.530 – 20.570.	
20.580	If the Command Display is not on the screen, press <COMND>.	COMMAND DISPLAY
20.590	Select [LIST ENTITIES IN IDF]	Additional ports/target appear.

Task 20 CONFIGURE HG/AM/CMPOINTS

Step	Action	Response
20.600	Key into REFERENCE PATH NAME	<u>EG</u> : NET>HNV1>
20.605	Key into pathname for IDF port	<u>EG</u> : HGPNTS for HG, AMPNTS for AM
20.610	Key into pathname For SEL LIST	<u>EG</u> : BXSL.EL
20.614	Select BUILD TYPES of entities to list	<u>EG</u> : HG or AM or CIU as applies. Additional targets appear.
20.618	Select types of entities	<u>EG</u> : ALL
20.620	Press <ENTER>	"BUSY", DISPLAY OF LIST FILE display
20.630	For a printout, press <COMND>	COMMAND DISPLAY
20.640	Select [PRINT FILE]	Ports/std. suffixes appear
20.650	Key into pathname for FILE port	<u>EG</u> : BXSL.EL
20.655	Key into Printer Print Device ID port	<u>EG</u> : \$P1 (default used)
20.660	Press <ENTER>	Point IDs are printed. (For more details, select PRINT ENTITIES)
20.670	Repeat Task 20 to build "master" points for other Data Hiway boxes.	<u>EG</u> : MC point
20.680	Go to Appendix D of this document to EXCEPTION BUILD other Data Hiway box points as required.	
20.690	Repeat Task 20 to build other "master" points for AM and CL.	<u>EG</u> : AM points (v/ratio is used in task 27 for CL). See Figure 7.
20.680	Go to Appendix D of this document to EXCEPTION BUILD other AM or CL points as required.	

NOTE

You should save HG and AM checkpoints to Zip disk at this time. To do so, go perform instructions in Appendix B (HG) or C (AM) of this document.

Table 2 AM, AM/CL Point VLRATIO (Regulatory) Example

TAG NAME	VLRATIO
UNIT ID	01
POINT DESCRIPTOR	CL Point
ENGR UNITS DESCRIP.	Fur/Fnt
POINT KEYWORD	CLPOINT
POINT ASSOCIATED DISPLAY	(blank) (R510 and later systems)
PRIMARY MODULE POINT ID	-----
POINT DISCLOSURE	[FULL]
PV ALGORITHM	[CL]
CONTROL ALGORITHM	[NULL]
OVERVIEW VALUE	25
I/O DISPLAY SUPPR.	[NOSUPPR]
POINT EXEC. PERIOD	[NOPERIOD]
PV DECIMAL FORMAT	[D1]
PV RANGE HIGH	100.0
PV RANGE LOW	0.0
PV RANGE EXT. HIGH	102.9
PV RANGE EXT. LOW	-2.9
NUMBER ASSC. CL BLKS	1

All others—default values.

Table 3 AM Points/AM Point (Regulatory) Example

TAG NAME	AM101
UNIT ID	01
POINT DESCRIP.	TEMPERATURE
ENGR UNITS DESCRIP.	KBTU/MCF
POINT KEYWORD	TEMPA
POINT ASSOCIATED DISPLAY	(blank) (R510 and later systems)
PRIMARY MODULE POINT ID	-----
POINT DISCLOSURE	[FULL]
PV ALGORITHM	[SUMMER]
CONTROL ALGORITHM	[NULL]
POINT EXEC. PERIOD	5SEC
PV DECIMAL FORMAT	[D2]
PV RANGE HIGH	1.0
PV RANGE LOW	0.0
PV RANGE EX. HI	1.0
PV RANGE EX. LOW	-1.0
ALGORITHM EQN TYPE	[EQA]
OVERALL GAIN	0.01
NUMBER INPUT CONN.	1
SOURCE POINT. PARAM.	HG1001.OP
ALM ENB STATE	DISABLE

All others—default values.

Table 4 Parameters for (Regulatory) MC Point HG0501 Example

TAG NAME	HG0501
UNIT ID	01
POINT DESCRIPTOR	MC REGULATORY POINT NO 1
POINT KEYWORD	MCPID
PRIMARY MODULE POINT ID	-----
POINT ASSOCIATED DISPLAY	(blank) (R510 and later systems)
POINT DISCLOSURE	[FULL]
HIWAY NUMBER	02
BOX TYPE	[MC]
BOX NUMBER	05
POINT DISCLOSURE	[FULL]
SLOT NUMBER	01
LOAD DESTINATION	[HG]
ALGORITHM	[PIDNORM]
E. U. DESCRIPTOR	GAL/HR
OVERVIEW VALUE	20
PV RANGE OPTION	[CLMPZER0]
ALARM FORMAT	[ALFMT00]
ALARM PRIORITY	[LOW]
DISPLAY SUPPRESSION	[NOSUPPR]
OPERATOR MODE CHANGE	[PERMIT]

All others—default values.

Table 5 Parameters for PLCG Digital Input Point PLC0801 Example

TAG NAME	PLC0801
UNIT ID	01
POINT DESCRIPTOR	PL DIGITAL POINT NO 1
POINT KEYWORD	PL DIGIN1
PRIMARY MODULE POINT ID	-----
POINT ASSOCIATED DISPLAY	(blank) (R510 and later systems)
HIWAY NUMBER	04
BOX TYPE	[DHP]
BOX NUMBER	8
POINT DISCLOSURE	[BRIEF]
LOAD DESTINATION	[HG_HIWAY]
INPUT SLOT NUMBER	1
INPUT SUBSLOT NUMBER	1
ALARM FORMAT	[STATE1]
ALARM PRIORITY	[HIGH]
UPPER BOX TEXT	ON
LOWER BOX TEXT	OFF
PC TYPE	[MODICON]
INPUT 1 SPECIFIER CODE	0

All others—default values.

7.5 Task 21 – Configure MC/AMC Logic Blocks

Keyboard: Engr's

Lock: ENGR

Reference: MC11-500

Personality Loaded: Universal

NOTE

If the system has no Multifunction Controllers, skip to Task 22 to configure NIMs, PMs, APMs, and HPMS. If neither Multifunction Controllers nor NIMs, go to Task 23.

Task 21 CONFIGURE MC/AMC LOGIC BLOCKS

Step	Action	Response
21.01	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
21.03	Select [SYSTEM STATUS] display.	
21.04	Select the Hiway Gateway Node.	Node box is selected
21.05	Select [NTWK/HWYSTATUS]	Multifunction Controller Status display
21.07	Select [BOX STATUS]	
21.09	Enter Box Number	<u>EG</u> : 5
21.11	Press <ENTER>	BOX STATUS DISPLAY
21.14	Select [IDLE]	Target changes color.
21.16	Select [EXECUTE COMMAND]	MF CTLR STATUS: IDLE
21.18	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
CONFIGURE LOGIC BLOCKS		
21.20	Select [LOGIC BLOCKS]	LBC-BOX SELECTION display appears
21.22	Key into PATHNAME port	<u>EG</u> : NET>HNV1>LB020520 (02 = Hiway 2; 05 = Box 5, 20 = revision level 20)
21.24	Press <ENTER>	Message: "SELECT FUNCTION"
21.25	Press <F2>	Message: "TEXT BUILDER IS SELECTED"
21.26	Select [CREATE]	LBC-TEXT BUILDER display
21.27	Key into BLOCK NUMBER	<u>EG</u> : 1
21.28	Press <ENTER>	Additional ports displayed
21.29	Key into ENTER ALGORITHM port	<u>EG</u> : 1
21.30	Press <ENTER>	Algorithm number changes to AND; additional ports appear
21.31	Key into INPUT PARAMETER #1	<u>EG</u> : LP(01).SOA
21.32	Key into INPUT PARAMETER #2	<u>EG</u> : LP(11).SOA
21.33	Key into INPUT PARAMETER #3	<u>EG</u> : LP(12).SOA

Task 21 CONFIGURE MC/AMC LOGIC BLOCKS

Step	Action	Response
21.34	Tab to NEXT BLOCK, [YES] , press <SELECT>.	Block 1 information appears in lower half of display. At top left, Block Number changes to 2.
21.35	Tab to ENTER ALGORITHM; key into port	<u>EG</u> : 3
21.36	Press <ENTER>	Algorithm number changes to XOR; Additional ports appear
21.37	Key into INPUT PARAMETER #1	<u>EG</u> : LB(102)
21.38	Key into INPUT PARAMETER #2	<u>EG</u> : LB(103)
21.39	Key into INPUT PARAMETER #3	<u>EG</u> : LB(104)
21.40	Tab to NEXT BLOCK, [YES] , press <SELECT>	Block 2 information is added to lower half of display. At top left, Block Number changes to 3.
21.41	Tab to ENTER ALGORITHM; key in	<u>EG</u> : 7
21.42	Press <ENTER>	Algorithm number changes to FLIP-FLOP; additional ports appear
21.43	Key into INPUT PARAMETER #1	<u>EG</u> : LB(102)
21.44	Key into INPUT PARAMETER #2	<u>EG</u> : LB(103)
21.45	Key into OUTPUT PARAMETER	<u>EG</u> : FL(001)
21.46	Tab to NEXT BLOCK [YES] , press <SELECT>.	Block 3 information is added to lower half of display
21.47	Hold <CTL> down and press <^> (function key = F12)	Message: "WRITING FILE NET> HMV1>LB020520.JS"
21.48	Press <F4> Wait	LBC Compile Display appears; Message: "LBC COMPILE STARTED" Message: "WRITING FILE NET> HMV1>LB020520.JL/.JO" Message: "COMPILE COMPLETED" when compile is finished.
21.49	Press <F5>	Message: "LBC DATA LOAD STARTED"
21.50	Press <F6>	Message: "READING DATA BASE"- then LBC Compare Display appears with last line "NO DIFFERENCES FOUND"
21.51	For other Multifunction Controllers, press <F1>. Then, repeat 21.22-21.50. The following steps add comments to the file. If you do not want to add comments, skip to 21.68.	Comments are used to document the logic block configuration.
21.52	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
21.53	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
21.54	Type in EDIT command.	<u>EG</u> : EDIT NET>H MV1>LB020520.JS (.JS = Source File)

Task 21 CONFIGURE MC/AMC LOGIC BLOCKS

Step	Action	Response
21.55	Press <ENTER>	Logic Block data from previous steps appears.
21.56	Move the cursor to the end-of-file: Press <F3>, Press <F2>	Jump Menu appears Cursor moves to end-of-file following existing text
21.58	Press <RETURN>	
21.60	Type in the comments, for example: — <RETURN> — LOGIC BLOCKS 001 THROUGH 003 ARE OVERTEMP/CONTROL ALARMS. <RETURN> — THE OUTPUT OF THIS CIRCUIT CONTROLS A SECONDARY CIRCUIT<RETURN> — CONSISTING OF LOGIC BLOCKS 010 THROUGH 017.<RETURN>	
21.62	Move the cursor to the beginning-of-file: Press <F3>, Press <F1>	Jump Menu appears Cursor moves to beginning-of-file
21.64	Update and exit from the file: Press <F1>, Press <F2>	Quit Menu appears Message: "FILE WAS UPDATED- X LINES WRITTEN"
21.66	Repeat 21.54-21.64 for other Logic Block Files.	
21.68	Hold down <CTL> and press <HELP>	ENGINEERING MAIN MENU

NOTE

If the system has a **Network Interface Module/ PM, APM, or HPM**, go to Task 22 near the back of this manual. The sample system did not have an LM or SM, but this would be the time to configure them also. When the NIM and UCN nodes have been configured, continue on to Task 23.

7.6 Task 22 – Configure UCN Nodes and Subsystems

CAUTION

The NIM(s) must be loaded with its operating personality and have a status of OK on the System Status Display before starting this procedure. The NIM was loaded in Task 16.

If you are configuring a PM or HPM, the procedure is nearly identical to the APM configuration described below except where noted. The sample system did not have an SM or LM. Refer to the manuals in Section 1, References, if necessary, to configure these.

Keyboard: Engr's

Lock: ENGR

Reference: PM09-640, AP09-640, and HP09-640

Personality Loaded: Universal

Task 22 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
22.110	Hold <CTL> down and press <HELP> If you don't have a NIM, skip to Task 23.	ENGINEERING MAIN MENU
\$NMhhNnn Where hh is the UCN number and nn is the NIM Node number		
CONFIGURE THE NIM UCN ADDRESSES		
22.115	Select [NETWORK INTF. MODULE]	NIM BUILD TYPE SELECT MENU
22.120	Select [UCN NODE CONFIGURATION]	UCN-NODE CONFIGURATION display
22.130	Key into NETWORK NUMBER port	EG: 03
22.140	Key into UCN NODE NUMBER port (NIM node number)	EG: 01 (primary NIM node # on 1st pass) EG: 02 (redundant NIM node number, if any, on 2nd pass)
22.150	Select an NODE TYPE option	EG: [NIM]
22.160	Select a NODE ASSIGNMENT option	EG: [THISNIM]
22.165	Press <ENTER>	PED reappears
22.169	Select a LOAD SCOPE option	EG: [NIMANDPM]
22.173	Select a NUMBER OF MESSAGE TEXTS	EG: [3]
22.180	Press <ENTER>	Message Text ports appear
22.184	Key in message text [RED TAG] [IN USE]	EG: [OUT/SERV]
22.188	Select an SOE SYNC option	EG: [ENABLE] (if NIM has EPNI board)
22.190	Press <ENTER>	PED reappears with values in cyan

NOTE

If you select any options outlined in yellow or if you key data into any port outlined in yellow, the content of the PED-set or PED changes when you press the <ENTER> key. You should press <ENTER> after each PED page.

Task 22 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
22.210	Press <PAGE FWD> to next PED page (if any) key in data, select options, and press <ENTER> after each page.	UCN NODE CONFIGURATION Ped
WRITE TO IDF		
22.320	Press <COMND>	COMMAND DISPLAY
22.400	If this is the <u>first</u> write to this IDF, go to step 22.460; if not, go to step 22.485.	IDF for this example is NIMIDF
22.460	Select [WRITE TO IDF]	Pathname ports and target appear.
22.470	Key into REFERENCE PATHNAME port	<u>EG:</u> NET>HNV1>
22.480	Key into pathname for IDF port	<u>EG:</u> NIMIDF
22.485	Press <ENTER>	NIM Config data written to IDF. UCN NODE CONFIGURATION display
22.488	Repeat steps 22.130 – 22.485 for redundant NIM (if any).	
22.490	Repeat steps 22.130 – 22.488 for each other NIM pair.	All configured NIMs on IDF
22.500	Press <COMND>	COMMAND DISPLAY
22.510	Select [LOAD MULTIPLE]	Additional pathname ports appear
22.515	Key into REFERENCE PATHNAME port	<u>EG:</u> NET>HNV1>
22.520	Key into pathname for IDF port	<u>EG:</u> NIMIDF
22.525	Press <ENTER> "OPERATION COMPLETE" COMMAND DISPLAY	"WAIT" in yellow; "PROCESSING ..."
22.530	Press <F1>	UCN NODE CONFIGURATION display
\$NMhhNnn Where hh is the UCN number and nn is the APMM Node number		
	UCN device* (PM/APM/HPM/LM/SM) NODE CONFIGURATION *NOTE: References on examples used (EG) are APM but can apply to any UCN device.	The UCN NODE CONFIGURATION display should be on the screen.
22.550	Key into NETWORK NUMBER port	<u>EG:</u> 03
22.560	Key into UCN NODE NUMBER port (APMM node number)	<u>EG:</u> 03 (Primary node # on first pass, redundant node #, if any, on second pass) <u>EG:</u> 04
22.570	Select [UCN NODE TYPE]	<u>EG:</u> [APM]
22.580	Select a NODE ASSIGNMENT option	<u>EG:</u> [THISNIM]
22.590	Press <ENTER>	PED reappears with values in cyan
22.595	Key in any other data/select options press <ENTER>	

Task 22 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
WRITE TO IDF		
22.610	Press <COMND>	COMMAND DISPLAY
22.620	If this is the <u>first</u> write to this IDF, go to step 22.630; if not, go to step 22.660.	IDF for this example is APMMIDF
22.630	Select [WRITE TO IDF]	Pathname ports and target appear.
22.640	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
22.650	Key into pathname for IDF port	<u>EG</u> : APMMIDF
22.660	Press <ENTER>	APM Config data written to IDF. UCN NODE CONFIG. PED display
22.670	Repeat steps 22.550 – 22.660 for the redundant APMM (if any).	
22.680	Repeat steps 22.550 – 22.670 for each other APMM.	All configured APMMs in IDF
22.700	Press <COMND>	COMMAND DISPLAY
22.710	Select [LOAD MULTIPLE]	Additional pathname ports appear
22.720	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>
22.730	Key into pathname for IDF port	<u>EG</u> : APMMIDF
22.740	Press <ENTER> Repeat steps 22.550 - 22.740 for other PMs, or HPMs (if any).	"WAIT" in yellow;"...OPERATION COMPLETE" Use different IDF names for PMs and HPMs
LOAD THE APM		
22.742	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
22.744	&Z6 copy into \$F1	
22.746	Select [SYSTEM STATUS]	SYSTEM STATUS display
22.748	Select the NIM Node	Node Box is selected
22.750	Select [NTWK/HWY STATUS]	UCN nn STATUS display
22.754	Select [nn APM nn] The status must be <i>ALIVE</i> (Select target 2nd time if loading the redundant node)	Target fills with color; selected node is highlighted.
22.758	Select [LOAD/SAVE RESTORE]	Additional targets appear
22.762	Select [PROGRAM LOAD]	Target turns white
22.766	Select [ENTER]	PGM SOURCE FOR NODE nn display
22.770	Select [DEFAULT SOURCE]	EXECUTE target appears
22.774	Select [EXECUTE]	DATA SOURCE FOR NODE nn display
22.778	Select [ALTERNATE SOURCE]	Targets [1], [2], etc. appear
22.782	Select <u>EG</u> : [1]	&Z6 and [EXECUTE COMMAND]

Task 22 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
22.786	Select [EXECUTE COMMAND]	" BUSY UCN nn " [nn APM nn LOADING] " * " "MOUNTAPM nn DATA BASE"
22.790	Select [MOUNT APMnn DATA BASE] (Primary node only)	Additional targets appear
22.794	Select [ABORT REQUEST] (Primary node only)	<u>EG</u> : [nn APM nn] [IDLE] [ALIVE/IDLE] [LOADED]
22.796	Repeat Tasks 22.754 – 22.794 for redundant APM (if any).	<u>EG</u> : [nn APM nn] [IDLE] [BACKUP] [LOADED]
22.798	Repeat Tasks 22.754 – 22.796 for other PMs, APMs, or HPMs.	
\$NMhhBuu Where hh is the UCN number and uu is the APMM Node number		
PM/APM/HPM NODE SPECIFIC CONFIGURATION		
22.800	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
22.802	Select [NETWORK INTF. MODULE]	NIM BUILD TYPE SELECT MENU
22.806	Select [NODE SPECIFIC CONFIG.]	NODE SPECIFIC CONFIGURATION MENU
22.808	Key into NETWORK NUMBER	<u>EG</u> : 03
22.814	Key into NODE NUMBER	<u>EG</u> : 03 (Primary APMM)
22.816	Key into NODE TYPE	<u>EG</u> : APM
22.818	Press <ENTER>	
22.820	Press <PAGE FWD> to next PED page, key in data, select options, and press <ENTER> after each page or gold port	NODE SPECIFIC CONFIG. PED
22.822	Select a CONTROL PROC. OPTION (Not Applicable for HPMs)	<u>EG</u> : [ON]
22.826	Key into NUM OF REG CONTROL SLOTS	<u>EG</u> : 25
22.830	Key into NUMBER OF FAST REG	<u>EG</u> : 5
22.834	Key into NUMBER OF REG PV SLOTS	<u>EG</u> : 20
22.840	Key into NUM OF FAST REG PV SLOTS	<u>EG</u> : 5
22.844	Key into NUMBER OF LOGIC SLOTS	<u>EG</u> : 10
22.846	Key into NUM OF FAST LOGIC SLOTS	<u>EG</u> : 3
22.850	Key into NUM OF DIG. COMP. SLOTS	<u>EG</u> : 15
22.852	Key into NMBER OF FAST DIG. COMP.	<u>EG</u> : 2
22.854	Key into NUM OF DEV. CTL SLOTS	<u>EG</u> : 4
22.856	Key into NMBER OF FAST DEV. CTL	<u>EG</u> : 2

Task 22 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
22.858	Key into SEQUENCE PROCESSING	<u>EG</u> : 1_PU
22.860	Key into NUM OF PROC MODULE SLOTS	<u>EG</u> : 10
22.862	Key into NUM OF NUMERICS SLOTS	<u>EG</u> : 48 (Multiples of 16 only)
22.864	Key into NUM OF STRINGS SLOTS*	<u>EG</u> : 16 (Multiples of 16 only)
22.866	Key into NUM OF TIMES SLOTS*	<u>EG</u> : 32 (Multiples of 32 only)
22.868	Key into NUM OF ARRAY SLOTS*	<u>EG</u> : 10
22.870	Select an SI DATA... SCAN Period*	<u>EG</u> : 1.0
22.874	Select a SCAN RATE	<u>EG</u> : [REG1LOG1]
22.880	Select a PACKAGING OPTION (Select REDUN if redundant nodes)	<u>EG</u> : [REDUN]
22.883	Select a Simulation Indicator option*	<u>EG</u> : ON
* Not applicable for PMs		
22.885	Press <PAGE FWD> to next PED page, key data, select options, and press <ENTER> after each page	I/O MODULE CONFIGURATION PED
22.900	Choose an I/O Module number, press <PAGE FWD> if necessary	<u>EG</u> : I/O MODULE NUMBER 1
22.910	Key into FILE NUMBER	<u>EG</u> : 1
22.920	Key into CARD NUMBER (Example is 1st card Number if REDUN is selected)	<u>EG</u> : 11 for APM ; (9 for PM, 03 for HPM)
22.930	Select a MODULE TYPE	<u>EG</u> : [HLAI]
22.932	If the system has redundant IOPs, select REDUN. If not, skip to 22.940.	<u>EG</u> : IOMB File and Card number ports
22.934	Enter redundant IOP File and Card No.	<u>EG</u> : File 2, Card No. 11
22.940	Choose another I/O Module number, press <PAGE FWD> if necessary	<u>EG</u> : MODULE NUMBER 2
22.950	Key into IOM-A FILE NUMBER	<u>EG</u> : 1
22.960	Key into IOM-A CARD NUMBER	<u>EG</u> : 12 for APM
22.970	Select MODULE TYPE	<u>EG</u> : [AO]
22.975	Select a FAILURE OPTION	<u>EG</u> : [HOLD]
22.980	Key in any other data/select options, press <ENTER>.	
22.990	Press <PAGE FWD> to next PED page (if any) key in data, select options, and press <ENTER> after each page	I/O MODULE CONFIGURATION PED
22.995	Repeat steps 22.940 through 22.990 to configure other I/O Modules.	

Task 22 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
WRITE TO IDF		
22.1000	Press <COMND>	COMMAND DISPLAY
22.1020	If this is the <u>first</u> write to this IDF, go to step 22.1030; if not, go to step 22.1060.	IDF for this example is APMCONFG
22.1030	Select [WRITE TO IDF]	Pathname ports and target appear.
22.1040	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
22.1050	Key into pathname for IDF port	<u>EG</u> : APMCONFG or HPMCONFIG
22.1060	Press <ENTER>	I/O Module Config data written to IDF. IO MODULE CONFIG. display
22.1070	Repeat steps 22.800 – 22.1060 for other APMs on this UCN (if any).	
22.1080	Repeat steps 22.800 – 22.1070 for each other PM, APM, or HPM primary (if any).	All configured PMs, APMs, HPMs on IDFs (use different IDF names for PMs and HPMs)
22.1100	Press <COMND>	COMMAND DISPLAY
22.1110	Select [LOAD MULTIPLE]	Additional pathname ports appear
22.1115	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>
22.1120	Key into pathname for IDF port	<u>EG</u> : APMCONFG
22.1125	Press <ENTER>	"WAIT" in yellow;.."OPERATION COMPLETE"
22.1150	Repeat steps 22.1100 – 22.1150 for PMs or HPMs (if any).	PMs, APMs, HPMs loaded with Node Specific Configuration Data
PM/APM/HPM POINT BUILDING		
22.1200	Press <F9>	NIM BUILD TYPE SELECT MENU
22.1220	Select [PROCESS POINT BUILDING]	NIM PROC. POINT BUILD TYPE display
22.1230	Select a Point Build Type	<u>EG</u> : [REGULATORY CONTROL] NIM-POINT ASSIGNMENT display
22.1240	Key into TAG NAME port	<u>EG</u> : FC2401
22.1244	Select a NODE TYPE	<u>EG</u> : APM
22.1250	Select a POINT FORM option	<u>EG</u> : [FULL]
22.1260	Key into the POINT DESCRIPTOR port	<u>EG</u> : TOWER FLOW CONTROL
22.1265	Key into the E. U. DESCRIPTOR port	<u>EG</u> : KB/D
22.1275	Key into the POINT KEYWORD port	<u>EG</u> : FLOW
22.1285	Key into the UNIT ID port	<u>EG</u> : 01
22.1290	Key into the NETWORK NUMBER port	<u>EG</u> : 03
22.1295	Key into the NODE NUMBER port	<u>EG</u> : 03
22.1300	Key into the SLOT NUMBER port	<u>EG</u> : 01
22.1310	Key in other data/select options press <ENTER>.	

Task 22 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
22.1315	Press <PAGE FWD>	Next PED page
22.1320	Select a Control Algorithm; press <ENTER>	<u>EG</u> : [PID]
22.1325	Press <PAGE FWD>	Next PED page
22.1330	Key into PV RANGE HIGH port	<u>EG</u> : 25.00000
22.1335	Select a PV SOURCE OPTION	<u>EG</u> : [ALL]
22.1340	Press <ENTER> after each page and after any entry boxed in yellow.	
22.1345	Press <PAGE FWD>	Next PED page
22.1350	Select a NORMAL MODE	<u>EG</u> : [AUTO]
22.1355	Select a NORMAL MODE ATTRIBUTE	<u>EG</u> : [OPERATOR]
22.1360	Key in any other data/select options; press <ENTER> after each page	
22.1365	Press <PAGE FWD>	Next PED page
22.1370	Key into SET POINT HIGH LIMIT port	<u>EG</u> : 25.00000
22.1380	Key into SET POINT LOW LIMIT port	<u>EG</u> : 0.000000
22.1390	Key into SET POINT port	<u>EG</u> : 15.00000
22.1395	Key in any other data/select options; press <ENTER> after each page.	
22.1400	Press <PAGE FWD>	Next PED page
22.1420	Select an ALGORITHM FORM option	<u>EG</u> : [IDEAL]
22.1430	Key into OVERALL GAIN VALUE port	<u>EG</u> : 0.699200
22.1440	Key into INTEG. TIME IN MINUTES port	<u>EG</u> : 0.100000
22.1445	Key in any other data/select options; press <ENTER> after each page.	
22.1450	Press <PAGE FWD>	Next PED page
22.1460	Key into INPUT CONN. SOURCE port	<u>EG</u> : - - . - -
22.1462	Key into OUTPUT DEST. SOURCE port	<u>EG</u> : !AO02S02.OP
22.1465	Key in any other data/select options; press <ENTER> after each page.	
22.1470	Press <PAGE FWD>	Next PED page
22.1485	Key in any other data/select options; press <ENTER> after each page.	
22.1490	Press <PAGE FWD>	Next PED page
22.1500	Key into PV HIGH TRIP POINT port	<u>EG</u> : 25.00000
22.1502	Key into PV LOW TRIP POINT port	<u>EG</u> : 0.000000
22.1504	Key into POSITIVE PV ROC TRIP port	<u>EG</u> : 32.00000

Task 22 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
22.1508	Press <ENTER> after each page and after each entry boxed in yellow.	
22.1510	Press <PAGE FWD>	Next PED page
22.1512	Key into NEG. PVROC TRIP POINT port	<u>EG</u> : 20.00000
22.1514	Key into DEV HIGH TRIP POINT port	<u>EG</u> : 5.000000
22.1516	Key into DEV LOW TRIP POINT port	<u>EG</u> : 5.000000
22.1518	Press <ENTER>	
22.1520	Press <PAGE FWD> to next PED page. Enter any other data/select options and press <ENTER> after each page until all pages are configured.	
22.1522	Press <F8>	PED STATUS display
22.1524	Check for proper status	"Display Set Entered?" Yes Errors? No
22.1526	If status is OK, go to step 22.1530; else select faulty item; enter corrections.	
WRITE TO IDF		
22.1530	Press <COMND>	COMMAND DISPLAY
22.1532	If this is the <u>first</u> write to this IDF, go to step 22.1534; if not, go to step 22.1560.	IDF for this example is APMPNTS
22.1534	Select [WRITE TO IDF]	Pathname ports and target appear.
22.1540	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
22.1550	Key into pathname for IDF port	<u>EG</u> : APMPNTS
22.1560	Press <ENTER>	NIM Point Config. data written to IDF. NIM Point PED display
22.1570	Repeat steps 22.1240 – 22.1560 to build other points of this type, or press <F9>. Then repeat steps 22.1230 – 22.1570 to build other types of NIM Points until all NIM points are built. If you want to clone points from a master point, go to Appendix D.	
22.1600	Press <COMND>	COMMAND DISPLAY
22.1610	Select [LOAD MULTIPLE]	Additional pathname ports appear
22.1620	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>
22.1630	Key into pathname for IDF port	<u>EG</u> : APMPNTS
22.1640	Press <ENTER> Build/load points for PMs and HPMs (if any) by repeating steps 22.1200 - 22.1640.	"WAIT" in yellow; "PROCESSING ..." "OPERATION COMPLETE" Use different IDF names for PM or HPM points.

Task 22 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
NIM LIBRARY CONFIGURATION \$NhhLIBn Where hh is the UCN number and n is the library number Note: The NIM Library is usually automatically loaded while compiling an APM Sequence program. The following steps illustrate how to access, manually load, or modify it.		
22.1710	If the COMND or Point CONF. display is on screen, press <F9> <u>twice</u> . Then go to step 22.1720. Else hold <CTL> down and press <HELP>	POINT TYPE BUILD MENU, then NIM BUILD TYPE SELECT MENU ENGINEERING MAIN MENU
22.1715	Select [NETWORK INTF. MODULE]	NIM BUILD TYPE SELECT MENU
22.1720	Select [LIBRARY CONFIGURATION]	NIM LIBRARY CONFIGURATION display
22.1722	Key into the NETWORK NUMBER port	<u>EG:</u> 03
22.1724	Key into the Library Number port	<u>EG:</u> 1
22.1728	Press <ENTER>	
22.1730	<PAGE FWD>	
22.1732	Key into the INDEX ENTRY ports	<u>EG:</u> TRUE for 002, FALSE for 003, etc.
22.1734	Press <ENTER>	
22.1736	<PAGE FWD>. Enter any other data, press <ENTER> after each page.	
WRITE TO IDF		
22.1740	Press <COMND>	COMMAND DISPLAY
22.1745	If this is the <u>first</u> write to this IDF, go to step 22.1755; if not, go to step 22.1765.	IDF for this example is LIBCNFG
22.1750	Select [WRITE TO IDF]	Pathname ports and target appear.
22.1755	Key into REFERENCE PATHNAME port	<u>EG:</u> NET>HNV1>
22.1760	Key into pathname for IDF port	<u>EG:</u> LIBCNFG
22.1765	Press <ENTER>	NIM Library Config data written to IDF. NIM LIBRARY CONFIGURATION display
22.1767	Repeat steps 22.1724 – 22.1765 to build Libraries 2/3 (if desired) for this UCN.	
22.1770	Repeat steps 22.1722 – 22.1767 to build Libraries for other UCNs.	
22.1800	Press <COMND>	COMMAND DISPLAY
22.1810	Select [LOAD MULTIPLE]	Additional pathname ports appear
22.1820	Key into Reference Pathname port	<u>EG:</u> NET>HNV1>
22.1830	Key into pathname for IDF port	<u>EG:</u> LIBCNFG
22.1840	Press <ENTER>	"WAIT" in yellow; "PROCESSING ..." "OPERATION COMPLETE"

Task 22 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
NOTE In the following steps you will not be able to save data for any IO Processor for which you have not previously built and loaded at least one data point. When saving data, all IOPs that are configured for a given PM, APM, or HPM will assume a "DB Invalid" status (and should not be validated until a point has been built and loaded).		

SAVE CHECKPOINT DATA FOR A NIM and UCN devices (PM/APM/HPM/LM/SM)		
22.1910	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
22.1920	CKPT Zip into \$F1	
22.1930	Select [SYSTEM STATUS]	SYSTEM STATUS display
22.1934	Select the NIM Node	Node Box is selected
22.1938	Select [NTWK/HWY STATUS]	UCN nn STATUS display
22.1945	Select the appropriate UCN device target.	EG: nn APM nn target fills with color
22.1950	Select [LOAD/SAVE RESTORE]	Additional targets appear.
22.1955	Select [SAVE DATA]	Target changes color; additional targets
22.1960	Select [ENTER]	SAVE UCN NODE display
22.1965	If you wish to save checkpoints on emulated disk, go to step 22.1970, then execute below: If you wish to save checkpoints on an HM, then you must copy null METADATA file to HM checkpoint volume. Mount copy &Z6.LCN Select [DEFAULT SOURCE] and then go to step 22.1980.	Note: Both NIM data and the associated APM data is saved by this procedure. CP \$F5>&Ixx>METADATA.FF NET>&Ixx>= -D
22.1970	CHKPOINT.LCN emulated disk into \$F1	
22.1975	Select [ALTERNATE SOURCE]	Targets [1] and [2] and so on appear
22.1978	Select [1]	<u>EG:</u> Volume ID = CKPT
22.1980	Select [EXECUTE COMMAND]	"BUSY UCN nn display and " * "
22.1999	Wait for status to change	"SAVED"
22.2000	Repeat steps 22.1945 – 22.1999 for other PMs, APMs, and HPMs.	

Section 8 – Procedure 5/Configure Custom Functions

Figure 8 shows a graphic representation of the Tasks (23-28) that must be performed to Configure Custom Functions.

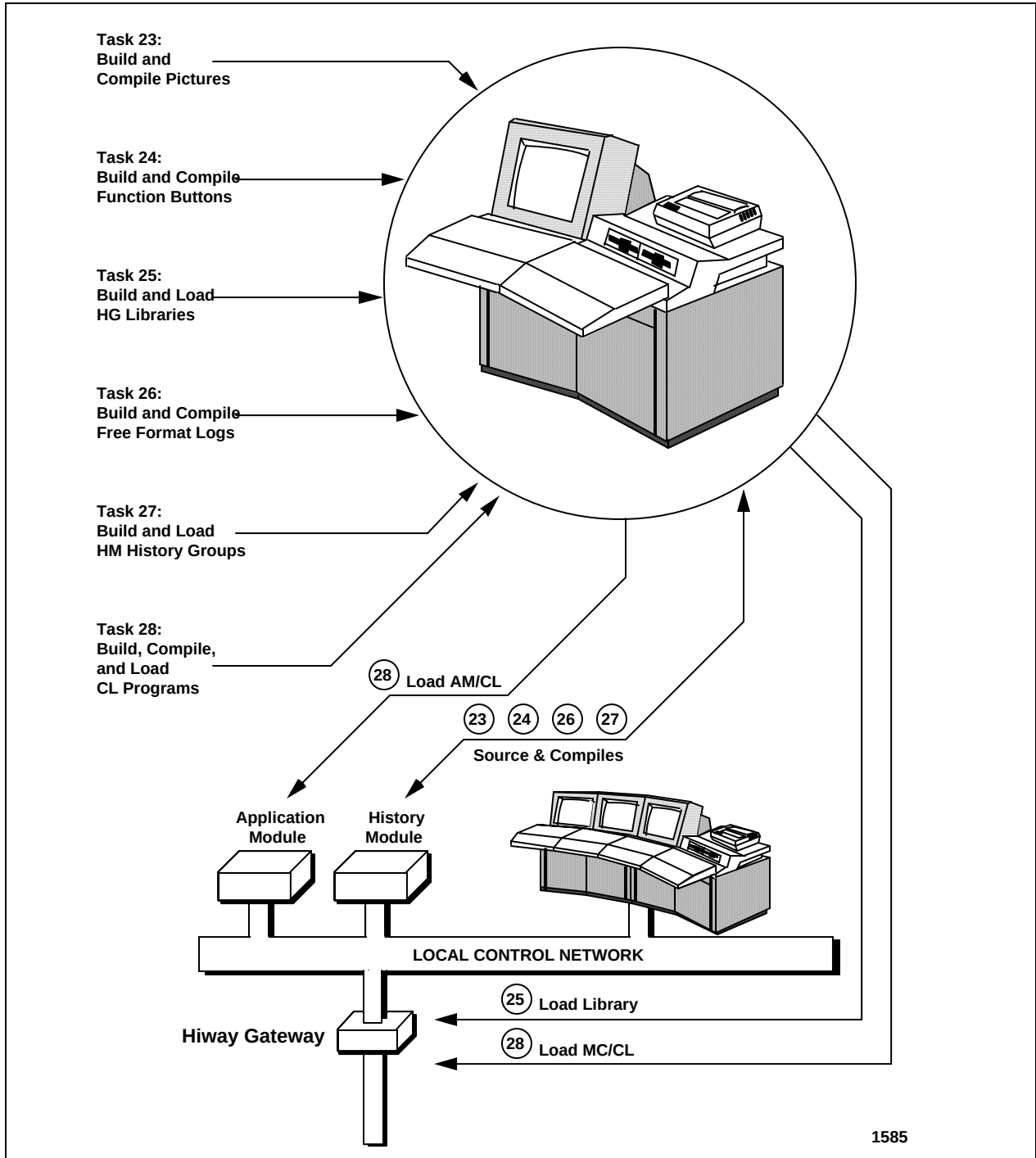


Figure 8 Configure Custom Functions

8.1 Task 23 – Build and Compile Pictures

Keyboard: Engr's

Lock: ENGR

Reference: SW09-650 and SW11-650

Personality Loaded: Universal

Task 23 BUILD AND COMPILE PICTURES

Step	Action	Response
23.10	Make sure points referenced by pictures have been built and loaded.	
23.14	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
23.18	Select [PICTURE EDITOR]	PICTURE EDITOR display
23.19	The following steps and Figures 9 and 10 show how to build a simple Custom Graphic Display.	If pictures were prebuilt, go to step 23.160 to read them in and then compile them on the HM.
23.22	Key in a Picture Editor command	<u>EG:</u> ADD SOLID
23.28	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
23.32	Move cursor to desired location	Hold <CTL> down, press <TAB> keys
23.40	Press <SELECT>	Cross marks the coordinate (see Figure 9 part a)
23.44	Repeat steps 23.32 – 23.40 until all of the coordinates are selected.	Note that deletes the last coordinate selected if you need to make a correction.
23.48	Press <ENTER>	Command Completed Note: The NEW command clears the screen if you want to redraw.
23.56	Repeat steps 23.22 – 23.48 to build additional objects.	<u>EG:</u> Another Solid. (Figure 9 part b)
23.60	Hold <CTL> down and press a color key (color keys are on the keypad).	<u>EG:</u> <F4> Status indicator at top center of screen changes to white.
23.65	Key in a Picture Editor command	<u>EG:</u> ADD LINE
23.70	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
23.75	Move cursor to desired location	<u>EG:</u> Hold <CTL> down; press TAB
23.80	Press <SELECT>	Cross marks the Coordinate
23.85	Repeat steps 23.75 – 23.80 until all of the coordinates are selected.	<u>EG:</u> Lines connect objects (See Figure 10 part c).
23.90	Press <ENTER>	Command Completed

Task 23 BUILD AND COMPILE PICTURES

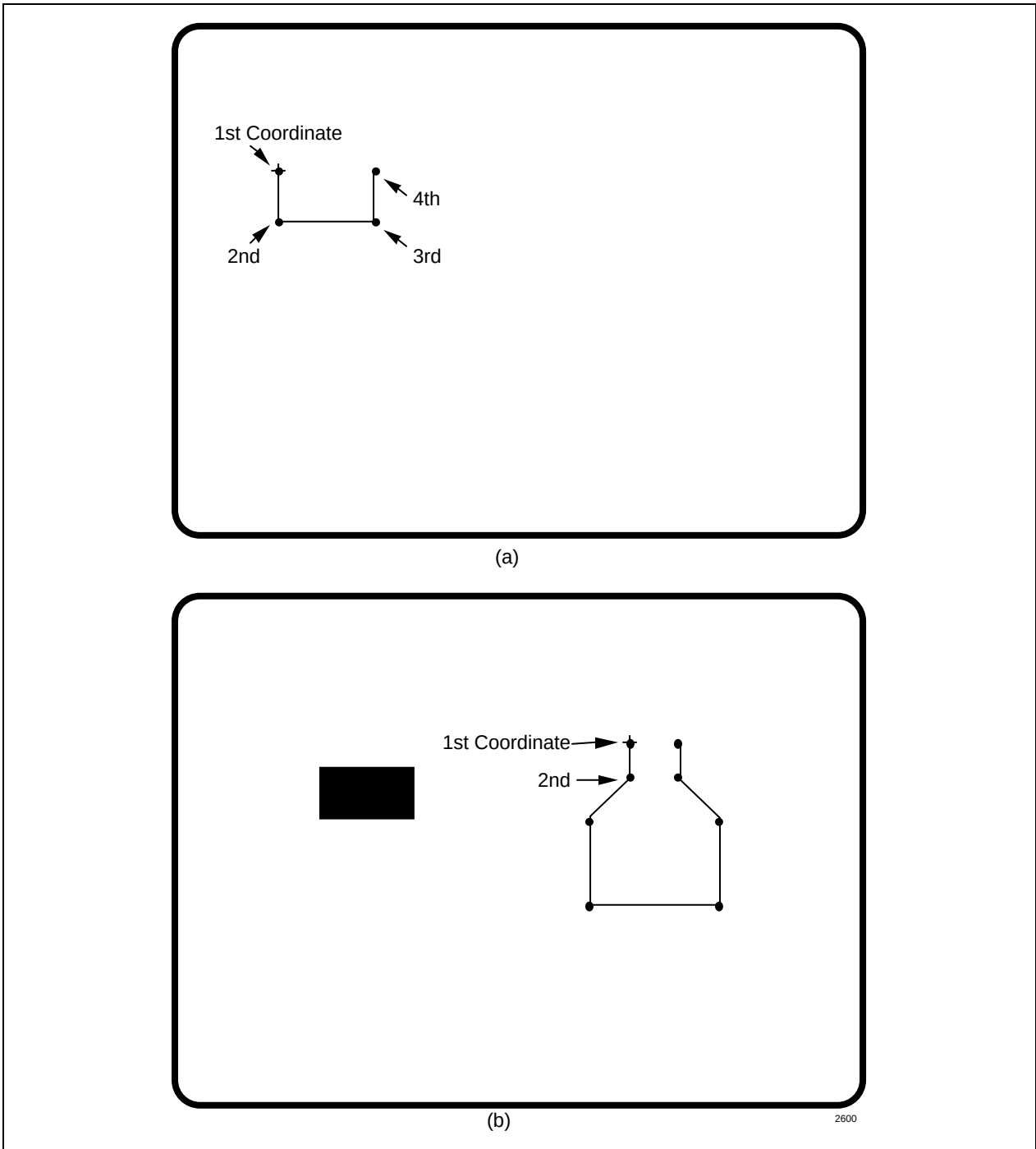
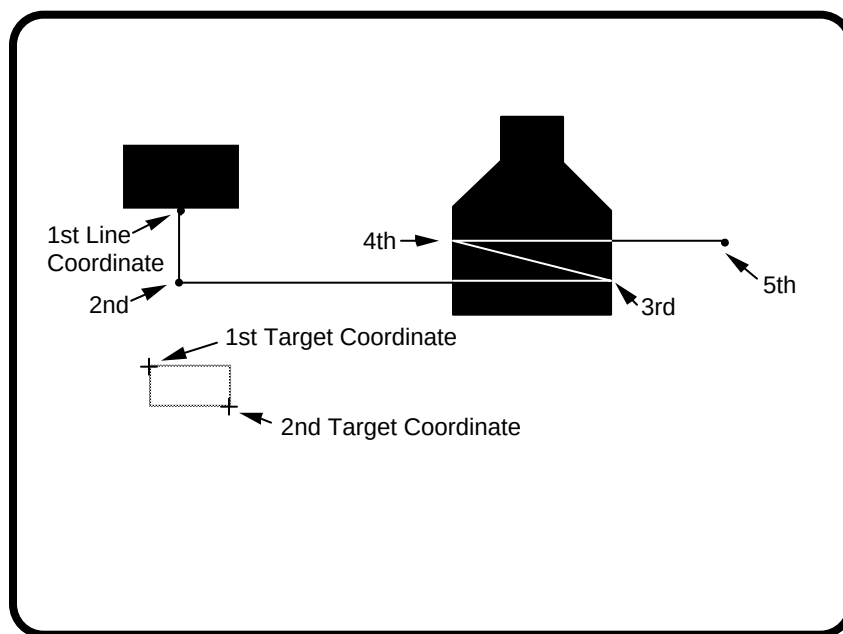
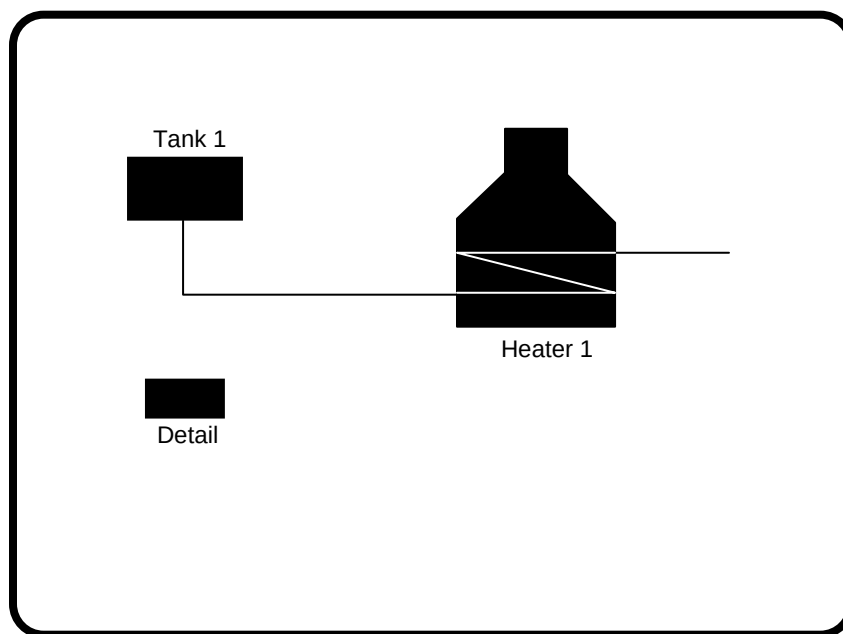


Figure 9 Picture Building

Task 23 BUILD AND COMPILE PICTURES



(c)



(d)

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Figure 10 Picture Building, continued

Task 23 BUILD AND COMPILE PICTURES

Step	Action	Response
23.94	Key in Picture Editor commands to add conditional behavior, and so on.	<u>EG</u> : SEL SOL; ADD COND
23.98	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
23.102	Move cursor to desired location	<u>EG</u> : Place cursor on solid rectangular object
23.106	Press <SELECT>, then press <ENTER>	Object turns white and blinks; Screen Form appears.
23.110	Key in screen form information (points you refer to must have been built and loaded in Task 21).	<u>EG</u> : (in condition block): IF HG1001.PV > 75 THEN SET RED ELSE SET CYAN
23.114	Press <ENTER>	Picture returns (or point Type/Format requested if point not found)
23.118	Key in commands to add targets	<u>EG</u> : ADD TARG
23.124	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
23.128	Move cursor to desired location.	<u>EG</u> : Place cursor under first solid rectangular object (see 9-2c)
23.132	Press <SELECT>	Cross marks 1st target coordinate
23.136	Move cursor; press <SELECT>	Rectangle drawn around coordinates
23.140	Press <ENTER>	Target screen form appears
23.145	Key into Action area	<u>EG</u> : DETAIL("HG1001")
23.150	Press <ENTER>	Picture returns (or point Type/Format requested if point not found)
23.154	Enter any other commands to complete desired custom display.	<u>EG</u> : ADD TEXT Label objects (see 9-2d)
23.156	Press <ENTER> to complete command	
23.159	Go to step 23.210 to compile the display	
Follow steps 23.160 – 23.190 to read in prebuilt pictures from floppies, cartridge, or Zip disks.		
23.160	Insert Zip disk with picture files in \$F1; key in the SET PATHNAME command.	<u>EG</u> : S P \$F1>VOL1>DISP1 (where VOL1 is a Zip volume name and DISP1 is a picture file on the Zip)
23.170	Press <ENTER>	Pathname appears at top of display
23.180	Key the READ command onto the command line.	<u>EG</u> : READ DISP1 (Assumes DISP1 is filename)
23.190	Press <ENTER>	Picture appears on US screen
23.210	Key in a COMPILE command	<u>EG</u> : COMPILE NET>HMV1>DISP1

Task 23 BUILD AND COMPILE PICTURES

Step	Action	Response
23.220	Press <ENTER>. Picture on the screen is compiled onto the HM To have a redundant schematic, you should compile it to other HMs, Zip disks, or floppies now.	Message: "WRITING SOURCE FILE" Message: "WRITING OBJECT FILE" No redundant schematics were configured on the example system.
23.230	Repeat steps 23.19 – 23.220 until each of your pictures is compiled.	
23.240	If you want to save picture source files insert Zip in \$F1.	<u>EG</u> : USER (volume created in task 4)
23.241	Press <ESC>	"ESCAPED FROM PICTURE EDITOR"
23.244	Key in a copy command	<u>EG</u> : CP NET>HNV1>DISP1.DS \$F1>PICT> DISP1 -D
23.248	Press <ENTER> Source file is copied	Message: "COPY COMPLETE"
23.249	Hold <CTL> down and press <HELP>	Return to Picture Editor

8.2 Task 24 – Build and Compile Buttons

Keyboard: Engr's

Lock: ENGR

Reference: SW11-570

Personality Loaded: Universal

Task 24 BUILD AND COMPILE BUTTONS

Step	Action	Response
Make sure all the points and pictures you need were created in previous tasks.		
24.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
24.30	Select [BUTTON CONFIGURATION]	BUTTON CONFIGURATION MENU
24.35	Select a button (<u>EG</u> : 7)	ACTION & LAMP DATA regions appear
24.40	Key into the Action region	<u>EG</u> : DETAIL("HG1001")
24.45	Press <ENTER>	BUTTON CONFIGURATION MENU appears
24.50	Select another button (<u>EG</u> : 8)	ACTION & LAMP DATA regions appear
24.55	Key into the Action region	<u>EG</u> : SCHEM("DISP1")
24.60	Press <ENTER>	BUTTON CONFIGURATION Menu
24.65	Select another button (<u>EG</u> : 9)	Action & Lamp Data regions appear
24.70	Key into the Action region	<u>EG</u> : SCHEM(R_STR)
24.75	Press <ENTER>	PARAMETER ENTRY display
24.80	Key into ports exposed by Actor.	<u>EG</u> : 10 into "Tip Length" <u>EG</u> : "ENTER SCHEMATIC NAME" into "Prompt 1–30 Char"
24.85	Press <ENTER>	BUTTON CONFIGURATION MENU
CONFIGURE A DISPLAY SET		
24.90	Select another button	<u>EG</u> : 10 ACTION & LAMP DATA regions appear
24.100	Key into the Action region	<u>EG</u> : CROSSCRN(1);SCHEM("DISP1"); CROSSCRN(2);GROUP(1,0)
24.140	Press <ENTER>	BUTTON CONFIGURATION MENU
24.150	Repeat steps 24.35 – 24.140 to configure rest of buttons in set.	Buttons for all stations in an Area are configured by this task.
NOTE If you want a button on the Operator Keyboard to call up either the Overview Display or the Area Trend Display, configure them here. For the Universal Work Station's Engineering Keyboard, <PAGE FWD> from the main Button Menu to the PF Button Menu and configure as desired.		
24.160	Key COMPILE xxx into the cmd line	<u>EG</u> : COMPILE NET>&D01>BUTTON
24.170	Press <ENTER>	"WRITING SOURCE/OBJECT FILE"
24.180	Key in a PRINT command	<u>EG</u> : PRINT 7..10 \$P1

Task 24 BUILD AND COMPILE BUTTONS

Step	Action	Response
24.190	Press <ENTER>	Button Config. record is printed

8.3 Task 25 – Build and Load HG Libraries

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

Task 25 BUILD AND LOAD HG LIBRARIES

Step	Action	Response
NOTE: If you don't have a Hiway Gateway, skip to Task 26		
HG LIBRARY \$HnnLIBx where nn = Hiway No. and x = Library No.		
25.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
25.15	Select [HIWAY GATEWAY]	HG-BUILD TYPE AND CONFIG. MENU
25.20	Select [HG LIBRARY]	HG LIBRARY CONFIGURATION display
25.30	Key into HIWAY NUMBER port	<u>EG</u> : 02 (configured in Task 9)
25.40	Key into LIBRARY NUMBER port	<u>EG</u> : 1
25.50	Key into INDEX ports	<u>EG</u> : ON (into "Index 001" port) <u>EG</u> : OFF (into "Index 002" port) <u>EG</u> : TRUE (into "Index 003" port) <u>EG</u> : FALSE (into "Index 004" port)
25.60	Key into other Index ports as required. Press <ENTER> after page is done.	PED reappears with values in cyan
25.70	Press <PAGE FWD> and key into Index ports as required.	New PED page appears.
25.80	When a given library is complete, press <COMND>	COMMAND DISPLAY
25.82	Select [WRITE TO IDF]	Pathname Ports and target appear.
25.83	Key in a REFERENCE PATHNAME	<u>EG</u> : NET>HNV1>
25.84	Key in an IDF Pathname Press <ENTER>	<u>EG</u> : HGLIB Library is saved in IDF. HG-LIB. CONFIG display
25.85	Press <COMND>	COMMAND DISPLAY
25.90	Select [LOAD]	
25.100	Press <ENTER>	"BUSY" HG-LIBR. CONFIGURATION display
NOTE You should save the HG checkpoint(s) on HM and perhaps Zip disk at this time. See Appendix B of this document for instructions.		
25.110	Repeat Task 24 for each other HG Library to be configured.	<u>EG</u> : Library numbers 2, 3, and 4

8.4 Task 26 – Build and Compile Free Format Logs

Keyboard: Engr's

Lock: ENGR

Reference: HM11-660

Personality Loaded: Universal

Task 26 BUILD AND COMPILE FREE FORMAT LOGS

Step	Action	Response
26.10	Make sure points referenced in the logs have been built and loaded in Task 20.	Prebuilt logs can be read in from removable media and compiled on the HM per step 26.90
26.14	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
26.18	Select [FREE FORMAT LOGS]	FFL BUILDER display. At top of screen, R- 0, 42 indicates the Edit region is now at top left of drawing area.
26.22	On Cmd line, type: SET GRID ON	
26.26	Press <ENTER>	Grid appears
26.34	Enter Command: ADD TEXT , press <ENTER>	"ENTER TEXT" System ready for text input
26.38	Move cursor to locations where you want text and type it in. 9-3 shows an example of text in a Free Format Log. See 9-3a.	EG: WEST AREA LOG (title) FURNACE INLET TEMPERATURE = COIL OUTLET TEMPERATURE =
26.42	Press <ENTER>	Text is added to display
26.46	Enter Command: ADD VALUE Press <ENTER>	"ENTER VALUE COORDINATES" System waits for you to specify Value coordinates
26.50	Move cursor to each point where you want a value; press <SELECT>. Note cursor location (For example, C-224, 944) on 2nd line at that time.	+ Marks Value locations. (see 9-3b)
26.54	Press <ENTER> EXPRESSION for Value at XXX, YYY. (For example, VALUE AT 224, 944)	Screen form appears requesting

Task 26 BUILD AND COMPILE FREE FORMAT LOGS

DD MMM YY HH MM 01

R 0, 42 C-- 0, 1040

WEST AREA LOG

60

FURNACE INLET TEMPERATURE =

COIL OUTLET TEMPERATURE =

PUMP MOTOR STATUS =

AVERAGE FLOW =

OIL FLOW =

50

10 20 30 40 50 60 70

ADD TEXT

Enter Text

a. Text Locations

DD MMM YY HH MM 01

R 0, 42 C-- 0, 1040

WEST AREA LOG

+

+

60

FURNACE INLET TEMPERATURE = +

COIL OUTLET TEMPERATURE = +

PUMP MOTOR STATUS = +

AVERAGE FLOW = +

OIL FLOW = +

50

10 20 30 40 50 60 70

ADD VALUE

Enter Value Coordinates

b. Value Locations. Selected locations are marked by +.

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Figure 11 Free Format Log Example

Task 26 BUILD AND COMPILE FREE FORMAT LOGS

Step	Action	Response
26.58	Enter EXPRESSION information for each value (one per pass)	<u>EG:</u> SYS_TIME (1st Value) SYS_TIME (2nd Value) HG1001.PV HG1002.PV and so on for others (points must exist)
26.62	Press <ENTER>	Screen form requests FORMAT
26.74	Enter Format information (one per pass)	<u>EG:</u> DATEMM-DD-YYENDDATE (1st Value) TIMEHH:MM:SSSENDTIME (2nd Value) R-ZZZZ9.99 (other Values)
26.78	Press <ENTER>	String of letters replaces cross at value location (For example, DTDTDT, R-ZZZZ9.99, and so on). EXPRESSION for the next value appears.
26.79	Repeat steps 26.58 – 26.78 until all Value information is entered	
26.80	If TYPE is requested, Enter TYPE information for that Value.	<u>EG:</u> DATE, TYPE, REAL, and so on. (If type is known to the system, this screen form is omitted.)
26.82	Enter command: SET GRID OFF , press <ENTER>	Grid disappears
26.90	If FFL was prebuilt you can read it in from Zip disk at this time.	<u>EG:</u> READ \$F1>VOL2>FFL001 (reads file FFL001 from Zip named VOL2)
26.94	Enter Command to compile log	<u>EG:</u> COMPILE NET>HNV1>FFL001
26.96	Press <ENTER>	"WRITING SOURCE/OBJECT FILE" Log named FFL001 is compiled on HM
26.100	Repeat steps 26.22 – 26.96 for each FFL.	
26.102	If you want to save a copy of the FFL, you can copy it to a Zip disk at this time. Insert Zip in \$F1.	<u>EG:</u> USER (volume created in task 4)
26.106	Press <ESC>	"ESCAPED FROM PICTURE EDITOR"
26.108	Key in a copy command	<u>EG:</u> CP NET>HNV1>FFL001.DS \$F1>PICT>FFL001 -D
26.110	Press <ENTER>	Message: "COPY COMPLETE" Source file is copied
26.114	Hold <CTL> down and press <HELP>	Return to FFL Builder display

8.5 Task 27 – Build and Load HM History Groups

Keyboard: Engr's

Lock: ENGR

Reference: HM12-500 and SW11-151

Personality Loaded: Universal

Task 27 BUILD AND LOAD HM HISTORY GROUPS

Step	Action	Response
	HM HISTORY \$CHuu(n)	uu = Unit No., n = Hist. Group
	Note: The HM Online Personality must be running for Task 27.	Check the System Status display for "HISTORY MODULES-STATUS OK"
27.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
27.20	Select [HM HISTORY GROUPS]	CH HEADER DISPLAY
27.30	Key into UNIT port	<u>EG:</u> 01
27.40	Key into HISTORIZATION GROUP	<u>EG:</u> 1
27.50	Key into ENTITY.PARAMETER table Suggestion: Move cursor to the port; then, press <CLR-ENT>.	<u>EG:</u> HG1001.PV
27.60	When all applicable entries to the ENTITY.PARAM table are complete, press <ENTER>	Values and ports turn cyan
27.70	Press <COMND>	COMMAND DISPLAY
27.80	Select [WRITE TO IDF]	Pathname ports and target appear.
27.90	Key in a REFERENCE PATHNAME	<u>EG:</u> NET>HNV1>
27.100	Key in an IDF Pathname Press <ENTER>	<u>EG:</u> HMHIS Library is saved in IDF. PED display
27.105	Press <COMND>	COMMAND DISPLAY
27.110	Select [LOAD]	
27.115	Press <ENTER>	PARAMETER ENTRY DISPLAY

8.6 Task 28 – Build, Compile, and Load CL

Keyboard: Engr's

Lock: ENGR

Reference: AM11-575 and SW27-500

Personality Loaded: Universal

Task 28 BUILD, COMPILE, AND LOAD CL

Step	Action	Response
CREATE CL SOURCE FILE		If you are not using CL, skip to Section 29.
28.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
28.20	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
28.30	Key in an EDIT command	<u>EG:</u> EDIT NET>CL>VLR_CALC.CL
28.40	Press <ENTER>	Message: "OPEN FILE - FILE NOT FOUND"
28.50	Key in CL source file statements	<u>EG:</u> See Figure 13. This example is similar to Fig. 3-4 in the <i>CL/AM Overview Manual</i> . The point named in the BLOCK header must be a point built in task 22 (For example, VLRATIO). All points referenced in the statement must exist (For example, HG1001, HG1002). CLSLOTS must equal 1 or be greater than 1. In this example, PVALGID = CL.
28.60	Move cursor to 1st letter on line 1: Press <F3>, Press <F1>	JUMP MENU Cursor moves to start of file.
28.70	Update and exit: Press <F1>, Press <F2>	QUIT MENU Source file is transferred from temporary file to user volume. Message: "FILE WAS UPDATED xx Lines...."
COMPILE THE CL/AM BLOCK		
28.90	Key in a Compile command	<u>EG:</u> CL VLR_CALC
28.100	Press <ENTER> If errors were detected, examine the .LE file. A caret (^) marks the error(s). Make corrections, then recompile the file.	Elapsed time for Pass 1, Pass 2, Pass 3 time for compilation = nn.n sec. "Errors detected:None" <u>EG:</u> PRINT NET>CL>VLR_CALC.LE
LINK (AND LOAD) THE CL BLOCK		
28.110	Key in a Link Command	<u>EG:</u> LK VLR_CALC VLRATIO
28.120	Press <ENTER>	Elapsed time for linker = nn.n seconds. "Errors Detected None"

Task 28 BUILD, COMPILE, AND LOAD CL

```
BLOCK VLR_CALC (POINT VLRATIO; AT PV_ALG)
--
-- CALCULATE INTERNAL VAPOR/LIQUID RATIO FOR COLUMN NO. 2.
--
EXTERNAL HG1001, HG1002
LOCAL WL, WV, RATIO
LOCAL CP = 0.806 --LIQUID HEAT CAPACITY
LOCAL LAMBDA = 653.0 --HEAT OF VAPORIZATION
--
-- CALCULATE INTERNAL LIQUID FLOW RATE FROM HEAT AND MATERIAL BALANCE
SET WL = ((HG1001.PV - HG1002.PV)
&          * CP/LAMBDA)
--
-- CALCULATE INTERNAL VAPOR FLOW RATE FROM MATERIAL BALANCE
SET WV = HG1001.PV + WL
--
-- CALCULATE VAPOR/LIQUID RATIO AND STORE AS PV
SET RATIO = WV/WL
--

IF BADVAL (RATIO) THEN (CALL SET_BAD (PVCALC);
&                      SET PVAUTOST = BAD)
&                      ELSE (SET PVCALC = RATIO;
&                             SET PVAUTOST = NORMAL)
--
END VLR_CALC
```

Figure 12 CL Source File Example

Section 9 – Procedure 6/Configure Area Databases

Figure 13 shows a graphic representation of Task 29 that must be performed to Build and Install the Area Database.

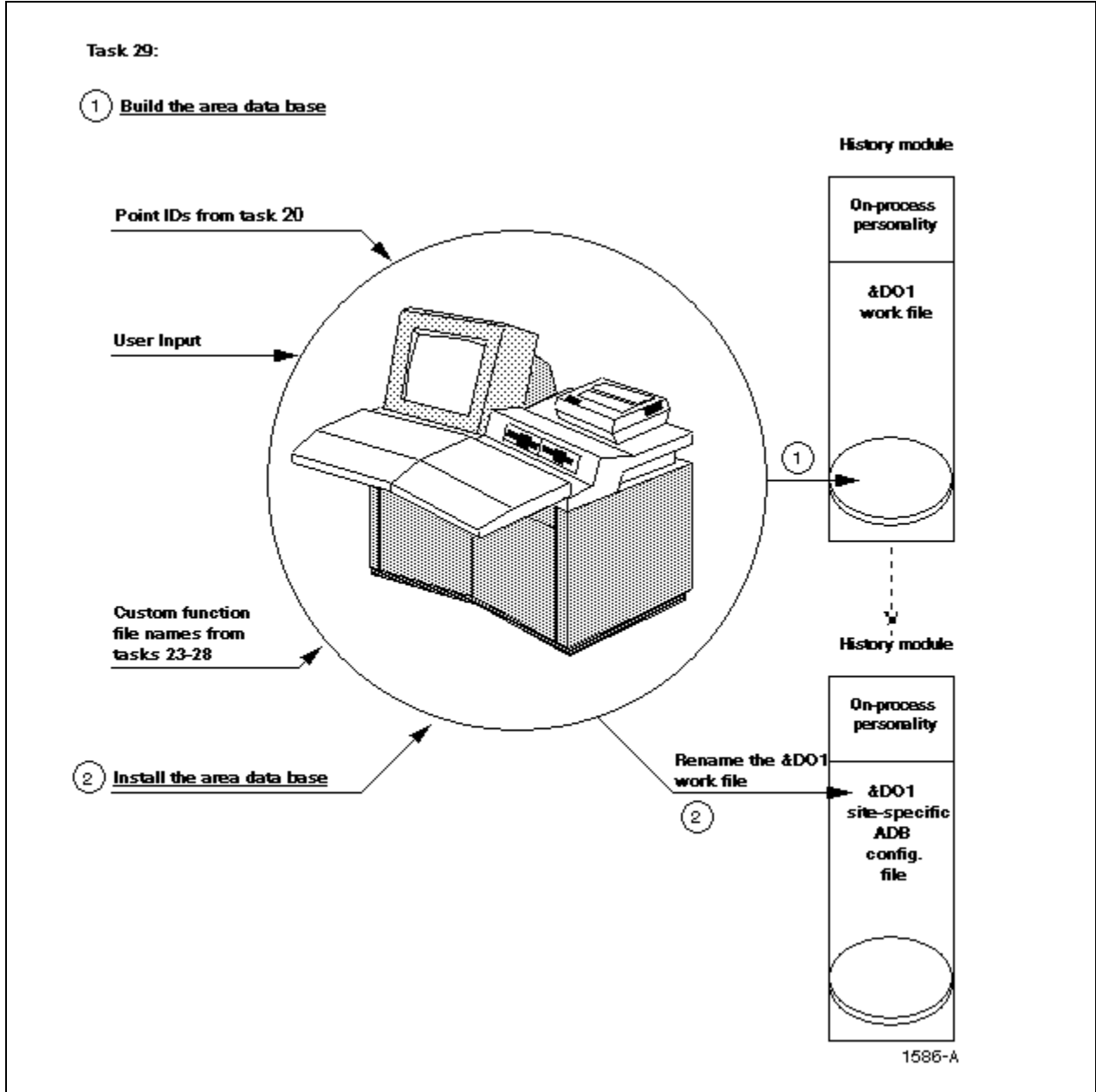


Figure 13 Build and Install the Area Database

9.1 Task 29 – Build, Select, and Install Area Databases

Keyboard: Engr's

Lock: ENGR

Reference: *Data Entity Builder Manual*

Personality Loaded: Universal

NOTE

1. Points to be referenced in the Area Database must already be configured in Task 21. Custom functions to be referenced in the Area Database must already be configured in Tasks 23-28.
2. In the steps that follow, the Area Database entity IDs required for reconstituting, loading, or reading a given part of the Area Database is listed just to the right of the bold heading for each part.

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
COPY BLANK AREA DATA BASE		
29.01	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
29.03	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
29.04	Key in a Copy File command to copy the blank Area Data Base file to an Area nn Volume on &D01. (nn = 01 for this example) Press <ENTER> after each Copy command above.	<u>EG:</u> CP NET>&ARG>AREA00.DA NET>&D01>AREA01 "COPY COMPLETE"
29.06	Key in an Unprotect command to the Area Data Base; then press <ENTER>	<u>EG:</u> UNPT NET>&D01>AREA01.DA "FILE UNPROTECT COMPLETE"
UNIT ASSIGN \$RUNTSTS		
29.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
29.20	Select [AREA DATA BASE]	AREA CONFIGURATION Menu
29.30	Select [UNIT ASSIGNMENT]	UNIT ASSIGNMENT DISPLAY CONFIG.
29.40	Select Unit ports that you want to be in the area.	<u>EG:</u> 01 Selected port(s) turn green
29.45	Press <ENTER>	UNIT ID targets appear
29.47	Enter Unit ID(s)	<u>EG:</u> 01
29.50	Select INITIALLY ASSIGNED? option	Select [YES] for each unit initially assigned to this area.
29.70	Press <ENTER>	PED reappears with values in cyan
SET A PATHNAME AND WRITE TO IDF		
29.80	Press <COMND>	COMMAND DISPLAY
29.90	Select [WRITE TO IDF]	Pathname ports and target appear
29.100	Key into Reference Path Name port	<u>EG:</u> NET>HNV1>

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.110	Key into pathname for IDF port	<u>EG</u> : BOILER1
29.120	Press <ENTER>	Unit Assignment data written to IDF
GROUP DISPLAY \$OGROUP(n) where n can be 1 to 400		
29.130	If UNIT ASSIGN display is on screen, and press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE].	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.140	Select [GROUP]	GROUP DISP. CONFIGURATION display
29.150	Key into GROUP NUMBER port	<u>EG</u> : 1
29.160	Key into GROUP TITLE port	<u>EG</u> : WEST GROUP
29.170	Key into the POINT IDS port	<u>EG</u> : HG1001
29.180	Select IN THE TREND SET? option	<u>EG</u> : [YES]
29.190	Repeat 29.170 – 29.180 for each other point in this group.	
29.200	Select a TREND TIMEBASE option	<u>EG</u> : The default option
29.201	Press <ENTER>	PED reappears with values in cyan
29.203	<PAGE FWD> to the Associated and Help display definitions. Key into ports as desired.	
29.210	Press <ENTER>	PED reappears with values in cyan
WRITE TO IDF WHEN PATHNAME IS ALREADY SET		
29.220	Press <COMND>	COMMAND DISPLAY with [WRITE TO IDF] selected
29.230	Press <ENTER>	GROUP DISPLAY CONFIG data to IDF
29.240	To configure more groups, hold <CTL> down and press <PAGE BACK>. Go repeat 29.140 – 29.230	
UNIT TREND DISPLAY \$OUNTRND(n) where n can be 1 to 36		
29.250	If GROUP DISP CONFIG is on screen and press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE].	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.255	Select [UNIT TREND]	UNIT TREND DISPLAY CONFIG display
29.260	Key into TREND DISP NUMBER port	<u>EG</u> : 1
29.265	Key into TREND TITLE port	<u>EG</u> : UNIT 1 TRENDS
29.270	Select TREND TIMEBASE option	<u>EG</u> : [2 HOURS]
29.275	Key in names of point(s) to be trended in cyan and/or yellow.	<u>EG</u> : HG1001 (CYAN)

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.280	Press <ENTER>	PED reappears with values in cyan
29.285	<PAGE FWD>	RELATED DISPLAY 001 DEF display
29.290	Key into SCREEN NUMBER port(s) and DISPLAY TYPE port(s).	<u>EG:</u> 1 <u>EG:</u> ALARMSUM
29.295	Press <ENTER>	PED reappears with values in cyan or SCHEMATIC NAME port appears if DISPLAY TYPE entered was SCHEM
29.300	Key into SCHEM. NAME port (if any)	<u>EG:</u> DISP1
29.305	Press <ENTER>	PED reappears with values in cyan
29.310	<PAGE FWD>	Next page appears
29.315	Repeat 29.290 – 29.310 to configure remaining Related Displays, and on Page 6, Assoc. Display and Help Display	Related, Associated, and Help displays configured
WRITE TO IDF WHEN PATHNAME IS ALREADY SET		
29.320	Press <COMND>	COMMAND DISPLAY with correct pathnames entered.
29.325	Press <ENTER>	UNIT TREND CONFIG. data to IDF
29.330	Repeat 29.260 – 29.330 to configure other Unit Trend displays. Use <PAGE FWD> and <PAGE BACK> as necessary.	All Unit Trend displays configured and stored in IDF
SYSTEM STATUS DISPLAY \$OSYSTAT		
29.350	If PED display is still on the screen Press <9> If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.360	Select [SYSTEM STATUS]	SYSTEM STATUS DISPLAY CONFIG display
29.364	If the screen is blank, press <PAGE FWD>	Next page appears
29.370	Key into Associated Display ports: SCREEN NO. port DISPLAY TYPE port Press <ENTER> CONSOLE NUMBER	<u>EG:</u> 0 <u>EG:</u> CONSTAT CONSOLE # port appears <u>EG:</u> 1
29.380	Key into HELP DISPLAY DEF ports: SCREEN NO. port DISPLAY TYPE port	<u>EG:</u> 1 <u>EG:</u> SYSSTAT
29.390	Press <ENTER>	PED reappears with values in cyan

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
WRITE TO IDF		
29.400	If not first write to IDF, press <COMND> and go to 29.410. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY with [WRITE TO IDF] selected. Example IDF is BOILER1.
29.410	Press <ENTER>	SYS STAT DISP CONFIG-data to IDF
FREE FORMAT LOGS \$OFFLOG(n) where n can be 1 to 100		
29.420	If PED display is on the screen, and press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION MENU AREA CONFIGURATION MENU
29.430	Select [FREE FORMAT LOG]	FREE FORMAT LOG CONFIG. display
29.440	Key into LOG NUMBER	EG: 1
29.450	Key into LOG NAME	EG: FFL001
29.470	Key into LOG DESCRIPTION	EG: WEST AREA LOG
29.480	Accept default for INVOC. TYPE	EG: DEMAND
29.490	Accept default for PRINTER ID	EG: 1
29.500	Press <ENTER>	PED reappears with values in cyan
WRITE TO IDF		
29.510	If not first write to IDF, press <COMND> and go to 29.515. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.515	Press <ENTER>	FFL CONFIG DATA to IDF
29.520	Repeat 29.440 – 29.515 for other Free Format Logs.	All FFL configuration data in IDF
HIWAY BOX ANNUN POLICY \$DMONBLK		
29.610	If PED display is on the screen, and press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.620	Select [PROC NTWK ANNUN POLICY]	PROCESS NETWORK DEV. ANNC. CONFIG display
29.630	Key into PROCESS NTWK NBR port	EG: 2
29.640	Select ADDRESS ANNUNC. options	EG: [YES] for all addresses
29.650	Press <ENTER>	PED reappears with values in cyan
29.660	Check PED set status (<CTL> <8>)	Display Set Entered?-Yes If errors, see: 29.1060 – 29.1080

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.670	If not first write to IDF, press <COMND> and go to 29.680. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
29.680	Press <ENTER>	HIWAY ANN PLCY CONFIG-data to IDF
PATHNAME CATALOG \$OABSTR		
29.690	If PED display is on the screen, and press <F9>. If Engineering Main Menu is on the screen, select [AREA DATABASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.700	Select [PATHNAME CATALOG]	PATHNAME CATALOG CONFIG. display
29.710	Key into BUTN CONF VOLUME ID port	EG: &D01
29.720	Key into BUTN CONF FILE NAME port	EG: BUTTON
29.724	Key into SCHEMATIC or FFL PATHNAMES—most critical pathname in top left-hand port, then left to right, then second line left to right, and so on.	EG: NET>HMV1 (your user volume), NET>&DSY, NET>&D01, NET>DIA1, and so on.
29.750	Key into MEM. RES SCHEMATIC OR FFL FILE NAME port and select [FFL] if you entered a FFL filename.	EG: FFL001 [FFL], DISP1, DISP2, DISP3, and so on.
29.760	Press <ENTER>	PED reappears with values in cyan
29.764	Use <PAGE FWD> to add more schematic/FFL names	All schematic/FFL file names entered
WRITE TO IDF		
29.770	If not first write to IDF, press <COMND> and go to 29.780 If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathnames set)
29.780	Press <ENTER>	PATHNAME CAT CONFIG-DATA to IDF

PROCESS MODULE GROUP DISPLAY \$OMODGRP(n) where n can be 1-50**NOTE**

If a MC process module point wasn't configured in Task 21, go to 29.890.

29.790	If PED display is on the screen, and press <F9>. If Engineering Main Menu is on the screen, select [AREA DATABASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.795	Select [PROCESS MODULE GROUP]	MODULE GROUP DISP CONFIG display

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.800	Key into GROUP NUMBER port	<u>EG</u> : 1
29.810	Key into GROUP TITLE	<u>EG</u> : MIX AND SOAK GROUP
29.815	Key into PROCESS MODULE ID ports	<u>EG</u> : SCALE1
29.820	Press <ENTER>	PED reappears with values in cyan
29.825	<PAGE FWD>	RELATED DISPLAY 001 DEF display
29.830	Key into Screen Number port(s) and Display Type port(s).	<u>EG</u> : 1 <u>EG</u> : SCHEM
29.835	Press <ENTER>	PED reappears with values in cyan and additional port(s) displayed.
29.840	Key into added port(s)	<u>EG</u> : DISP1
29.845	Press <ENTER>	PED reappears with values in cyan
29.850	<PAGE FWD>	Next page appears
29.855	Repeat 29.830 – 29.850 to configure ASSOCIATED DISPLAY and HELP DISPLAY	RELATED, ASSOCIATED, and HELP displays configured
WRITE TO IDF		
29.865	If not first write to IDF, press <COMND> and go to 29.870. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] (with path name BOILER1 set)
29.870	Press <ENTER>	PED display
29.880	Repeat 29.800 – 29.870 to configure other Process Module Groups. Use <PAGE FWD> and <PAGE BACK> as necessary.	All Process Module Groups configured and stored in IDF
MODULE SUMMARY DISPLAY \$OUNITSM(n) where n can be 1 to 36		
29.890	If PED display is on the screen, and press <F9>. If Engineering Main Menu is on the screen, select [AREA DATABASE]	AREA CONFIGURATION MENU AREA CONFIGURATION MENU
29.900	Select [MODULE SUMMARY]	UNIT SUMMARY DISP CONFIG display
29.910	Select an option for each Line	<u>EG</u> : [HEADER] for Line 01, page 1
29.920	Press <ENTER>	<u>EG</u> : TITLE port appears for Line 1
29.930	Key into each Title port	<u>EG</u> : WEST UNIT 1
29.940	Press <ENTER>	PED reappears with values in cyan
29.950	Use <PAGE FWD> to get to more pages to configure and repeat steps 29.910 – 29.940 for each.	UNIT SUMMARY DISP CONFIG display – n
29.960	<PAGE FWD> to RELATED DISP DEF	RELATED DISP DEFINITION display – 1

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.970	Key into each pair of SCREEN NUMBER and DISPLAY TYPE ports.	<u>EG:</u> 0 <u>EG:</u> ALARMSUM
29.980	Press <ENTER>	PED reappears with values in cyan
29.990	If desired, <PAGE FWD> to next page and repeat steps 29.970 – 29.980.	RELATED DISP DEFINITION display – n
29.1000	<PAGE FWD> to ASSOCIATED DISPLAY DEFINITION	ASSOCIATED & HELP DISPLAY DEFINITION
29.1010	Key into each pair of SCREEN NUMBER and DISPLAY TYPE ports.	<u>EG:</u> 0 <u>EG:</u> UNITALRM
29.1020	Press <ENTER>	Unit ID port appears.
29.1030	Key into Unit ID port	<u>EG:</u> 01
29.1040	Press <ENTER>	PED reappears with values in cyan
CHECK PED SET STATUS		
29.1050	Press <F8>	PED Set Stat display
29.1060	If "Display Set Entered?" is Yes, and no missing data or errors are indicated, go to 29.1090; else 29.1070.	If "No," check "Missing Data?" and "Errors?"
29.1070	Select a Display Title option, correct data and press <ENTER>	Applicable PED appears.
29.1080	Repeat step 29.1060	
WRITE TO IDF		
29.1090	If not first write to IDF, press <COMND> and go to 29.1100. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set).
29.1100	Press <ENTER>	MOD SUM DISP CONFIG data to IDF
ANNUNCIATOR DISPLAY \$ANNDATA		
29.1110	If PED display is on the screen, and press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.1115	Select < ANNUNCIATOR >	ANNUNCIATOR DISPLAY CONFIG. Display
29.1120	Key into BOX NUMBER port	<u>EG:</u> 1
29.1125	Key into BOX TITLE port	<u>EG:</u> AIR FLOW Note: Annunciator display boxes have two lines of four characters each.
29.1128	Select a Point Usage	<u>EG:</u> [ENTITY ID]
29.1130	Key into NUMBER OF POINTS port	<u>EG:</u> 1

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.1131	Press <ENTER>	
29.1132	Key into POINT NAMES ports	<u>EG</u> : HG1001
29.1135	Key into SCREEN NUMBER port	<u>EG</u> : 1
29.1140	Key into DISPLAY TYPE	<u>EG</u> : SCHEM
29.1145	Press <ENTER>	Additional port(s) appear
29.1155	Key into added port(s)	<u>EG</u> : DISP1
29.1160	Press <ENTER>	PED reappears with values in cyan
WRITE TO IDF		
29.1165	If not first write to IDF, press <COMND> and go to 29.1170 If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.1170	Press <ENTER>	ANNUNC. CONFIG DATA to IDF
29.1175	Repeat 29.1110 – 29.1170 for other boxes on Annunciator	All Annunciator Box configuration data in IDF
PROCESS JOURNALS \$OPRCJOR(n)		
29.1200	If PED display is on the screen, and press <F9>. If Engineer Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.1202	Select [PROCESS JOURNAL]	PROCESS JOURNAL CONFIG. display
29.1204	Key into JOURNAL NUMBER port	<u>EG</u> : 1
29.1206	Key into JOURNAL NAME port	<u>EG</u> : PRJNL001
29.1208	Key into JOURNAL TITLE port	<u>EG</u> : WEST AREA PROCESS ALARMS
29.1212	Select JOURNAL TYPE	<u>EG</u> : [ALARM]
29.1214	Select INVOCATION TYPE	<u>EG</u> : [PERIODIC]
29.1216	Select INITIALLY ACTIVE option	<u>EG</u> : [NO]
29.1220	Select DAYS OF WEEK option	<u>EG</u> : ALL (default)
29.1222	Key into START TIME port	<u>EG</u> : 0:10
29.1224	Key into PERIOD port	<u>EG</u> : 12:0
29.1226	Key into PRINTER ID port	<u>EG</u> : 1
29.1228	Press <ENTER>	PED reappears with values in cyan
29.1230	Press <PAGE FWD>	Next page of PED appears
29.1232	Key into TIME SPAN port	<u>EG</u> : 12
29.1234	Key into NUMBER OF UNITS port	<u>EG</u> : 1
29.1236	Press <ENTER>	PED reappears with Unit ID Box(s)

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.1240	Key into (Unit ID) box	<u>EG</u> : 01
29.1243	Press <ENTER>	PED reappears with values in cyan
29.1246	If not first write to IDF, press <COMND> and go to 29.1248. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.1248	Press <ENTER>	PROC. JOURNAL CONFIG data to IDF
29.1250	Repeat 29.1204 – 29.1248 to configure remaining journals.	
SYSTEM JOURNALS \$OSYSJOR		
29.1260	If PED display is on the screen, press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.1300	Select [SYSTEM JOURNAL]	SYSTEM JOURNAL CONFIG. display
29.1310	Key into JOURNAL NUMBER port	<u>EG</u> : 1
29.1320	Key into JOURNAL NAME port	<u>EG</u> : SYJNL001
29.1330	Key into JOURNAL TITLE port	<u>EG</u> : SYSTEM JOURNAL NO. 1
29.1340	Select JOURNAL TYPE	<u>EG</u> : [SYSTAT]
29.1350	Select INVOCATION TYPE	<u>EG</u> : [DEMAND]
29.1360	Key into PRINTER ID port	<u>EG</u> : 1
29.1370	Key into TIME SPAN port	<u>EG</u> : 24
29.1390	Press <ENTER>	PED reappears with values in cyan
29.1400	If not first write to IDF, press <COMND> and go to 29.1410. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.1410	Press <ENTER>	SYS. JOURNAL CONFIG-data to IDF
29.1420	Repeat 29.1310 – 29.1410 to configure remaining journals.	
REAL-TIME JOURNALS \$RTJNL		
29.1430	If Command display is on the screen, press <F9>. If Engineering Main Menu is on the screen, select [AREA DATABASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.1440	Select [R-T JOURNAL ASSIGNMENT]	R-T JOURNAL ASSIGNMENTS DISPLAY CONFIGURATION
29.1450	Key into PRINTER ID ports	<u>EG</u> : 1 (into all ports)

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.1460	Select JRN TYPE-INIT ACT? options	<u>EG</u> : [No] for SYS STATUS CHANGES
29.1470	Press <ENTER>	PED reappears with values in cyan
29.1490	If not first write to IDF, press <COMND> and go to 29.1500. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.1500	Press <ENTER>	REAL-TIME JRNL config-data to IDF
OVERVIEW DISPLAY \$OOVERVW		
29.1510	If PED display is on the screen, press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.1520	Select [OVERVIEW]	AREA OVERVIEW DISP CONFIG display
29.1530	Key into GROUP NUMBERS ports	<u>EG</u> : 1 (into Index 01)
29.1540	Key into NUMBER OF FIELDS port	<u>EG</u> : 1
29.1550	Press <ENTER>	FIELD DEFINITION ports appear
29.1560	Key into FIELD DEF ports: FLD TITLE TOP LEFT ELEMENT WIDTH HEIGHT	<u>EG</u> : WEST GROUP <u>EG</u> : 1 <u>EG</u> : 1 <u>EG</u> : 1
29.1570	Press <ENTER>	PED reappear with values in cyan
29.1573	<PAGE FWD> to the Related/Assoc. and Help Display definitions. Key into ports as desired.	
29.1578	Press <ENTER>	PED reappears with values in cyan
29.1580	Check PED-set status: Press <F8>	"Display Set Entered?"-Yes If errors, see: 29.1060 – 29.1080
29.1590	If not first write to IDF, press <COMND> and go to 29.1600. If first write to IDF, go perform 29.80 - 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.1600	Press <ENTER>	OVERVIEW DISP CONFIG data to IDF
AREA TREND DISPLAY \$OTREND		
29.1610	If PED display is on the screen, press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.1615	Select [AREA TREND]	AREA TREND OV. DISP CONFIG display
29.1620	Key into TREND TITLE port	<u>EG</u> : WEST AREA TRENDS

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.1625	Select TREND TIMEBASE option	<u>EG</u> : [8 HOURS]
29.1630	Key in names of points to be trended in cyan and yellow.	<u>EG</u> : HG1001 (cyan)
29.1635	Press <ENTER>	PED reappears with values in cyan
29.1640	<PAGE FWD>	RELATED DISP 001 DEF display
29.1645	Key into each SCREEN NUMBER port and each Display Type port.	<u>EG</u> : 1 <u>EG</u> : SCHEM
29.1650	Press <ENTER>	PED reappears with values in cyan and SCHEMATIC NAME appears (or other port depending on the DISPLAY TYPE entered).
29.1655	Key into each additional name port	<u>EG</u> : DISP1
29.1660	Press <ENTER>	PED reappears with values in cyan
29.1665	<PAGE FWD>	Next page appears
29.1670	Repeat 29.1650 – 29.1665 to configure remaining Related Displays, and on page 7, Associated and Help display	RELATED, ASSOCIATED, and HELP displays configured
WRITE TO IDF WHEN PATHNAME IS ALREADY SET		
29.1675	Press <COMND>	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathnames set)
29.1680	Press <ENTER>	AREA TREND CONFIG. data to IDF
STANDARD LOGS \$OLOG(n) where n can be 1 to 100		
29.1900	If PED display is on the screen, press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.1905	Select [STANDARD LOG]	STANDARD LOG CONFIGURATION display
29.1910	Key into LOG NUMBER port	<u>EG</u> : 1
29.1920	Key into LOG NAME port	<u>EG</u> : BLR1RPT
29.1930	Key into LOG DESCRIPTION port	<u>EG</u> : DAILY BOILER LOG
29.1940	Enter other data as required.	<u>EG</u> : Use default values.
29.1950	Key into NUM VALUES PER POINT port	<u>EG</u> : 1
29.1960	Key into NUMBER OF POINTS port	<u>EG</u> : 1
29.1965	Press <ENTER>; press <PAGE FWD>	INDEX ports appear.
29.1970	Key into INDEX ports until all Point Ids have been keyed in.	<u>EG</u> : HG1001 (for Index 001) (from step 20.30)
29.1980	Press <ENTER>	PED reappears with values in cyan

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
WRITE TO IDF		
29.2000	If not first write to this IDF, press <COMND> and go to 29.2005. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; with correct pathnames.
29.2005	Press <ENTER>	Standard Log config data to IDF
PRINTED TREND \$OPRTRND(n) where n can be 1 to 100		
29.2010	If PED display is on the screen, press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.2012	Select [PRINTED TREND]	PRINTED TREND DISP CONFIG display
29.2014	Key into TREND NUMBER port	<u>EG</u> : 1
29.2016	Key into TREND NAME port	<u>EG</u> : BOILER 1
29.2018	Key into TREND TITLE port	<u>EG</u> : BOILER 1 TREND RECORD
29.2020	Select INVOCATION Type	<u>EG</u> : [PERIODIC]
29.2024	Select INITIALLY ACTIVE option	<u>EG</u> : [NO]
29.2028	Select START TIME port	<u>EG</u> : [0]:[10]
29.2032	Key into PERIOD port	<u>EG</u> : 1:30
29.2036	Key into PRINTER ID port	<u>EG</u> : 1
29.2040	Key into TIME SPAN port	<u>EG</u> : 2
29.2044	Key into GROUP NUMBER port	<u>EG</u> : 1
29.2050	Press <ENTER>	PED reappears with values in cyan
WRITE TO IDF		
29.2070	If not first write to this IDF press <COMND> and go to 29.2080. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (pathname is set)
29.2080	Press <ENTER>	PRINTED TREND CONFIG. data to IDF
29.2085	Repeat 29.2014 – 29.2080 for other Printed Trends.	All printed trends configured and stored in IDF.
LCN NODE ANNUNCIATION POLICY \$ANPOLC		
29.2090	If PED display is on the screen, press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.2094	Select [LCN NODE ANNUNC. POLICY]	LCN NODE ANNUNC. POLICY display

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.2102	Use <PAGE FWD> to get to all node numbers you wish annunciated. Select [ADDRESS ANNUNCIATED] option	EG: 01, 02, 24, 25, 30, 38, 39, 40, 43, and 46 EG: [YES] for each of above nodes
29.2106	Press <ENTER>	PED reappears with values in cyan
29.2108	If not first write to IDF, press <COMND> and go to 29.2112. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (pathname is set)
29.2112	Press <ENTER>	LCN NODE ANNUN config data to IDF.
SEQUENCE OF EVENTS JOURNAL \$SOEJOR(n)		
29.2120	If PED display is on the screen, press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.2124	Select [SOE JOURNAL ASSIGNMENT]	SEQUENCE OF EVENTS JOURNAL display
29.2128	Key into JOURNAL NUMBER	EG: 1
29.2130	Key into JOURNAL NAME	EG: SOEJNL01
29.2134	Key into JOURNAL TITLE	EG: BOILER1 SEQ OF EVENTS
29.2138	Key into HM NODE NUMBER	EG: 43
29.2140	Press <ENTER>	PED reappears with values in cyan
29.2144	If not first write to IDF, press <COMND> and go to 29.2148. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (pathname is set)
29.2148	Press <ENTER>	SOE JOURNAL config data to IDF
REPORT DISPLAY \$OREPORT(n) where n can be 1 to 100		
29.2150	If PED display is on the screen, press <F9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION MENU AREA CONFIGURATION MENU
29.2154	Select [REPORT]	REPORT DISPLAY CONFIGURATION display
29.2156	Key into REPORT NUMBER port	EG: 1
29.2160	Key into REPORT NAME port	EG: WESTRPT
29.2164	Key into REPORT TITLE port	EG: WEST AREA REPORT
29.2170	Select INVOCATION TYPE option	EG: [DEMAND]
29.2175	Key into NO. ITEMS IN REPORT port	EG: 1
29.2180	Press <ENTER>	ITEMS IN REPORT port appears

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.2185	Key into Items In Report port Note: Items must be previously built as trends or logs.	<u>EG:</u> BLR1RPT (configured in 29.1920)
29.2190	Press <ENTER>	PED reappears with values in cyan
WRITE TO IDF		
29.2194	If not first write to IDF, press <COMND> and go to 29.2198. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.2198	Press <ENTER>	REPORT CONFIGURATION data to IDF

NOTE

Nodes associated with the area being multiple loaded must have a status of OK before the load multiple can be successfully performed.

LOAD MULTIPLE ENTITIES FROM IDF**SELECT AN AREA**

29.2205	Press <COMND>	COMMAND DISPLAY
29.2208	Select [SELECT AREA]	Area Ports/Target appear
29.2212	Key into REFERENCE PATH NAME port	<u>EG:</u> NET>H MV1>
29.2216	Select an area Current AREA DATA BASE pathname port is filled in automatically	<u>EG:</u> [1] <u>EG:</u> NET>&D01>AREA01
29.2218	Press <ENTER>	“OPERATION COMPLETE”
29.2220	Select [LOAD MULTIPLE]	Additional Ports/Target appear

INSTALL THE AREA DATABASE**NOTE**

The Area Database is installed to a volume and file with the volume name of &Dnn and with the file name of AREAnn.

For &Dnn, the directory names of &D01 through &D10 were automatically created during the HM Volume Configuration task, Task 10. For AREAnn, nn is the index number from the Area Names configuration list that you configured in Task 7.

For example, the first area name you entered into the list is automatically accessed by AREA01, the second name by AREA02, and so on.

29.2230	Key into REFERENCE PATH NAME port	<u>EG:</u> NET>H MV1>
29.2240	Key into pathname for IDF port	<u>EG:</u> BOILER1

Task 29 BUILD, SELECT, AND INSTALL AREA DATABASES

Step	Action	Response
29.2250	Key into current AREA DATA BASE pathname port	<u>EG</u> : NET>&D01>AREA01
29.2260	Press <ENTER> Wait—Entities are listed twice	"PROCESSING ENTITIES..." "OPERATION COMPLETE"
29.2270	Press <COMND>	COMMAND DISPLAY
29.2280	Select [INSTALL AREA]	Additional targets appear.
29.2290	Key a (destination) pathname into Current AREA DATA BASE p/nm port	<u>EG</u> : NET>&Dnn>AREAnn (where nn is equal to 01–10 in both cases)
29.2295	Key a (source pathname) into the Pathname for WORKFILE port.	<u>EG</u> : NET>&D01>AREA01
29.2300	Press <ENTER>	"OPERATION COMPLETE"

NOTE

1. Most of the Command Display functions deal with IDFs; however, Exception Build files are more portable. If you want to create Exception Build files from your IDFs, follow the instructions in Appendix E.
2. After the Area Database is completely built, the IDFs and Area Database files can be copied to Zip disk and then deleted from the User Volume NET>HVM1>Boiler1.
3. For security, you should copy the following onto disk: Area Database (&D01), Pictures, Free Format Logs, Button source files (.DS and .KS files), and custom files (CUS/CLX, and so on).
4. Go perform the EC in Task 30, steps 30A.10 - 30A.50 to save all the &ASY files. Use the resulting disk as the new backup if you have to reload from disk or to perform On-Line Network Reconfiguration.

Section 10 – Procedure 7/Load Area Databases

10.1 Task 30 – Load Operator or Universal Personality to Universal Station

Keyboard: Both

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 30 LOAD OPERATOR/UNIVERSAL PERSONALITY TO UNIVERSAL STATION

Step	Action	Response
30.01	If the other US is a U ^X S, skip to 30.150 to load the U ^X S personality.	See 1.1.1 for U ^X S preparation. (Or load U ^X S with a standard personality by following the US procedure starting at 30.10.)
30.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
30.20	Select [CONSOLE STATUS]	CONSOLE STATUS AND ASSG. display
30.30	Under the STN header, select the number of the US you wish to load.	<u>EG</u> : 2
30.60	Select [LOAD/DUMP]	LOAD/DUMP NODE Targets appear
30.64	Select [MANUAL LOAD]	NODE LOAD targets appear
30.70	Select [OPERATOR PROGRAM] or [UNIVERSAL PROGRAM]	PGM SOURCE FOR NODE nn display (US must have enough memory to load UP)
30.80	Select [DEFAULT SOURCE]	EXECUTE COMMAND target appears
30.90	Select [EXECUTE COMMAND]	DATA SOURCE FOR NODE nn
30.100	Select [DEFAULT SOURCE]	EXECUTE COMMAND target appears
30.110	Select [EXECUTE COMMAND]	CONSOLE STATUS AND ASSG. Display "PRESS ENTER TO EXECUTE"
30.120	Press <ENTER>	STN NN STATUS goes to LOAD
30.130	Wait for several minutes. Watch the Console Status display at the Station being used.	"BUSY: US nn" Console Status Display appears with a status of READY on Station NN. STN nn STATUS eventually goes to OK . Type goes to OPR for Operator Personality or UNVL for Universal Personality.
30.140	Repeat Task 30 to load other Universal Stations.	If no other US or U ^X S stations, this completes Task 30.
30.150	To load a U ^X S, station with the U ^X S personality, press (LOAD) on the U ^X S Operator's Keyboard.	"W, N, 1, 2, 3, 4, X?"

Task 30 LOAD OPERATOR/UNIVERSAL PERSONALITY TO UNIVERSAL STATION

Step	Action	Response
30.160	Press (W) and then press (ENTER)	U ^X S loads from internal Winchester disk, then accesses LCN for NCF and area database. U ^X S Station Status goes to OK; Type = U ^X S.

Section 11 – Procedure 8/Modify Database

11.1 Task 31 – Change Network Configuration

Keyboard: Engr's

Lock: ENGR

Reference: SW11-605

Personality Loaded: Universal

Task 31 CHANGE NETWORK CONFIGURATION

Step	Action	Response
MAKE A BACKUP COPY OF &ASY		
31.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
31.30	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
31.35	&Z1 into \$F1	
31.36	Place a blank Zip disk in \$F2 (must be Unprotected).	
31.40	Key in an EC command	EG: EC \$F1>&EC>ASY_BKUP.EC \$F2 \$P1 (printer \$P1 is optional)
31.45	Press <ENTER>	Message: "THIS EC WILL...." "....." "..... CONTINUE (Yes/No)?"
31.50	Key in YES , then press <ENTER>	Message: "THIS EC... IS THIS CORRECT?"
31.55	Key in YES , then press <ENTER>	Message: "MOUNT BACKUP MEDIA...."
31.60	Press <ENTER> (the EC prepares to update the Backup Log).	"UNPT NET>&ASY>BKUP_LOG.XX...." "DO YOU WANT TO RE-INITIALIZE...."
31.65	Key in YES , then press <ENTER> (the EC formats the Zip in \$F2, and begins copying files to it).	"CR \$F2>&ASY> -MF 1000" "COPYING" "EC COMPLETE"
31.68	Remove the Zip from \$F2. Label it BACKUP &ASY and write the date on it.	
31.70	Make backup copies of other files that you created (for example insert the AREA Zip in \$F2 and copy the area files, button files, and custom software files.	EG: CP NET>&D01>*. * \$F2>&D01>= -D CP NET>&CUS>*. * \$F2>&CUS>= -D CP NET>&CLX>*. * \$F2>&CLX>= -D

NOTE

Refer to the *Network Data Entry* manual (see References) for information on network reconfiguration.

Section 12 – Error Recovery Procedures

12.1 Error Conditions, Reasons, and Recovery

Task ERROR CONDITIONS AND RECOVERY

Task #	Error Conditions	Reason and Recovery Procedure
Many	US Node fails with an Internal File Manager error. HG Fails	Copy to existing, protected file. 1. Installing to a protected NCF.CF. 2. HG checkpoint to a protected file. 3. Picture Editor/Button Configurator read or write to protected file. 4. Using DEB commands to write to a protected IDF or Entity List File. Manually reset the HG and then repeat Task 17.
2	Universal Personality will not load.	One of the nodes is active. Check that <u>all</u> on the NET are in the RESET state. Then repeat Task 2. You may have to press RESET on the HM an extra time because it may have autobooted.
14	1. A change to &ASY was made to a copy of &ASY on the HM. 2. The changed &ASY on the HM <u>was not</u> copied to a Zip. 3. A load of the Universal Personality is being attempted, using Zips that include the CNCF Zip that existed before changing the &ASY on HM. 4. The CNCF Zip will not read in and the Station dies.	The time-stamp on the CNCF Zip disk is not the same as the time-stamp for &ASY in one of the nodes. To recover, do the following: 1. Load the Universal Personality, using the &Z1 Zip disk and answering the NCF? prompt with N (NET) for HM as the source of the NCF. 2. If recovery step 1 works, copy the &ASY from HM to the CNCF Zip you were trying to load from. 3. Try the new CNCF Zip to load the Universal Personality from Zips. If successful, mark the same date on it as exists on the &ASY on the HM.
15	Got a message of "HM INIT DID RESPOND FOR 1 MINUTE(REVIEW)"	Power to the Winchester disks is off, even though power to the HM node is on. Turn the power to Winchester disks to the "on" position. Repeat Tasks 14 and 15.

Task ERROR CONDITIONS AND RECOVERY

Task #	Error Conditions	Reason and Recovery Procedure
18	On-Process Personality isn't successfully loaded into a module or gateway.	Select the red failure message at the top of the display, to call up the module or gateway status display. On that display, select the node number for the node you are trying to load. This causes an error message to appear. If the message is TARGET NODE FAILED DURING LOAD, check the module or gateway to see if power is on, all boards have a green light on, and the node number appears in the red digits on the front of the unit. If, instead, the red digits show -189 or -190, reset the module or gateway and try to load it again. If another error code appears, call a hardware technician. If you were attempting to load a node or gateway using an HM as the program and data source, and the error message is FILES NOT FOUND, check the HM's status as described in the preceding paragraph. If it has failed, turn its power (power to both the module and disk drive) off, wait five seconds, and turn power back on. In about two minutes, its status on the module status display should become OK. If so, reset the node you were trying to load and try again. If you get the same error message again: - Use the Universal Personality's Command Processor functions to make sure that the volumes mentioned in Task 18 are all on the source HM. - Try to load the node using Zip disks. If this works, you may have to delete and recopy all the volumes on the HM. Get a hardware technician's assistance.

NOTE

If you do not find a description of your problem in these error recovery procedures, see the *TPS Messages Directory* (see References).

Appendix A – Load Operator or Universal Personality Into Another US

A.1 Task A – Load Op./Univ. Personality to Other US

Keyboard: Both

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task A LOAD OPERATOR/UNIVERSAL PERSONALITY TO OTHER US

Step	Action	Response
10	&Z1 into \$F1, CNCF into \$F2	
20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
30	Select [CONSOLE STATUS]	Console Status display <u>Other US Status seen on This US</u> PWR_ON or QUALIF
40	Select the number of the other Universal Station.	<u>EG: 2</u> (for STN #2, node no. 2) LOAD or DUMP NODE targets appear
50	Select [LOAD/DUMP]	LOAD/DUMP targets on screen
55	Select [MANUAL LOAD].	Node Load targets appear
60	Select [OPERATOR PROGRAM] or [UNIVERSAL PROGRAM]	PGM SOURCE FOR NODE NN
70	Select [ALTERNATE SOURCE]	Boxes with 1 and 2 appear
90	Select box [1]	VOL ID: "&Z1" and targets appear
100	Select [EXECUTE COMMAND]	DATA SOURCE FOR NODE NN display
110	Select [ALTERNATE SOURCE]	Boxes with 1 and 2 appear
120	Select box [2]	VOL ID: "CNCF" and targets appear
130	Select [EXECUTE COMMAND]	CONSOLE STATUS AND ASSGN display PRESS ENTER TO EXECUTE
140	Press <ENTER>	Status of other Station = NET LOAD "SINGLE LOAD BUSY: US nn" "MOUNT STD ABS for US nn"
150	Select [MOUNT STD ABS for US nn]	SOURCE targets appear
160	Select [ALTERNATE SOURCE]	Boxes with 1 and 2 appear
165	Select box [1]	VOL ID: "&Z1" and targets appear
170	Select [EXECUTE COMMAND]	CONSOLE STATUS AND ASSGN. Display "BUSY: US NN" in upper middle "MOUNT \$Dnn> -USnn"

Task A LOAD OPERATOR/UNIVERSAL PERSONALITY TO OTHER US

Step	Action	Response
200	Insert into \$F2 the AREA Zip disk you made back in Task 4.	
210	Select [MOUNT \$Dnn> -USnn]	SOURCE targets appear
220	Select [DEFAULT SOURCE]	VOL ID and targets appear
230	Select [EXECUTE COMMAND] It may be necessary to repeat the two previous steps.	CONSOLE STATUS AND ASSGN display "BUSY: US NN" "NTWRK VIEW <>" "MOUNT STD ABSTS FOR US NN"
250	Select [MOUNT STD ABSTS FOR US]	SOURCE targets appear
260	Select [ALTERNATE SOURCE]	Boxes with 1 and 2 appear
265	Select box [1]	VOL ID: "&Z1 and targets appear
270	Select [EXECUTE COMMAND] System Status Display appears on Other US.	CONSOLE STATUS AND ASSGN. Display "BUSY: US NN" in upper middle Status for Other US is "WARNING" because minimum data was available. US will be loaded with full database in a later task.

Appendix B – Save HG or PLCG Checkpoints

Perform the task described below if you wish to SAVE checkpoint data for an HG or PLCG and its Boxes. You must have created a Checkpoint Zip disk (CKPT) in Task 4 before performing this task. You should save the HG and Box checkpoint data on a disk first, and then save it on the HM. No HM volume has to be created as the HG checkpoint volume is preconfigured.

B.1 Task B – Save Checkpoint Data for Hiway Gateway and Its Boxes

Keyboard: Engr.

Lock: ENGR

Reference: SW11-601 and SW09-605

Personality Loaded: Universal

Task B SAVE CHECKPOINT DATA FOR HIWAY GATEWAY AND ITS BOXES

Step	Action	Response
SAVE HG/BOX CHECKPOINT		
10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
20	Your HG checkpoint disk into \$F1	<u>EG</u> : Volume ID = CKPT
30	Select [SYSTEM STATUS]	SYSTEM STATUS display
34	Select HG Node	Node Box is selected
40	Select [NTWK/HWY STATUS] Hiway/box status must be running	HIWAY STATUS display
44	Select [BOX CMND]	
48	Enter Box Number	<u>EG</u> : 05
52	Press <ENTER>	BOX STATUS DISPLAY
56	Select [ENABLE PROC]	Target changes color
60	Select [EXECUTE COMMAND]	Wait...Box Status: OK
62	Repeat steps 44 - 60 for other boxes	All Box Status = OK
64	Select [SAVE DATA]	[ALL BOXES] target appears.
68	Select [ALL BOXES]	SELECT SAVE BOX PATH display
70	If you wish to save checkpoints on Zip disk, go to step 80. If you wish to save checkpoints on an HM, select [DEFAULT SOURCE] and then go to step 100.	
80	Select [ALTERNATE SOURCE]	Boxes [1] and [2] appear
90	Select [1]	<u>EG</u> : Volume ID = CKPT
100	Select [EXECUTE COMMAND]	HIWAY STATUS display

Task B SAVE CHECKPOINT DATA FOR HIWAY GATEWAY AND ITS BOXES

Step	Action	Response
110	Select [START FUNCTION]	Asterisk appears by box number as data is saved. "Busy HWY nn
120	Wait for status of last applicable box to change to "SAVED."	

Appendix C – Save AM Checkpoint

Perform parts of the task described below to SAVE checkpoint data for an Application Module, A^XM, or a Computing Module. You must have created a checkpoint Zip disk (CKPT) in Task 4 before performing this task. You should always save the AM checkpoint data on a Zip disk first, and then save it on the HM. No HM volume has to be created as the AM and CM checkpoint volumes are preconfigured.

C.1 Task C – Save Checkpoint Data for AM or CM Modules

Keyboard: Engr.

Lock: ENGR

Reference: SW11-601

Personality Loaded: Universal

Task C SAVE CHECKPOINT DATA FOR AM OR CM MODULES

Step	Action	Response
10	Checkpoint Zip disk into \$F1	<u>EG</u> : Volume ID = CKPT
20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
30	Select [SYSTEM STATUS]	SYSTEM STATUS display
34	Select AM Node	Node Box is selected (Similar for Computing Module)
40	Select [NODE STATUS]	Application Modules display
50	Select NODE number	<u>EG</u> : 40
60	Select [SAVE DATA]	CHKPT DEST FOR NODE NN display
70	If you are saving checkpoint data on a Zip disk, go to step 90.	
80	If you have saved checkpoint data on a Zip disk and now wish to save it on HM, select [DEFAULT SOURCE] and go to step 110.	
90	Select [ALTERNATE SOURCE]	Boxes [1] and [2] appear.
100	Select [1]	
110	Select [EXECUTE COMMAND]	STATUS changes from “OK” to “SAVE” “OK” for the selected node in step 50.
120	Repeat steps 60 – 110 to save checkpoint data to the HM.	

Appendix D – Exception Building

The steps listed describe the basic exception build procedure of cloning a point from a master point that you have previously built using Parameter Entry Displays. Refer to Section 7 in the Data Entity Builder manual for variations.

D.1 Task D – Build Points by Exception for HG, AM, A^xM, CM, PLCG, and PM/APM/HPM Points

Keyboard: Engr.

Lock: ENGR

Reference: SW11-506 and SW11-511

Personality Loaded: Universal

Task D BUILD POINTS BY EXCEPTION FOR HG, AM A^xM, CM, PLCG AND PM/APM/HPM POINTS

Step	Action	Response
CREATE THE EXCEPTION BUILD SOURCE FILE		
05	If the appropriate Data Entity Builder Menu function is already selected, go to step 10; otherwise go to step 15.	EG: [HIWAY GATEWAY], [APPLICATION MODULE], [NETWORK INTERFACE MODULE], and so on.
10	Press <COMND>, then go to step 30	COMMAND Display
15	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
20	Select [BUILDER COMMANDS]	COMMAND Display
30	Select [PRINT ENTITIES]	PATHNAMES display
40	Key into REFERENCE PATHNAME port	EG: NET>HMV1>
50	Select [PRINT IDF ENTITIES]	Additional pathnames ports
60	Key into PATHNAME FOR IDF port	EG: HGPNTS (or AMPNTS, and so on.)
70	Key master point ID into PATHNAME FOR SELECTION LIST	EG: HG1001 (or AM101, and so on.)
80	Key a file name into PRINT DEVICE ID or DESTINATION PATHNAME	EG: HGEXCBLD.EB (or AMEXCBLD.EB, or PMEXCBLD.EB, and so on.)
100	Press <ENTER>	"BUSY...OPERATION COMPLETE"
110	Press <ESC>	"ESCAPED FROM DEB"
120	Key in an Edit command and the pathname of the source file	EG: ED NET>HMV1>HGEXCBLD.EB or ED NET>HMV1>AMEXCBLD.EB
130	Press <ENTER>	Point parameters displayed
140	Tab* down to &N item; tab over and type new point name over old point name.	EG: &N HG1002
150	Tab to other parameters that you want to change; key in changes.	EG: SLOTNUM = 02 EG: EUDESC = BBLS/HR

Task D BUILD POINTS BY EXCEPTION FOR HG, AM A^xM, CM, PLCG AND PM/APM/HPM POINTS

Step	Action	Response
160	Press <F3>	JUMP Menu
170	Press <F2>	Jump to End-Of-File
180	Press <RETURN> three times	Inserts three blank lines
190	Key in an end of source file indicator	<u>EG</u> : &E
BUILD ADDITIONAL POINTS		
200	If you want to clone another similar point, copy the parameters as follows.	If no other points need to be cloned, go to step 310.
210	Press <F3>	JUMP Menu
220	Press <F1>	Jump to Beginning-Of-File
230	Tab down to &T item	
240	Press <F2>	BLOCK Menu
250	Press <F1>	"MARK AT LINE: N"
260	<PAGE FWD>, <TAB> to end of parameters	<u>EG</u> : All parameters highlighted
*Press <HELP> to display the Text Editor Help functions; <Page FWD> to see more. Press <HELP> to return to the editor.		
270	Press <F2>	"BUSY"... "LINES WRITTEN: NN"
280	Place cursor on blank line just above the &E.	
285	Press <F4>	"LINES READ: NN" (parameters copied)
290	Tab down/over to &N item; Type new point name over old point name.	<u>EG</u> : &N HG1003
300	Tab to other parameters that you want to change; key in changes	<u>EG</u> : SLOTNUM = 03
305	If there are other similar points to clone, repeat Steps 210 – 300.	
WRITE SOURCE TO SOURCE FILE		
310	Press <F1>	QUIT Menu
320	Press <F2>	Message : "FILE UPDATED NN LINES WRITTEN"
PROCESS THE EB SOURCE FILE		
330	Hold <CTL> down and press <HELP>	COMMAND DISPLAY
340	Select [EXCEPTION BUILD]	Ports/targets appear
350	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HMV1>
360	Key into pathname for SOURCE file port	<u>EG</u> : HGEXCBLD.EB
370	Key into pathname for IDF port	<u>EG</u> : HGPNTS
375	Key into pathname for SELECTION LIST port	<u>EG</u> : blank to select all points in Exception Build File

Task D BUILD POINTS BY EXCEPTION FOR HG, AM A^xM, CM, PLCG AND PM/APM/HPM POINTS

Step	Action	Response
380	Key into OPT path for TEMPLATE port	<u>EG</u> : blank to select all points in Exception Build File
390	Press <ENTER>	"PROCESSING ENTITY: nnnnnnn"... "OPERATION COMPLETE"
395	Wait until the COMMAND DISPLAY reappears.	Points built by exception are added to the named IDF
SAVE/LOAD CLONED POINTS		
460	Select [LOAD MULTIPLE]	Additional ports/target appear
470	Key into REFERENCE PATH NAME port	<u>EG</u> : NET>HVM1>
475	Select [with OVERWRITE]	
480	Key into pathname for IDF port	<u>EG</u> : HGPNTS, AMPNTS, APMPTNS , and so on.
485	Key into pathname for SELECTION LIST port	<u>EG</u> : blank for all points in IDF
487	Press <ENTER>	"WAIT" in yellow; "ENTITY NAMES". Pt. Config data written to IDF. "OPERATION COMPLETE"
495	If errors occur, select the point on the error display. Make corrections; then repeat Steps 460 – 495.	

Appendix E – Create Exception Build Files from IDFs

The steps listed describe the basic procedure to generate exception build files from Intermediate Data Files (IDFs). Refer to Section 7 in the *Data Entity Builder* manual for basic information on the Command Menu functions. This procedure does not destroy the IDFs. Intermediate Data Files have the suffix: .db. Exception Build Files have the suffix .eb.

E.1 Task E – Create Exception Build Files from IDFs

Keyboard: Engr.

Lock: ENGR

Reference: SW11-506 and SW11-511

Personality Loaded: Universal

Task E CREATE EXCEPTION BUILD FILES FROM IDFS

Step	Action	Response
05	If any Data Entity Builder Menu function is already selected, go to step 10; otherwise go to step 15.	<u>EG:</u> Any ped or the Command Display.
10	Press <COMND>, then go to step 30	COMMAND Display
15	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
20	Select [BUILDER COMMANDS]	COMMAND Display
30	Select [PRINT ENTITIES]	PATHNAMES display
40	Key into REFERENCE PATHNAME port	<u>EG:</u> NET>HMV1>
50	Select [PRINT IDF ENTITIES]	Additional pathnames ports
60	Key into pathname for IDF port	<u>EG:</u> HGPNTS (or AMPNTS, and so on)
65	If you do not remember all the IDF names, this procedure will list them— press <ESC>. Key in a Data Out command to your printer. Press <ENTER>. Key in a List command with a .db suffix, press <ENTER>. Key in another DO command to stop printing. Press <ENTER>. To return to the Command Display, Hold <CTL> down and press <HELP>	<u>EG:</u> DO \$P1 "DATA OUT COMPLETE" <u>EG:</u> LS NET>HMV1>*.db DB files are listed and printed. <u>EG:</u> DO "DATA OUT COMPLETE"
70	Leave Pathname for SELECTION LIST blank	
80	Key a file name into PRINT device ID or DESTINATION pathname	<u>EG:</u> HGPNTS.EB (or AMPNTS.EB, or PMPNTS.EB, HPMPNTS, and so on)
100	Press <ENTER>	"WAIT...PROCESSING ENTITY: nnnn OPERATION COMPLETE"

Task E CREATE EXCEPTION BUILD FILES FROM IDFS

Step	Action	Response
110	Repeat steps 60 - 100 for other IDFs that you want to convert to .EB files. If you want to see which exception build files have been generated, press <ESC>. Key in a List command with an .eb suffix (EG: LS NET>HVM1>*eb).	To review the contents of an .eb file, escape as before, then key in an edit command <u>EG</u> : ED NET>HVM1>HGPNTS.EB
114	To exit the Text Editor, press F1 twice	"Do you really wish to abort this edit?"
118	Key in the letter y	Command Processor Menu
122	To return to the Command Menu, Hold <CTL> down and press <HELP>	

Process Solutions

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